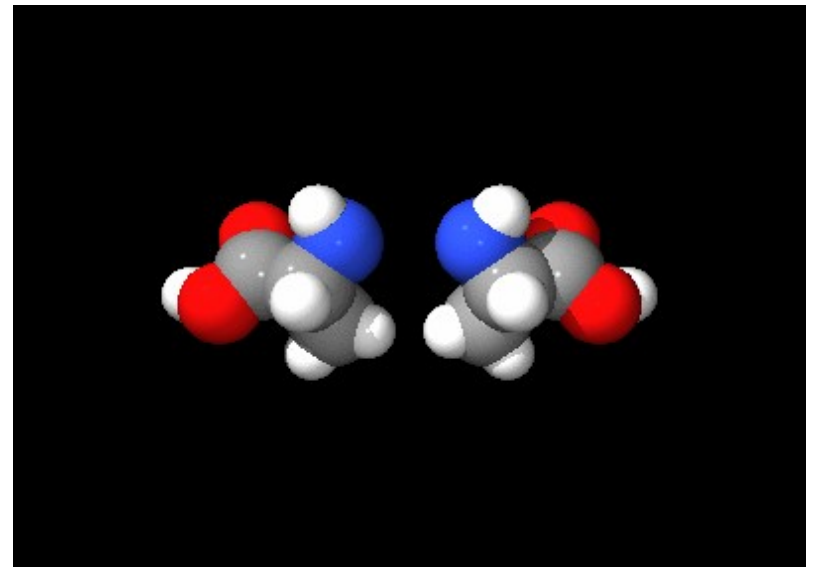
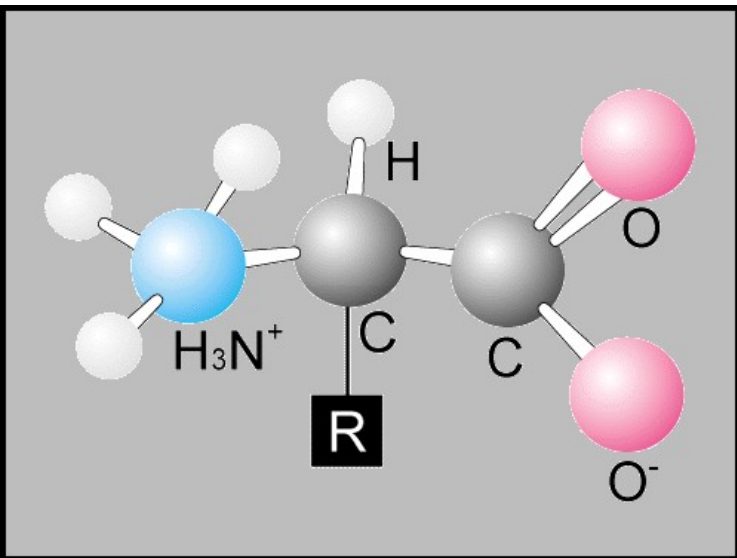


4. Amino Acids and Proteins



Learning objectives

At the end of this session the students will be able to:-

- Describe the contribution of each type of R group of the protein amino acids to their chemical properties.
- Explain and illustrate the primary, secondary, tertiary, and quaternary structure of proteins.
- Describe the most important structural similarities and differences between myoglobin and hemoglobin.
- Explain the role of 2,3-bisphosphoglycerate (BPG) in oxygen binding and delivery.

Functions of amino acids

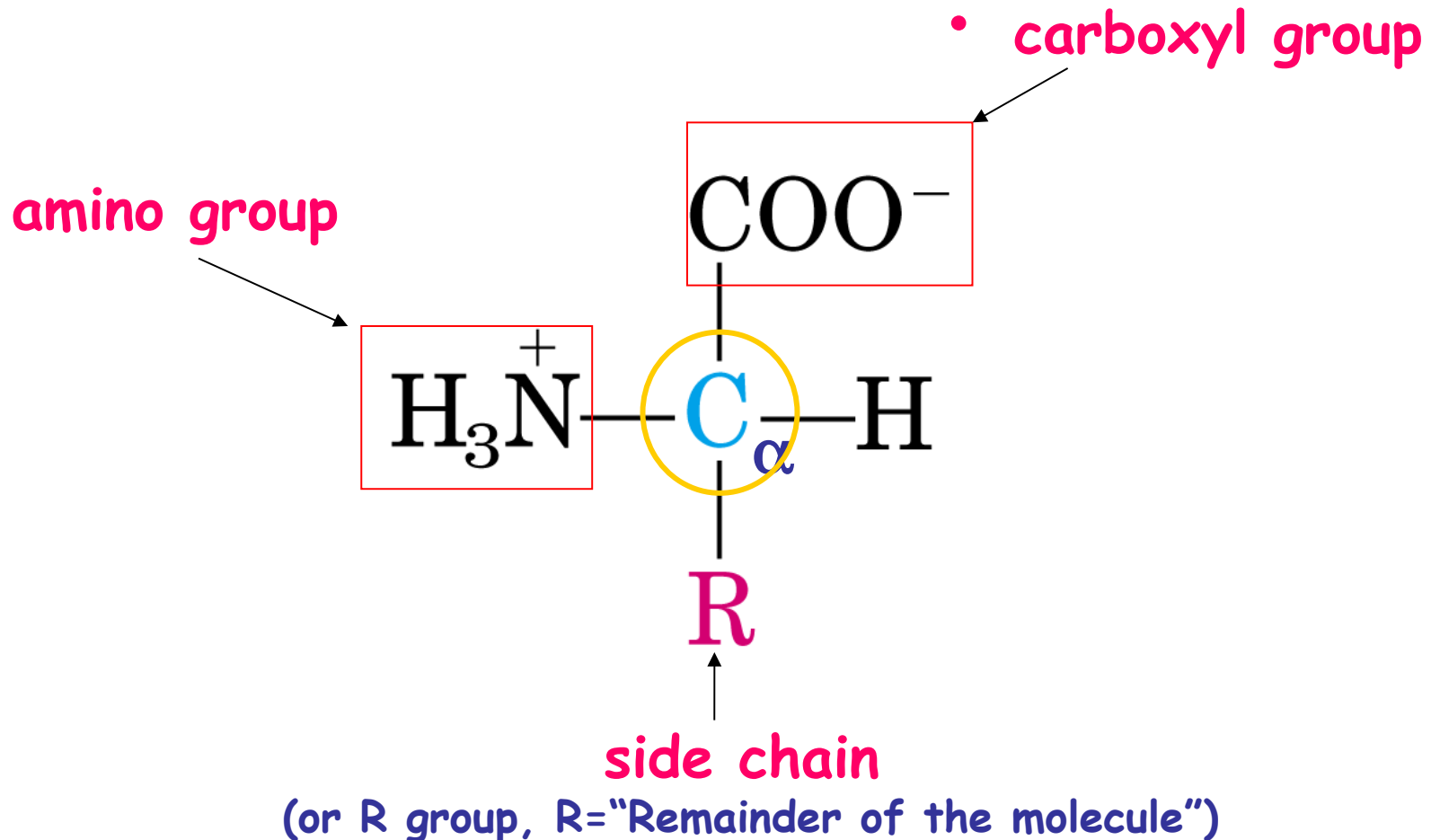
- A variety of roles in metabolism
 - the building blocks of proteins
 - the properties of proteins are determined by the physical and chemical properties of their monomer units, the amino acids.
 - forming parts of coenzymes
 - as precursors for the biosynthesis of molecules such as heme

Standard and nonstandard amino acids

- More than **300** different amino acids have been described in nature.
 - **Standard α -amino acids:**
 - Only **20** *are commonly found as constituents of proteins*
 - **Nonstandard amino acids:**
 - Amino acid derivatives found in protein
 - Non-protein amino acid

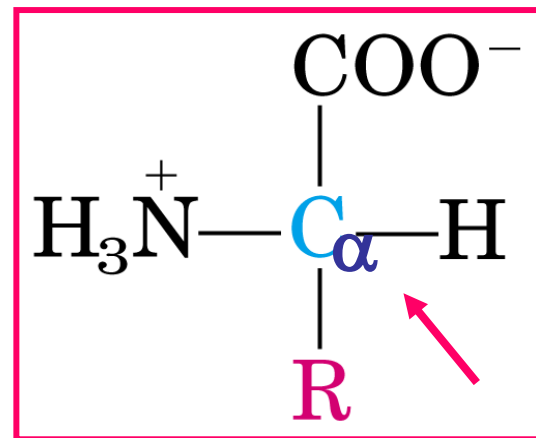
1. General structure of α -amino acids

(Very important!)



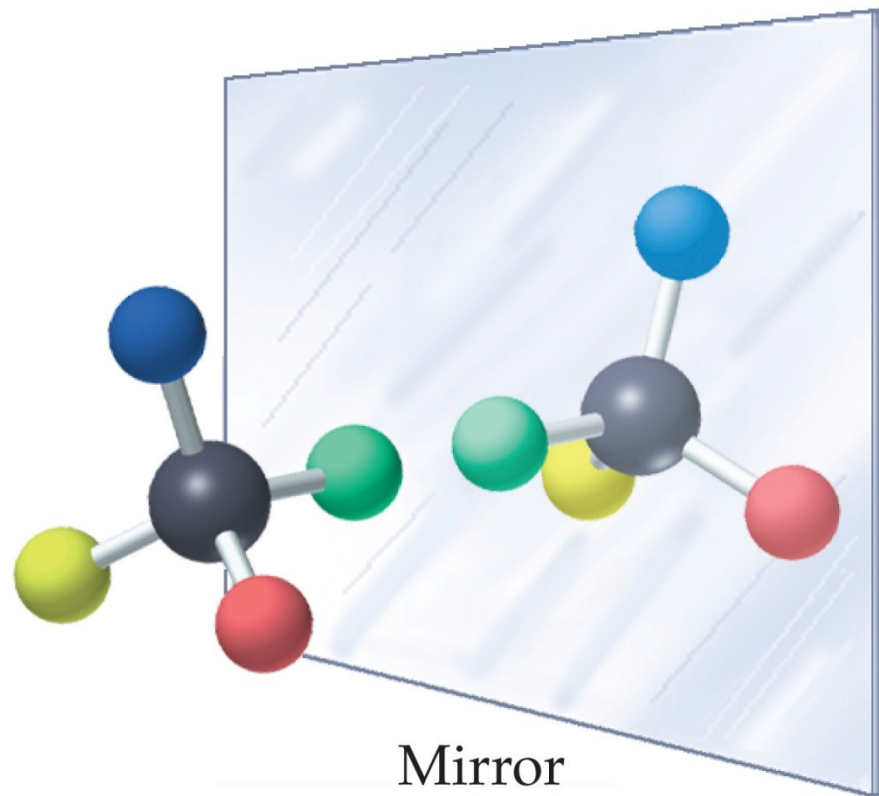
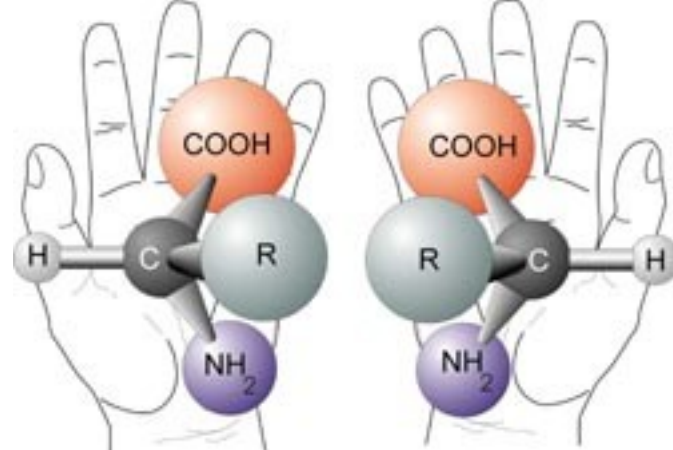
2. Stereochemistry of amino acids

- Configuration
- For all the common amino acids except *Glycine*, the α -carbon is bonded to four different groups.
- The α -carbon atom is thus a **chiral center**.



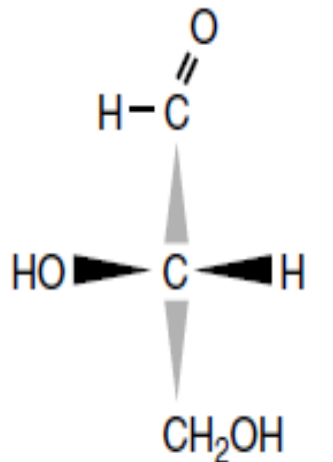
Chiral molecule

- Chiral molecule:
 - if a molecule has an atom bonded to **four different groups**, it can be chiral.
- Enantiomers:
 - a **chiral** molecule, has left- and right-handed isomers, called enantiomers.

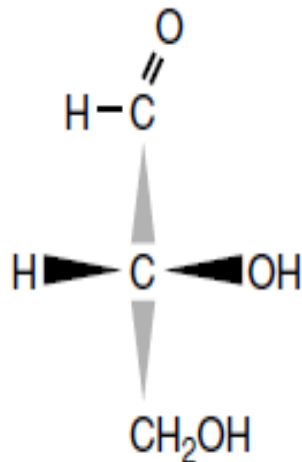


- The two enantiomers of each amino acid are designated by **D,L system** according to the D- and L-glyceraldehyde.

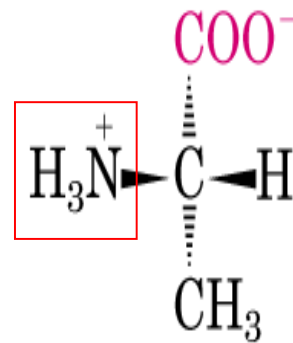
– D: Dextrorotation; L: Levorotation



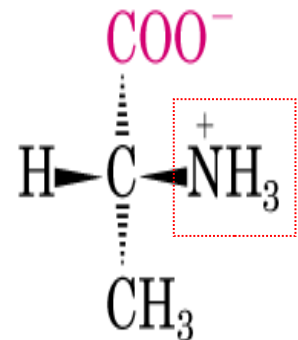
L-Glyceraldehyde



D-Glyceraldehyde



L-Alanine



D-Alanine

- Only the **L-amino acids** have been found in proteins.
- (D-isomers have been found only in small peptides of bacteria cell walls or in some peptide antibiotics).

3. The Classification of 20 standard Amino Acids (*very important!!!*)

- The name and abbreviation of amino acids
 - All the AAs were given a trivial (common) name.
 - Glutamate from wheat gluten.
 - Tyrosine from cheese (“tyros” in Greek).
 - Each AA is given a 3 letter abbreviation and 1 letter symbol.
 - They often take the first three letters and the first letter. When there is confusion, an alternative is used.
- They should be remembered !

Names and Abbreviations of the Standard Amino Acids

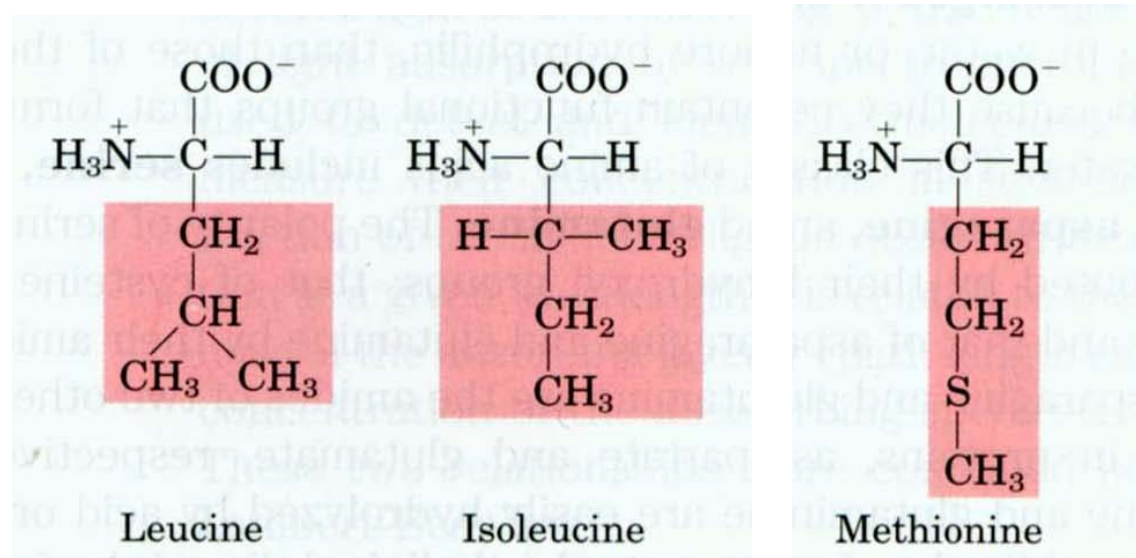
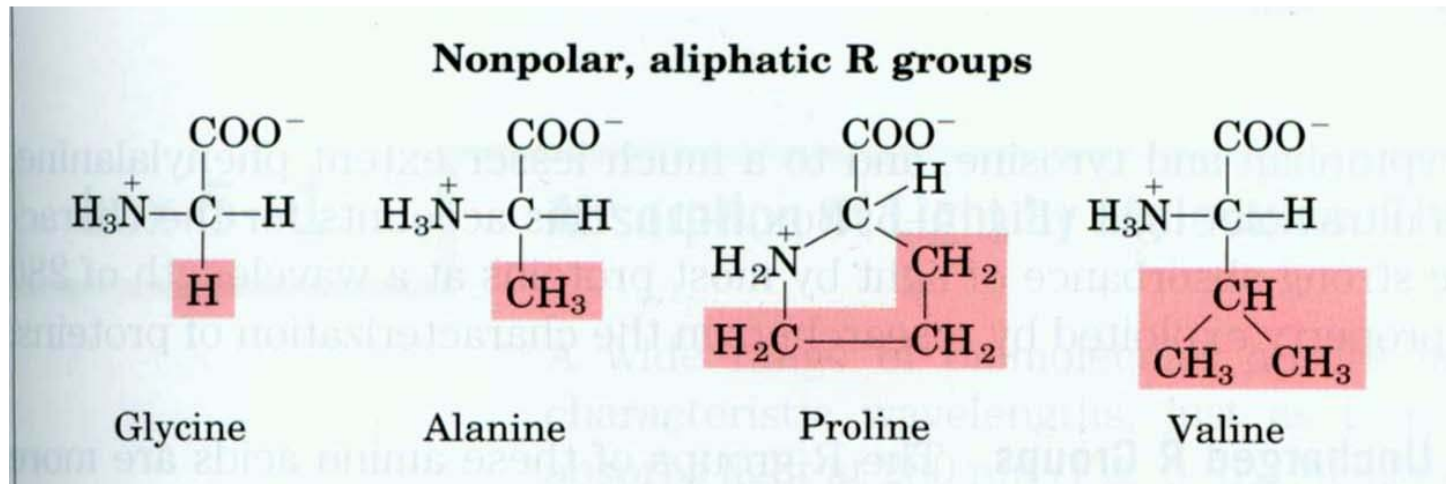
Amino Acid	Three-Letter Abbreviations	One-Letter Abbreviations
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartic acid	Asp	D
Cysteine	Cys	C
Glutamic acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

Classification of Amino Acids

- Based on **polarity**
 - Nonpolar, aliphatic amino acids
 - Polar, uncharged amino acids
 - Aromatic amino acids
 - Acidic amino acids
 - Basic amino acids

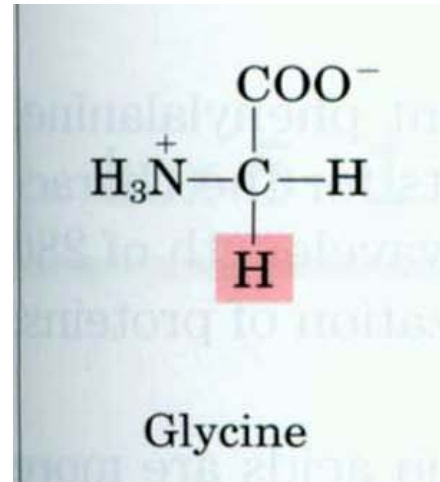
3.1 Nonpolar, aliphatic amino acids

•Gly, Ala, Pro, Val, Leu, Ile and Met



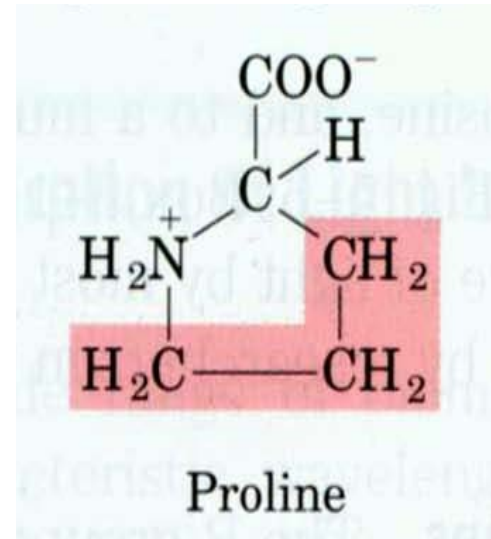
- **Gly:**

- **R group: hydrogen.**
- simplest structure.
- its very small side chain makes **no real contribution** to hydrophobic interactions.
- functions as an inhibitory neurotransmitter in the brainstem and spinal cord.
- **symmetric, not chiral.**



- **Pro:**

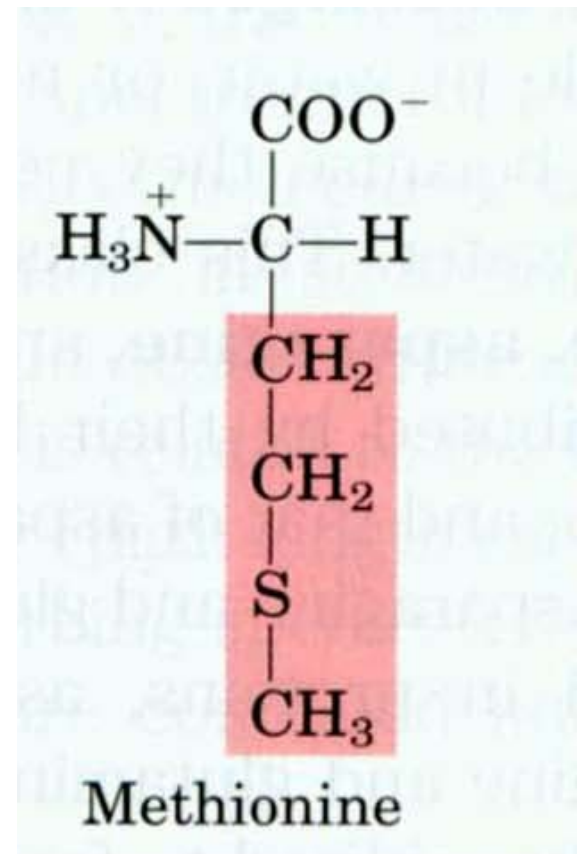
- **Imino acid**. i.e. secondary amino group
- **five-membered ring structure, rigid in conformation.**
- whose presence in a protein **disrupts** normal secondary structure.



...cont'd

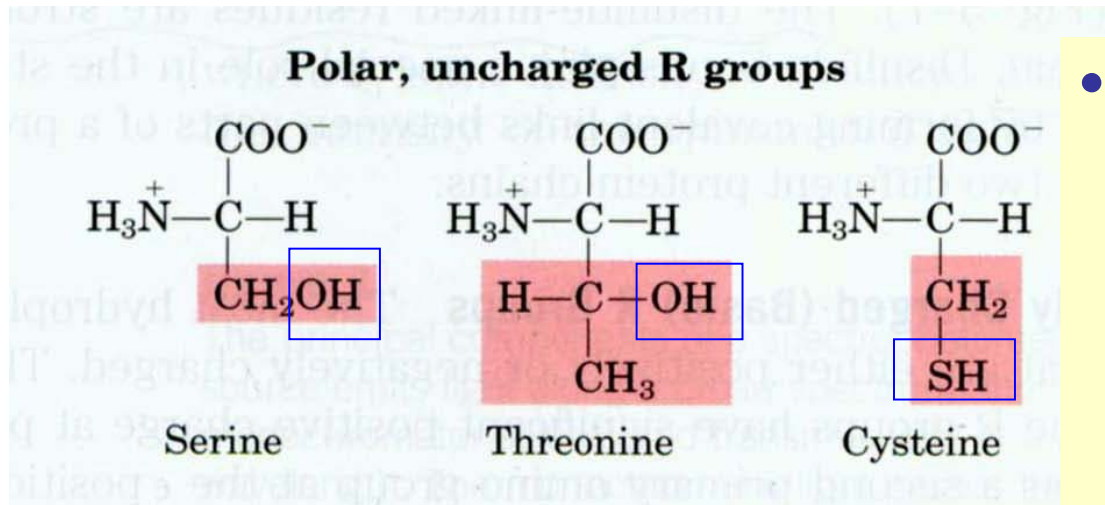
- **Methionine**

- one of the two **sulfur-containing** amino acids,
- has a nonpolar **thioether group** in its side chain.
- is part of S-adenosyl-methionine (SAM), a methyl donor in biochemical pathways

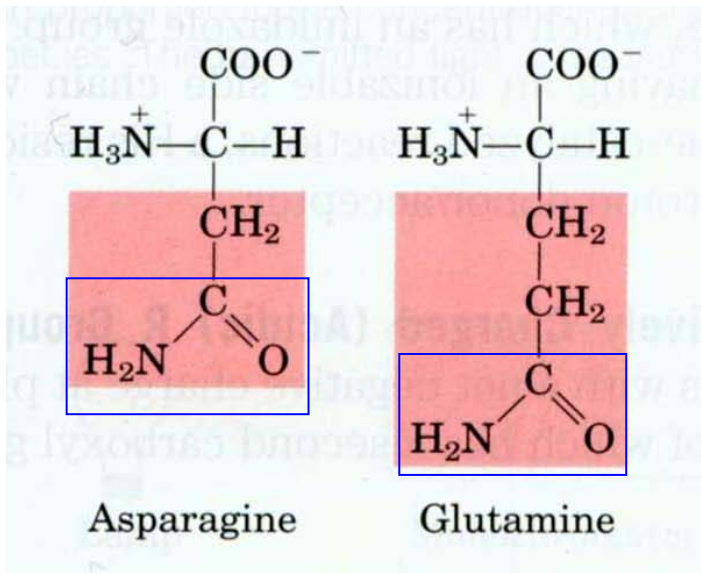


3.2 Polar, uncharged amino acids

➤ Ser, Thr, Cys, Asn, and Gln.

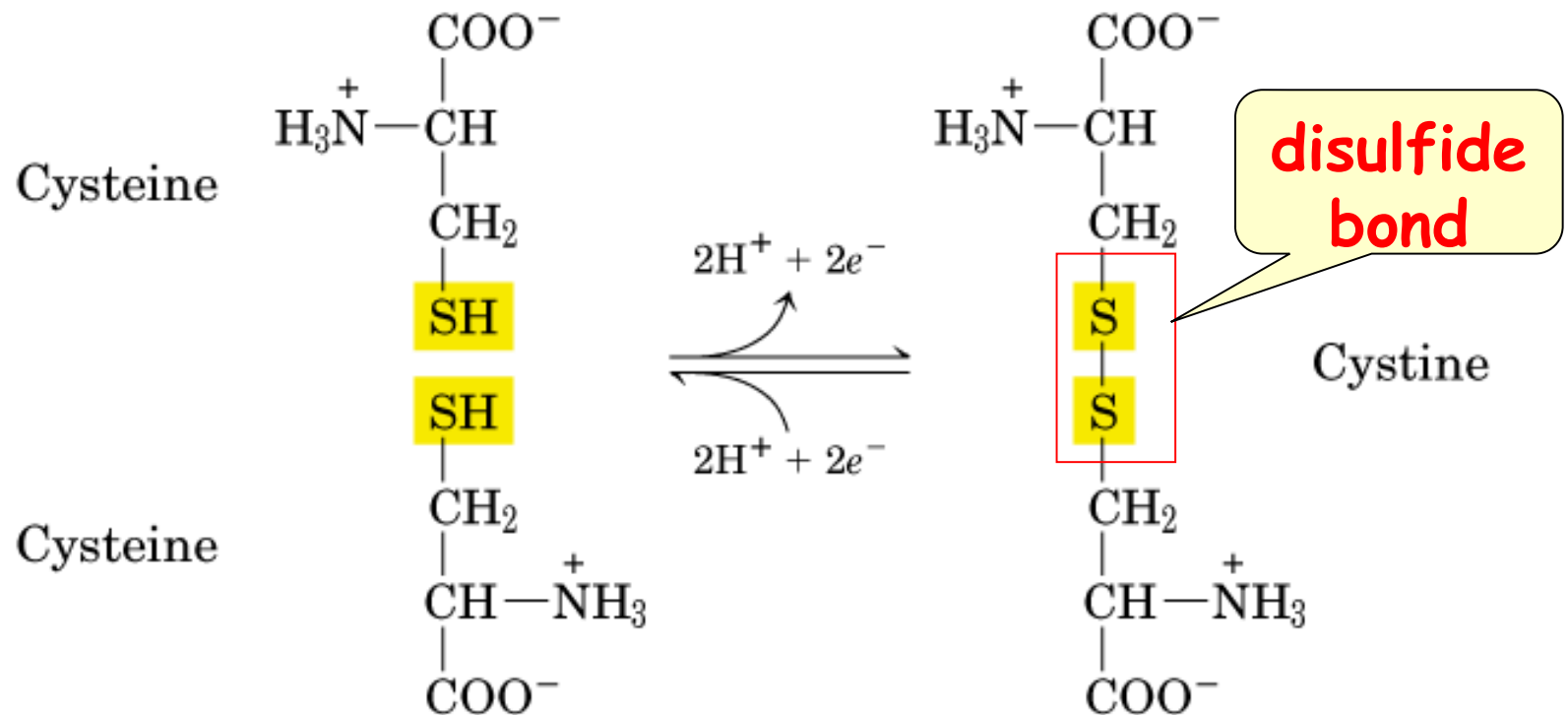


- R group: **hydrophilic**
- Hydroxyl groups
- Sulfhydryl group
- Amide groups



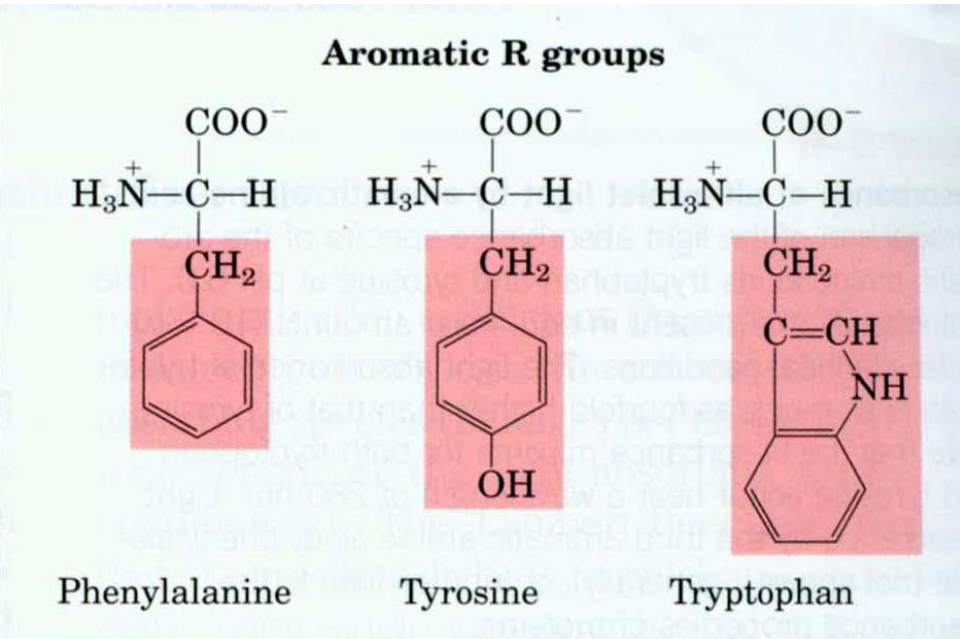
➤ Disulfide bonds

- -SH group of two Cys in proteins can be oxidized to form a covalent disulfide bond.
- **Disulfide bonds:** play a special role in the structures of many proteins by forming covalent links.



3.3 Aromatic amino acids

- **Phe, Tyr and Trp.**
 - Phe and Tyr: benzene rings.
 - Tryptophan: indole ring.

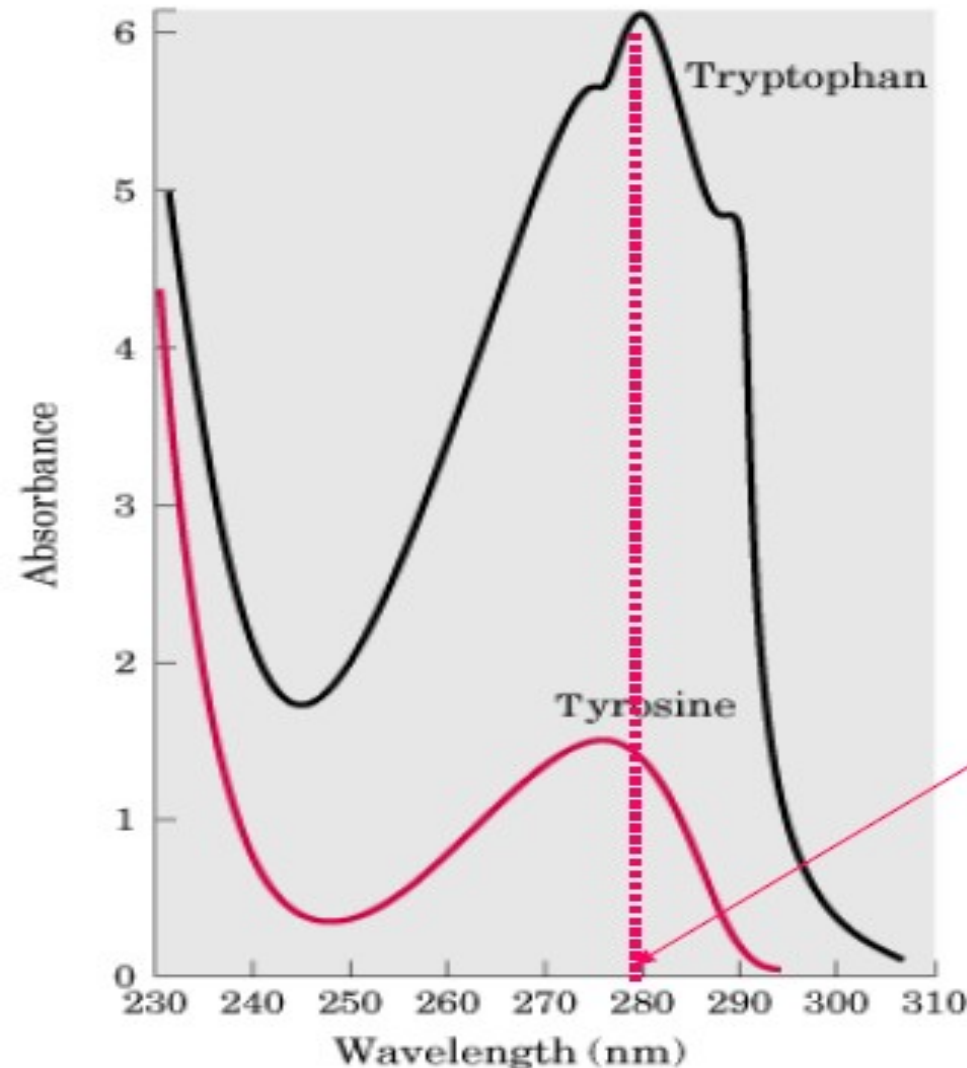


Phenylalanine and tyrosine are *precursors* for catecholamine's.

Tryptophan can form serotonin and niacin.

- The **-OH group in Tyr** is an important functional group in proteins. (phosphorylation, hydrogen bond, etc), **polar**

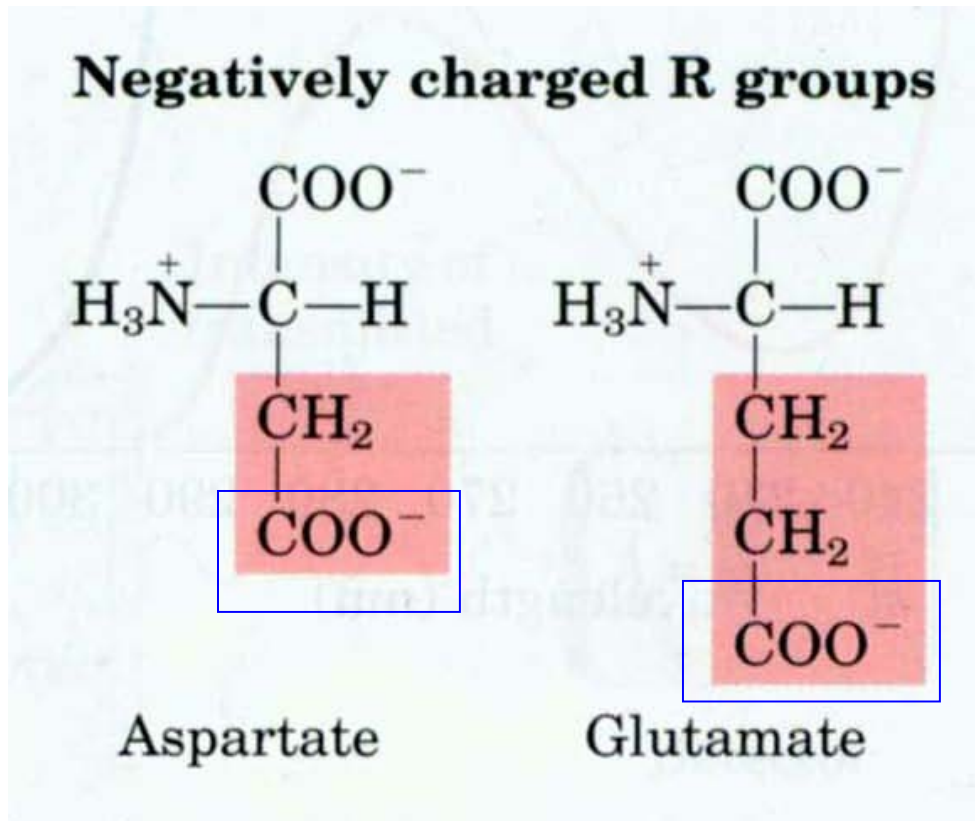
- They are jointly responsible for the **light absorption** of proteins at **280 nm**.



- Proteins in solution absorb UV light with absorbance maximum at **280nm**.
- **Measuring protein content** by **photospectrometry**.

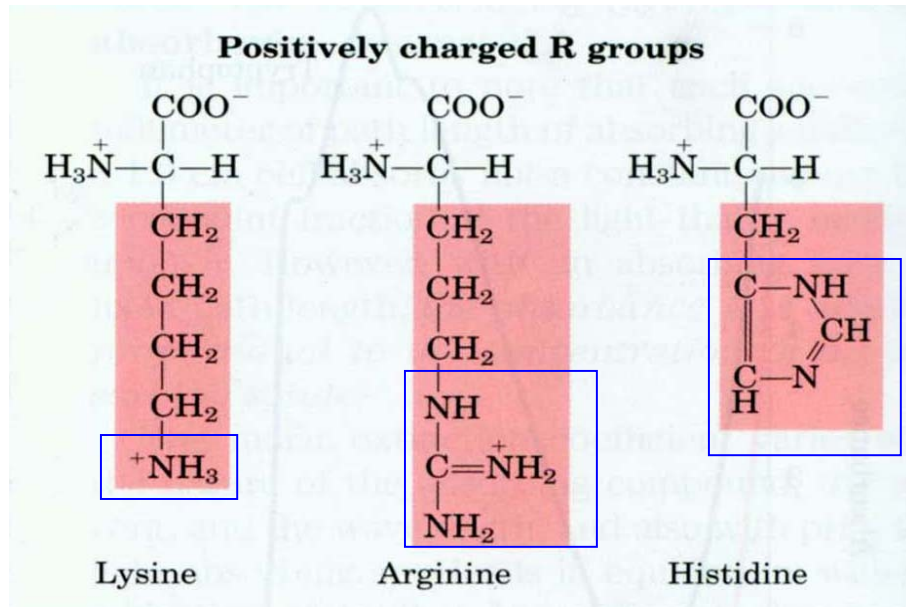
3.4 Acidic amino acids

- **Asp and Glu**
 - Have **carboxyl** in their R groups.
 - a net negative charge at pH 7.0



3.5 Basic amino acids

➤ Lys, Arg, and His.



➤ R groups

➤ Amino

➤ Guanidino

➤ Imidazole

➤ Positive charged R groups at pH 7.0

✓ **Histidine** is the **only common amino acid** having an ionizable side chain with **a pK_a near neutrality**.

✓ In many enzyme-catalyzed reactions, a His residue facilitates the reaction by serving as **a proton donor/acceptor**.

Note these structural features

1. All 20 are α -amino acids
2. For 19 of the 20, the α -amino group is primary; for proline, it is secondary (imino acid)
3. Except glycine, the α -carbons for 19 of them are asymmetric (or chiral).

- **Nonpolar Amino Acids**

Ala Val Leu Ile Gly Pro Met Phe Trp

- **Polar, Uncharged Amino Acids**

Ser Thr Cys Asn Gln

Tyr

Aromatic

- **Acidic Amino Acids**

Asp Glu

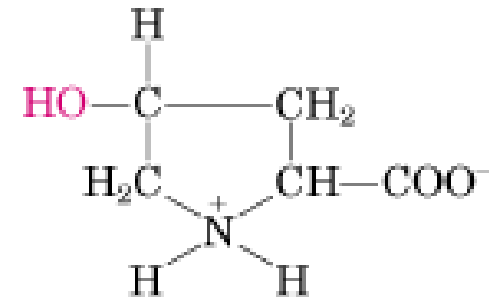
- **Basic Amino Acids**

Lys Arg His

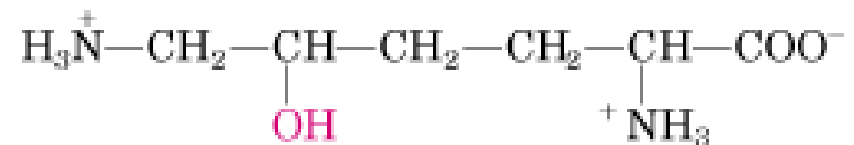
3.6 Nonstandard amino acids

- Amino acid derivatives found in proteins

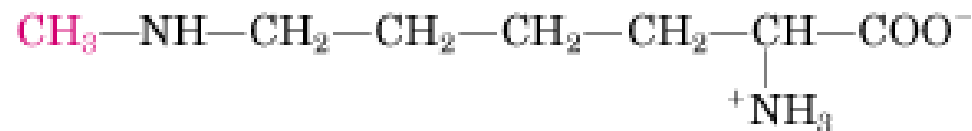
- 4-Hydroxyproline and 5-hydroxylysine in collagen.
- 6-N-Methyllysine in myosin.



4-Hydroxyproline

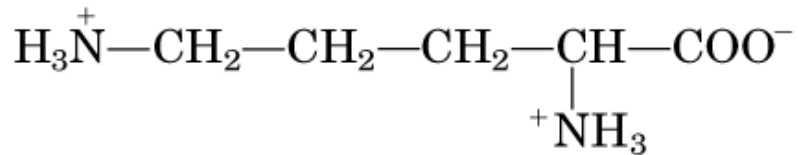


5-Hydroxylysine

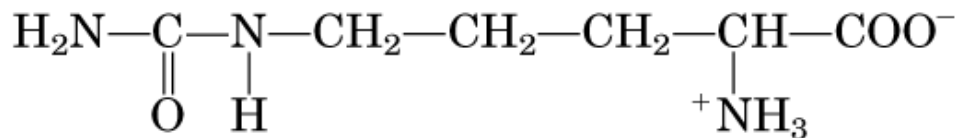


6-N-Methyllysine

➤ Many additional nonstandard amino acids are found **in cells**, but **not in proteins**



Ornithine



Citrulline

➤ **Ornithine and citrulline**

➤ **Intermediates in amino acid metabolism.**

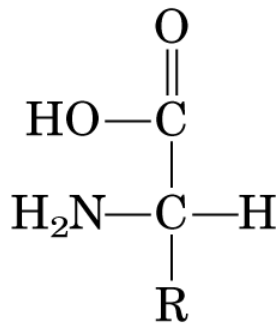
3.7 Essential and non-essential amino acids

- Essential amino acids (or indispensable amino acids):
 - Cannot be synthesized by the humans, must be supplied in the diet
 - 8: Phe, Val, Thr, Trp, Ile, Met, Leu, Lys
- Semi-essential amino acids:
 - 2: His and Arg
 - Required by infants and growing children

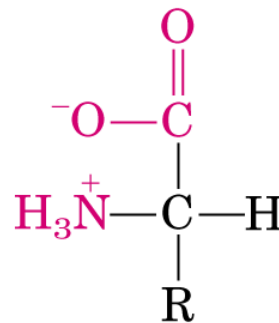
4. Acid/base properties of AAs

(Very important!)

- Amino acid has both a basic amine group and an acidic carboxylic acid group.
- In neutral solution (pH 7.0), the amino acid contains a negative charge and a positive charge. It is called a *zwitterion* (German for "hybrid ion").



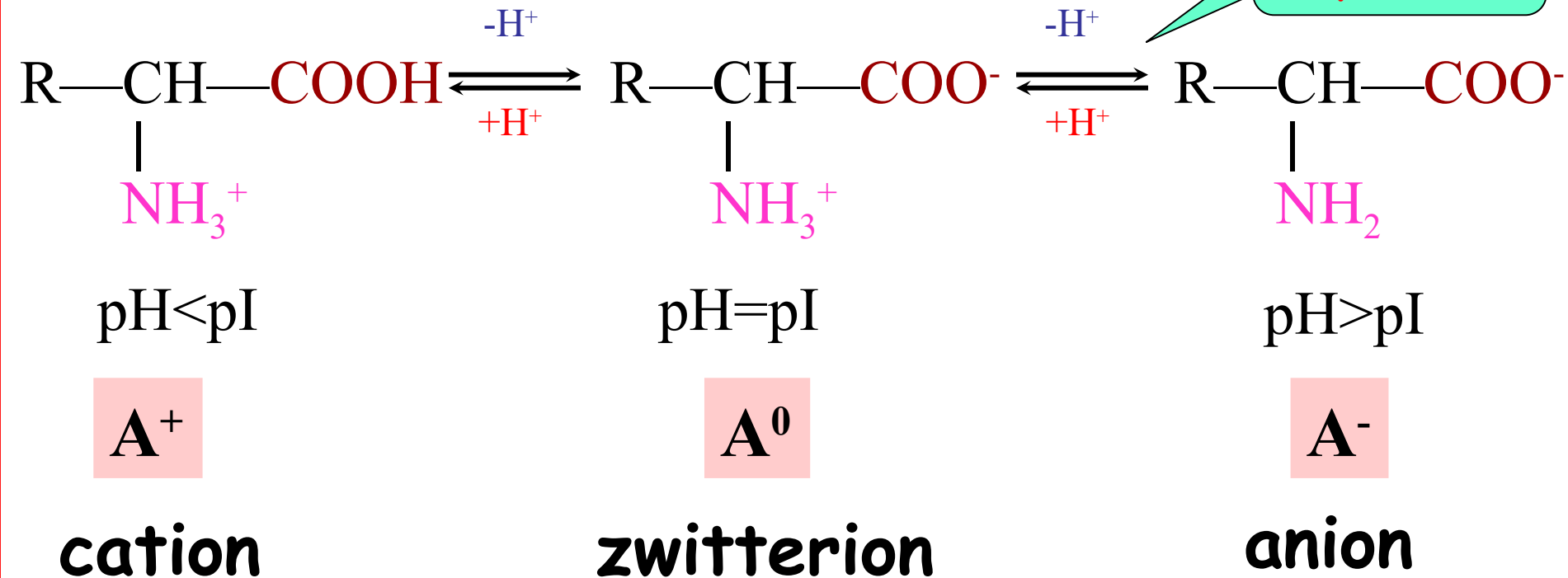
Nonionic
form



Zwitterionic
form

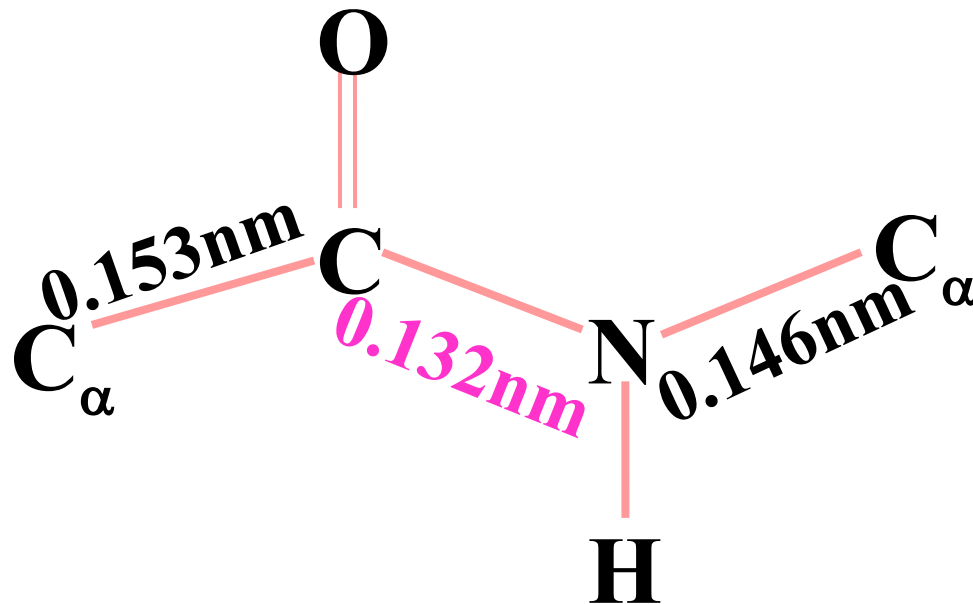
- AAs ionize to various states depending on pH values.
- **pI**: there is a **specific pH** (designated **isoelectric point, pI**) at which an AA has equal positive and negative charge (**no net electric charge**) .

Very important!!!



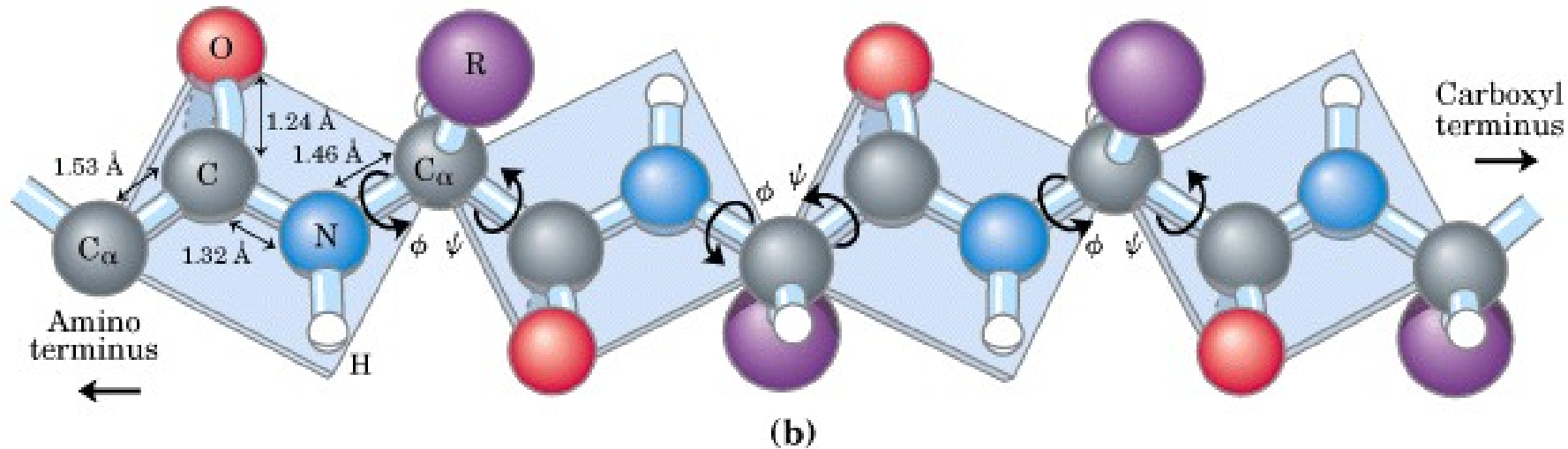
Characteristics of the peptide bond

- The peptide bond **have partial double bond feature**
 - about **0.132nm** (C-N single bond, 0.149nm; C=N double bond, 0.127nm),
 - *rigid and unable to rotate freely.*



➤ The peptide bond is planar, trans-configuration and uncharged.

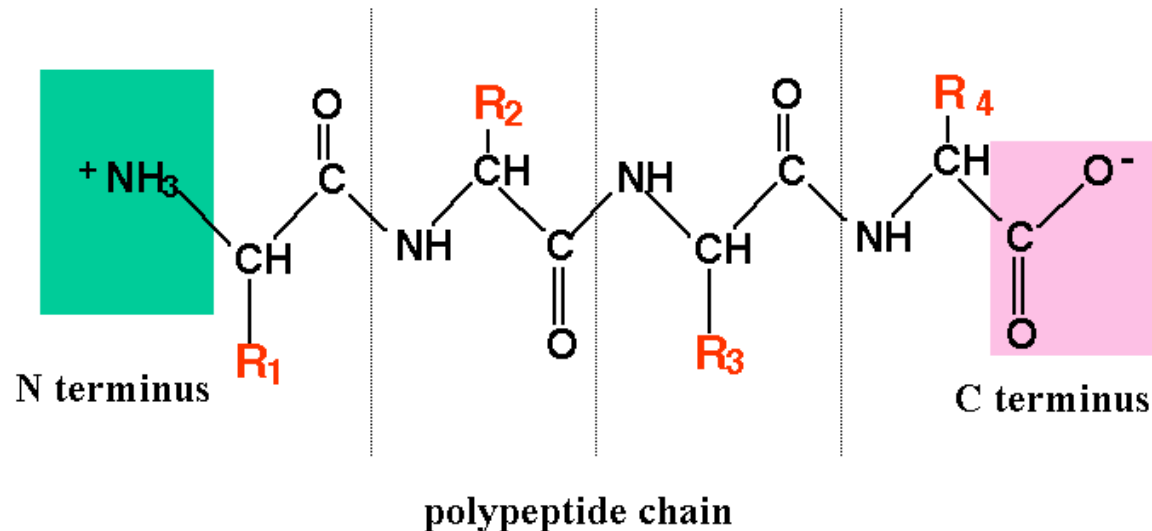
➤ **Peptide plane:** the six atoms attached to the peptide bond are coplanar, the carbonyl oxygen and the amide hydrogen are in trans positions.



The peptide chain is directional.

- **Amino-terminal or N-terminal:** the end having a free α -amino group.
- **Carboxyl-terminal or C-terminal:** the end having a free α -carboxyl group.
- By convention, the **N-terminal** is taken as the beginning of the peptide chain, and put **at the left (C-terminal at the right)**.

Peptide = chain of amino acids



Polymers of amino acids

- Peptides can be classified according to how many amino acids they contain
 - Dipeptide: 2 amino acid residues, tripeptide: 3 residues, and so on
 - Oligopeptide: 12~20 residues
 - Polypeptide: many residues

5. Proteins

- The name of the protein derives from the Greek *protos*, meaning “first” or “foremost”.
- Proteins mediate virtually every process that takes place in a cell, exhibiting an almost endless diversity of functions.

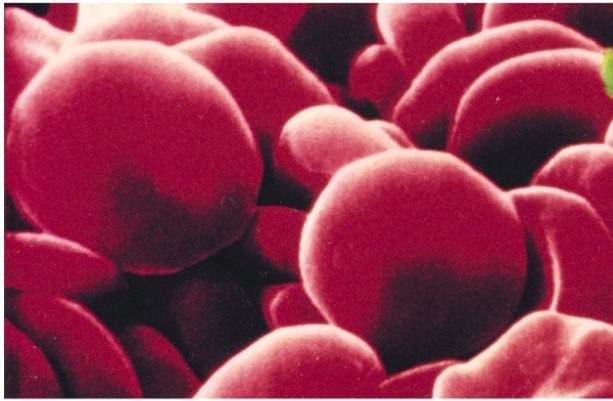
Functions

- Involved in almost everything
 - structure (keratin, collagen, elastin)
 - enzymes
 - carriers & transport (membrane channels)
 - receptors & binding (defense)
 - contraction (actin & myosin)
 - signaling (hormones)
 - storage (bean seed proteins)
 - immune response (immunoglobulin)



(a)

- Fireflies emit light catalyzed by **luciferase**



(b)

- Erythrocytes contain a large amount of **hemoglobins**, the oxygen-transporting proteins.



(c)

- Rhinoceros, the protein **keratin** is the chief structural components of hair, horn, wool, nails and feathers.

TABLE**Biological functions of proteins**

Function	Examples	Occurrence or role
Catalysis	lactate dehydrogenase	Oxidizes lactic acid
	cytochrome c	Transfers electrons
	DNA polymerase	Replicates and repairs DNA
Structure	viral-coat proteins	Sheath around nucleic acid of viruses
	glycoproteins	Cell coats and walls
	α -keratin	Skin, hair, feathers, nails, and hooves
	β -keratin	Silk of cocoons and spiderwebs
	collagen	Fibrous connective tissue
	elastin	Elastic connective tissue
Storage	ovalbumin	Egg-white protein
	casein	A milk protein
	ferritin	Stores iron in the spleen
	gliadin	Stores amino acids in wheat
	zein	Stores amino acids in corn
Protection	antibodies	Form complexes with foreign proteins
	fibrinogen	Involved in blood clotting
	thrombin	Involved in blood clotting
Regulation	insulin	Regulates glucose metabolism
	growth hormone	Stimulates growth of bone
Nerve impulse transmission	rhodopsin	Involved in vision
	acetylcholine receptor protein	Impulse transmission in nerve cells
Movement	myosin	Thick filaments in muscle fiber
	actin	Thin filaments in muscle fiber
	dynein	Movement of cilia and flagella
Transport	hemoglobin	Transports O_2 in blood
	myoglobin	Transports O_2 in muscle cells
	serum albumin	Transports fatty acids in blood
	transferrin	Transports iron in blood
	ceruloplasmin	Transports copper in blood

5.1 The chemical composition and classification of proteins

- The element composition in proteins are:
C: 50%; H: 7%; O: 23%; N: 16%; S: 0~3%
- $\text{Protein(g)} = \text{Nitrogen in Protein(g)} \times 6.25$

- This approach is based on **two assumptions**:
 - Dietary carbohydrates and fats do not contain nitrogen;
 - Nearly all of the nitrogen in the diet is present as amino acids in proteins.

- The classification of proteins
 - Proteins can be classified as simple proteins and conjugated proteins
 - Simple proteins
 - Composed of only amino acids.
 - Conjugated proteins
 - Contain permanently associated chemical groups (called as prosthetic group) in addition to amino acids.

Conjugated Proteins

Class	Prosthetic group(s)	Example
Lipoproteins	Lipids	β_1 -Lipoprotein of blood
Glycoproteins	Carbohydrates	Immunoglobulin G
Phosphoproteins	Phosphate groups	Casein of milk
Hemoproteins	Heme (iron porphyrin)	Hemoglobin
Flavoproteins	Flavin nucleotides	Succinate dehydrogenase
Metalloproteins	Iron	Ferritin
	Zinc	Alcohol dehydrogenase
	Calcium	Calmodulin
	Molybdenum	Dinitrogenase
	Copper	Plastocyanin

The shape and the size of proteins

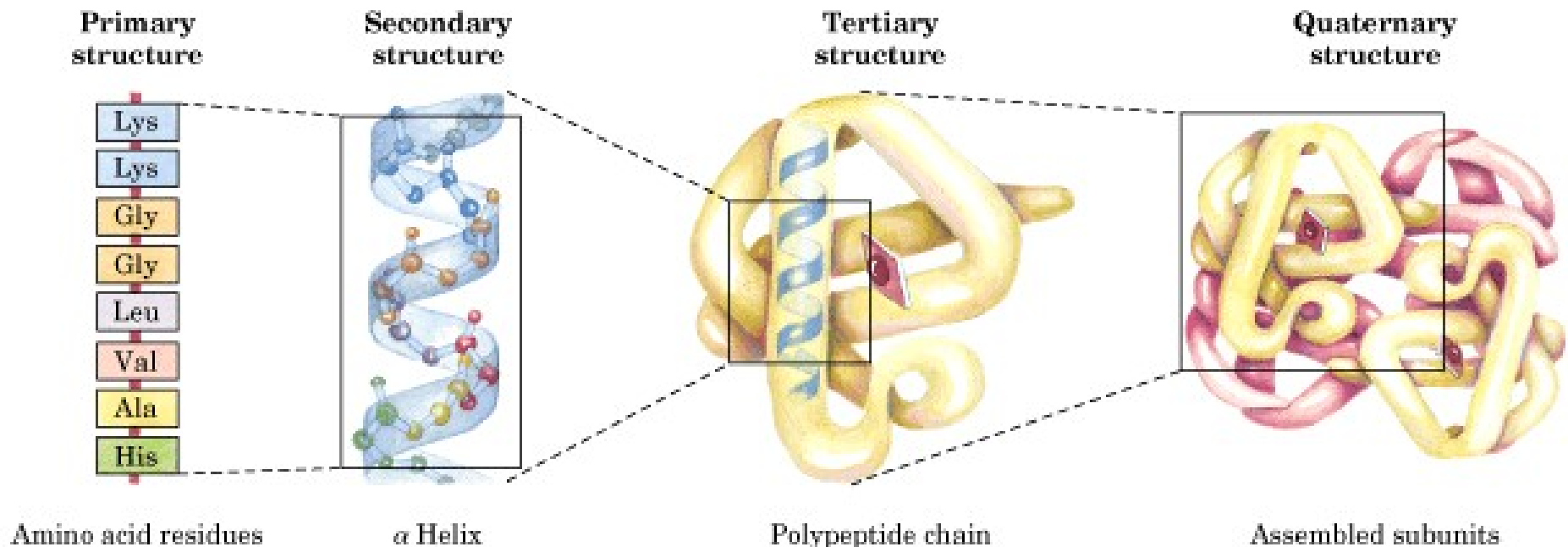
- Proteins are described as fibrous proteins or globular proteins by shape.
 - Fibrous proteins mainly function as structural components in living organisms, such as collagen. They are less soluble in water.
 - For example: collagens, elastins, keratins.
 - Globular proteins are generally well soluble in water and have various functions such as enzyme catalysis.
 - For example: albumins, globulins, histones and so on.

- The size of proteins are relatively large.
 - Proteins have molecular weight varied from 6,000 ~ 1,000,000 daltons, or larger.
 - The unit of weight is the dalton, one-twelfth the weight of an atom of ^{12}C .
 - Monomeric proteins: proteins consist of a single polypeptide chain.
 - Oligomeric/multimeric proteins: have two or more polypeptides associated noncovalently.
 - Each polypeptides is called as a subunit of the protein.

5.2 The Structure of Proteins

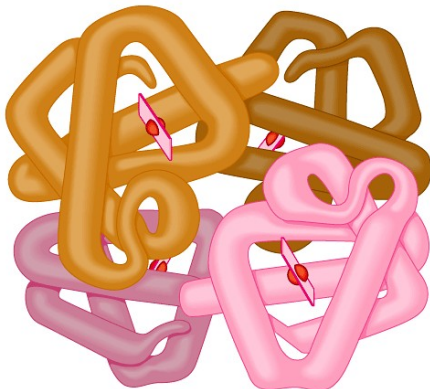
(Very important !!!)

•Protein structures have conventionally been considered at four different levels.

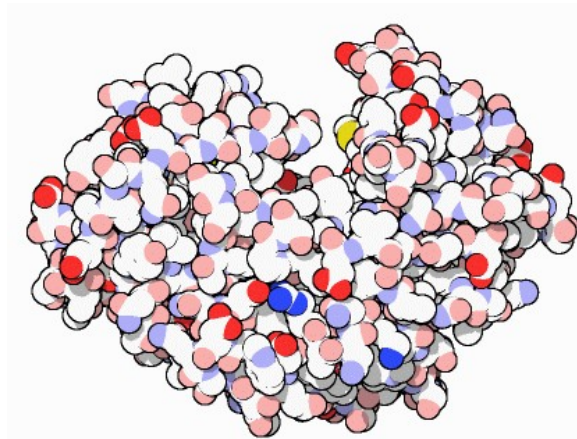


• Each protein usually has *one native conformation*

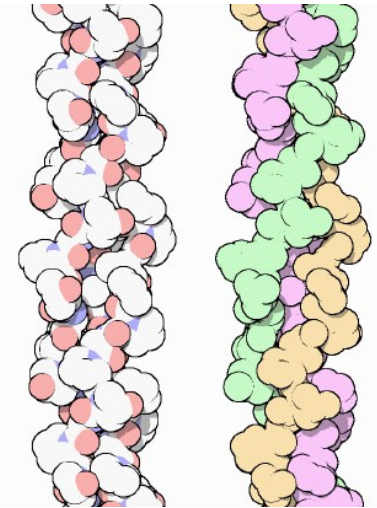
- Under physiological conditions of solvent and temperature, each protein folds spontaneously into one **3D conformation**, called the native conformation.
- This conformation is usually the **most stable thermodynamically**, and Usually **only the native conformation is functional**.



hemoglobin



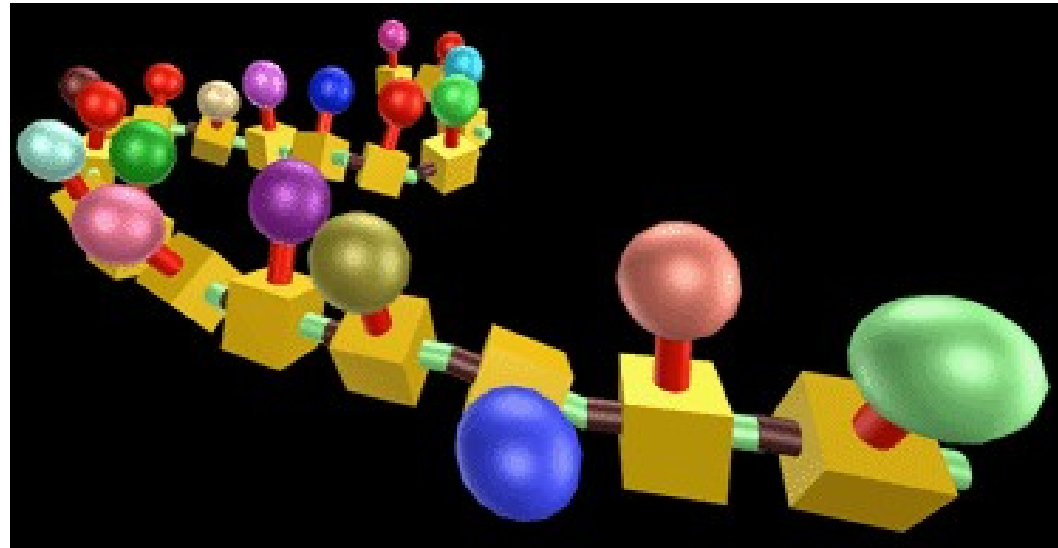
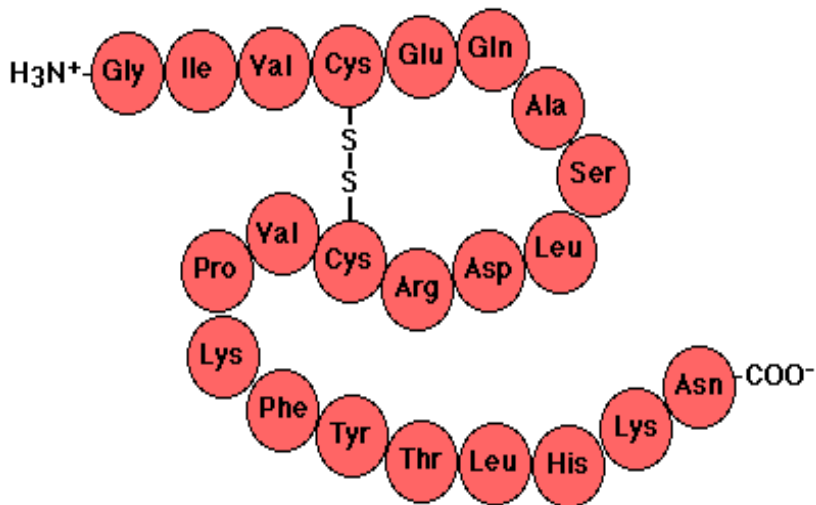
pepsin



collagen

5.2.1 The primary structure

- The primary structure of the protein is *the AA sequence in a protein*.
- The covalent bonds to maintain the primary structure
 - peptide bonds, disulfide bonds
- The peptide chain is known as the *backbone*, and the "R" groups are known as *side chains*.
- The primary structure is usually shown using abbreviations (three letters or one letter) for the amino acid residues, from N-terminal (left) to C-terminal (right).





The Nobel Prize in Chemistry 1958

"for his work on the
structure of insulin"



Frederick Sanger

United Kingdom

**University of Cambridge
Cambridge, United Kingdom**

b. 1918



The Nobel Prize in Chemistry 1980

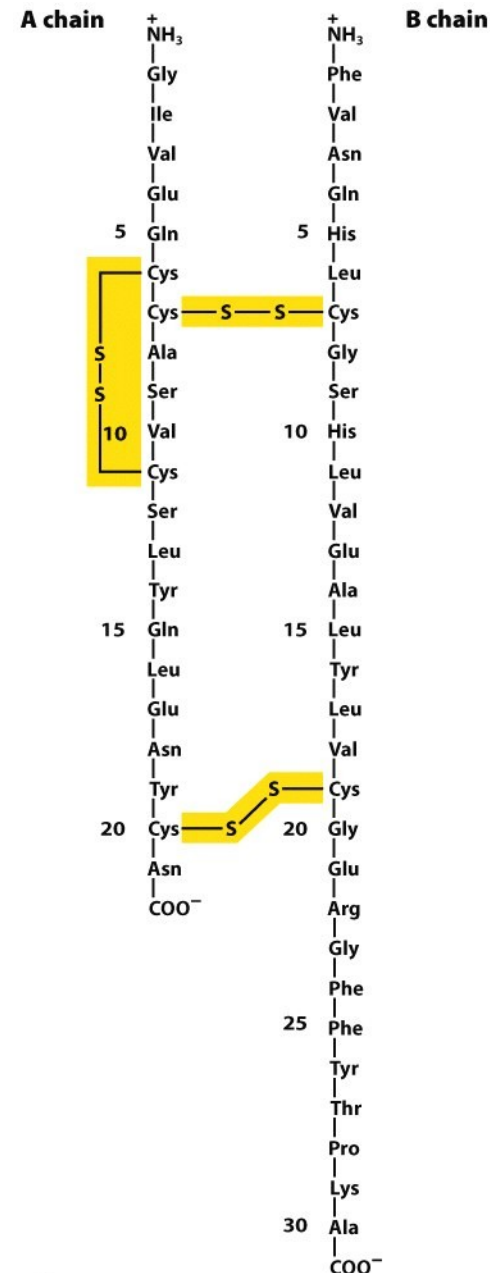
"for his contributions concerning
the determination of base
sequences in nucleic acids"



- Insulin consists of **two polypeptide chains, A and B**, held together by two disulfide bonds.

- **A chain:** 21 residues
- **B chain:** 30 residues

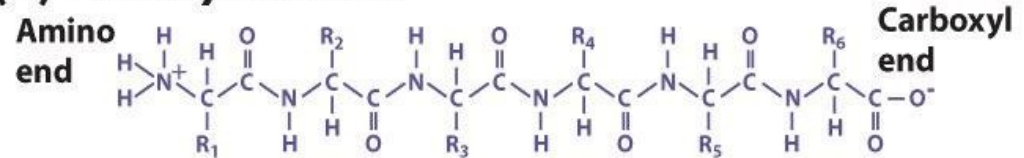
- The sequence shown is that of **bovine insulin**



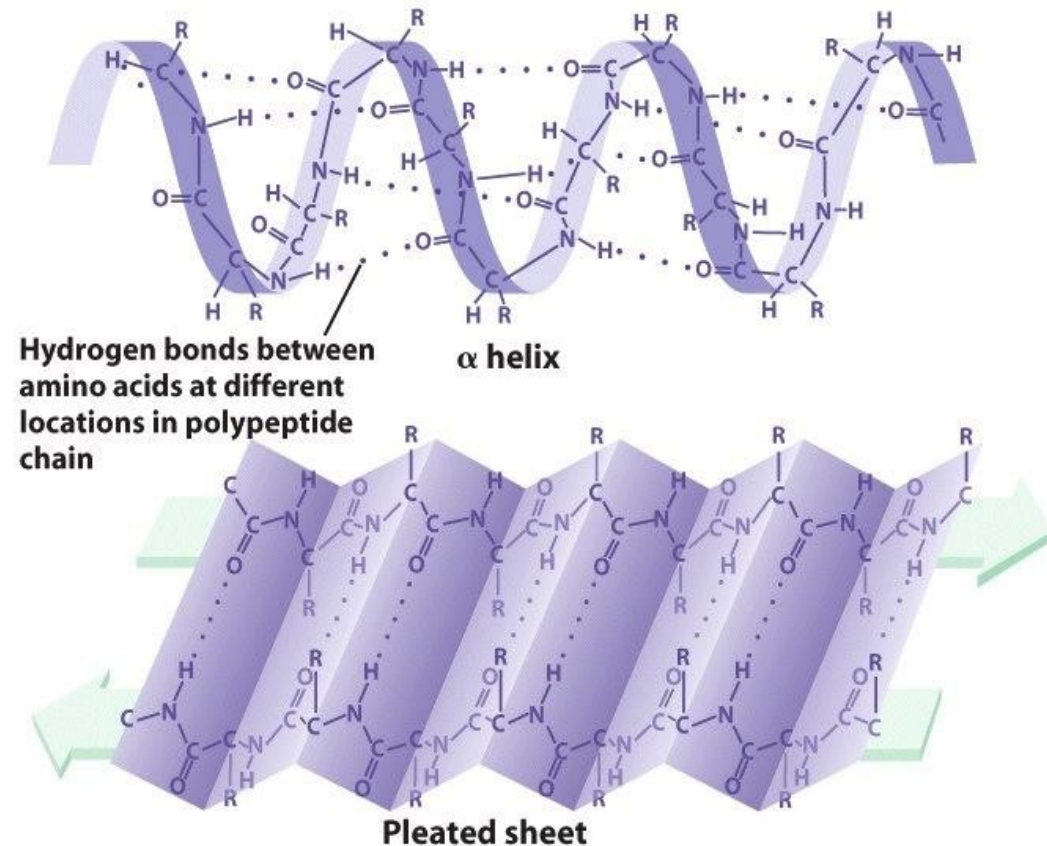
5.2.2 The secondary structure

- **Definition:**
 - the local spatial arrangement of its main-chain atoms without regard to the conformation of its side chains or to its relationship with other segments.

(a) Primary structure



(b) Secondary structure



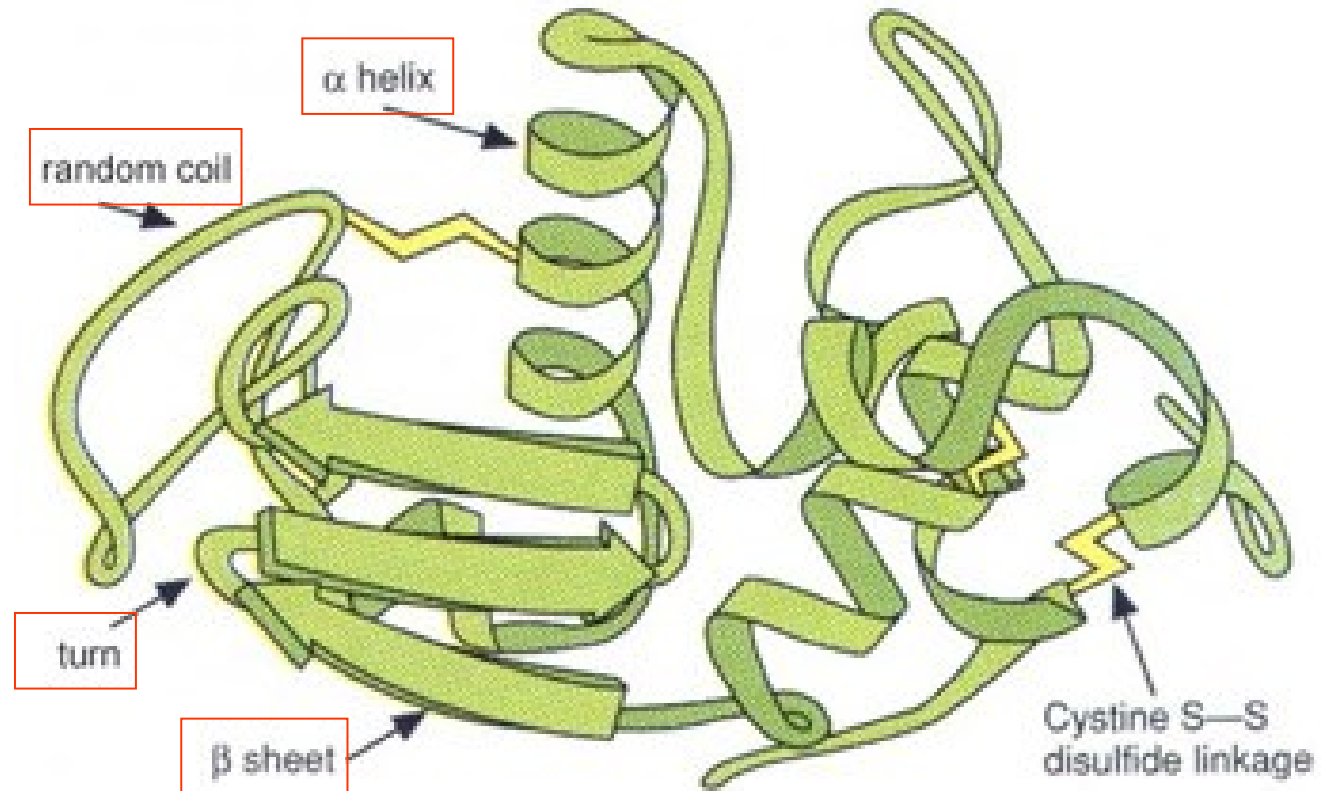
- **Classes**

- Three common secondary structures

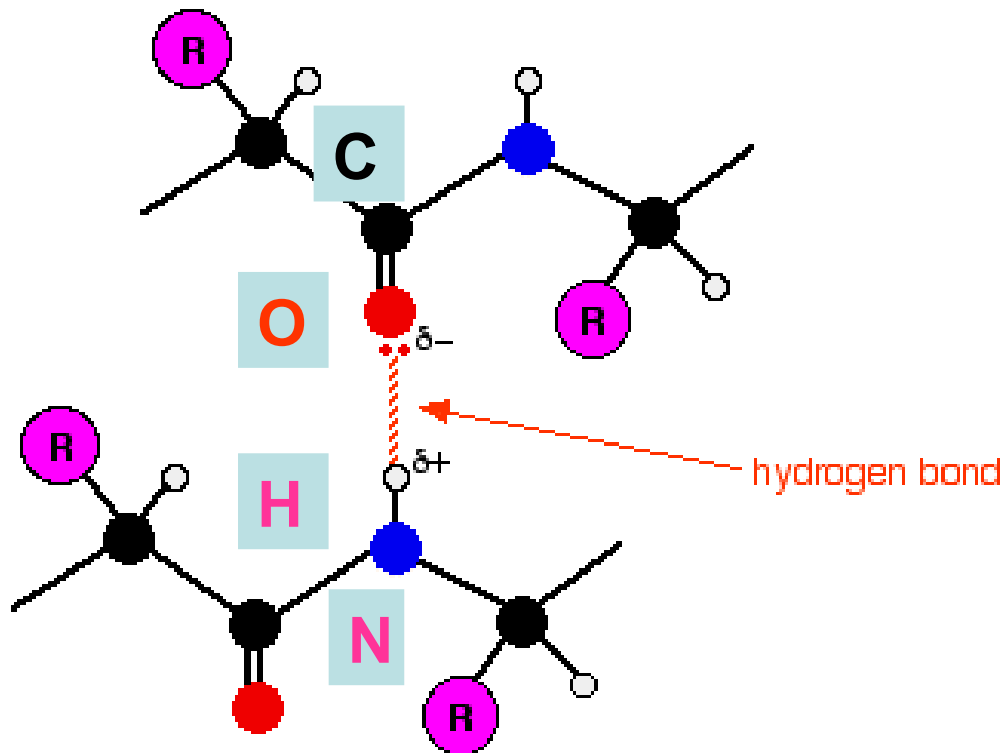
- *α -helices, β -pleated sheets, turns.*

- **Random coil**: which cannot be classified as one of the standard three classes is usually grouped into a category called "**random coil**".

- These secondary structures are held together by **hydrogen bonds**.

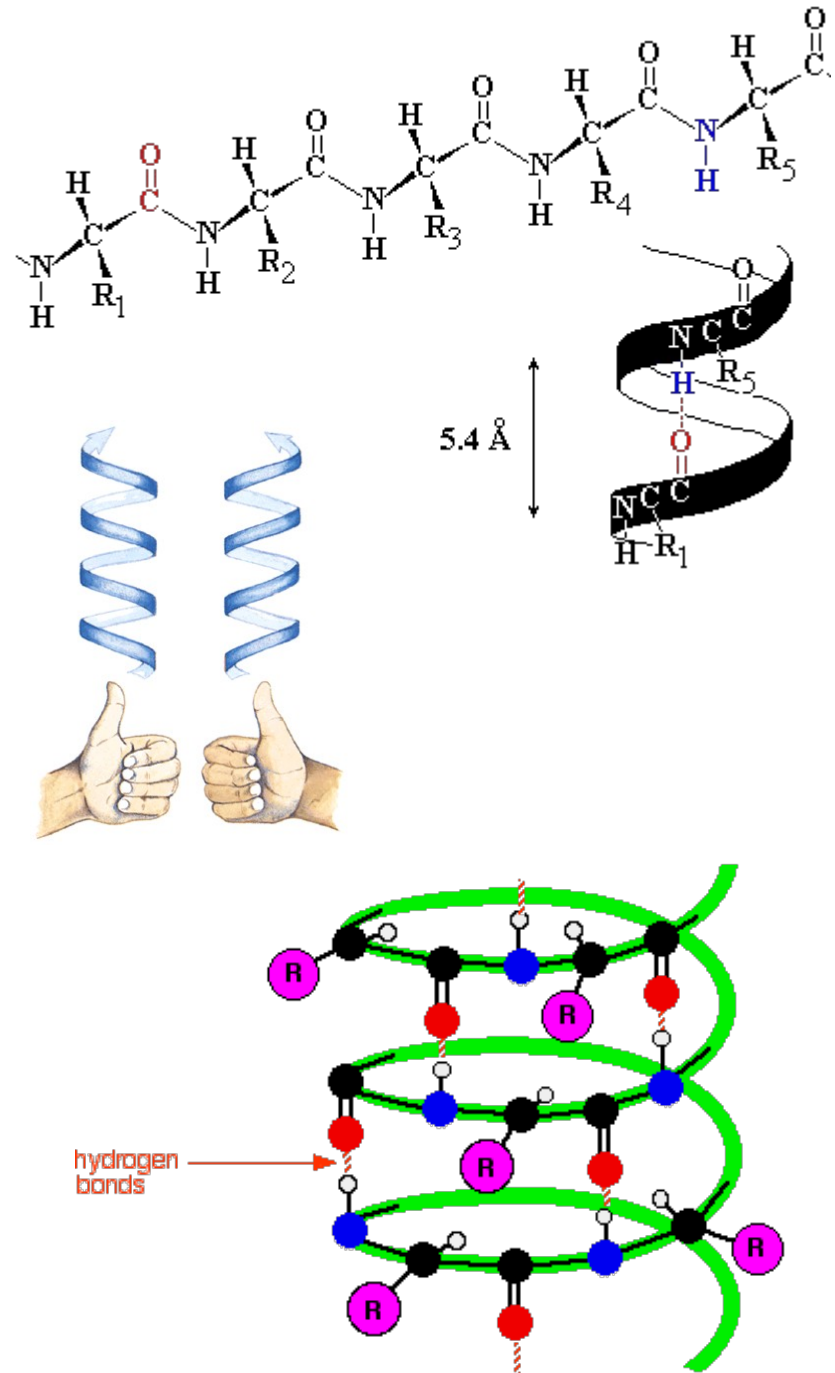


- The **hydrogen bonds** in protein form between one of an oxygen (**O**) atom and the hydrogen (**H**) attached to a nitrogen (**N**) atom.



α -helix

- Proposed by Pauling and Corey, 1951
- The polypeptide backbone is tightly wound around the long axis (rodlike).
- C=O group of amino acid #1 form **hydrogen bond** with the H-N group of amino acid #4 (and C=O #2 with H-N #5).
- $\text{N}-\text{H} \cdots \text{O}=\text{C}$
- **R groups protrude** outward from the helical backbone.
- 3.6 amino acids per turn.
- **Right handed**.

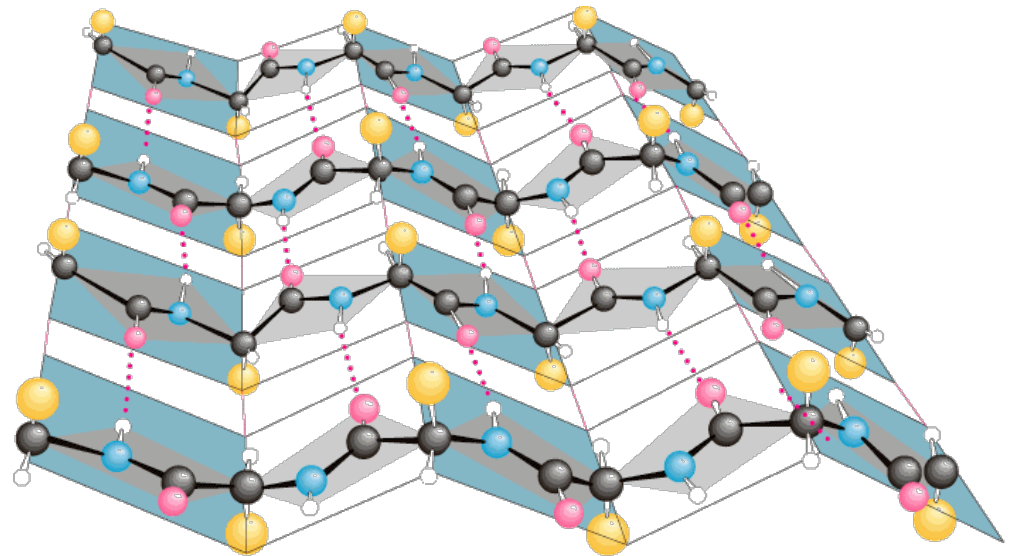
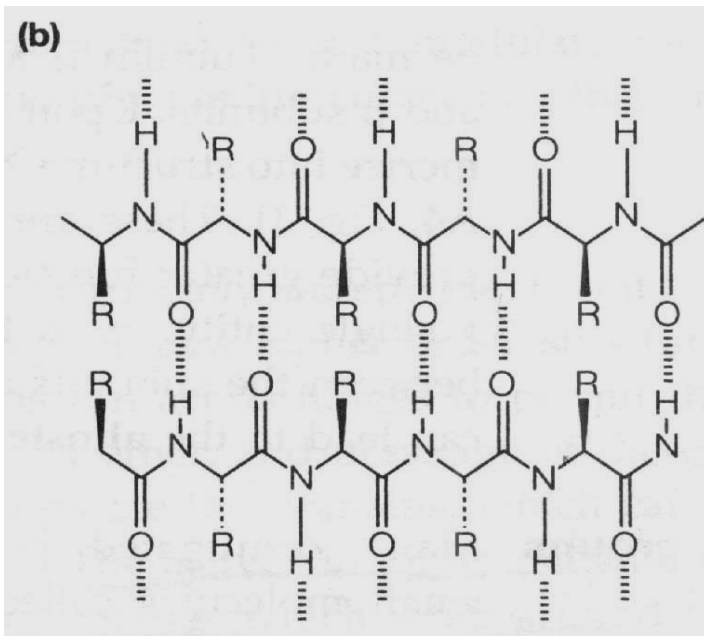


Amino acids affect α -helix stability

- Proline and glycine are sometimes known as "**helix breakers**" because they disrupt the regularity of the α helical backbone conformation.
 - **Pro**: inability to form hydrogen bond from its amide nitrogen.
 - **Gly**: more conformational flexibility.
 - **Acidic or basic** amino acids also interfere with α -helix structure.

β -pleated sheet

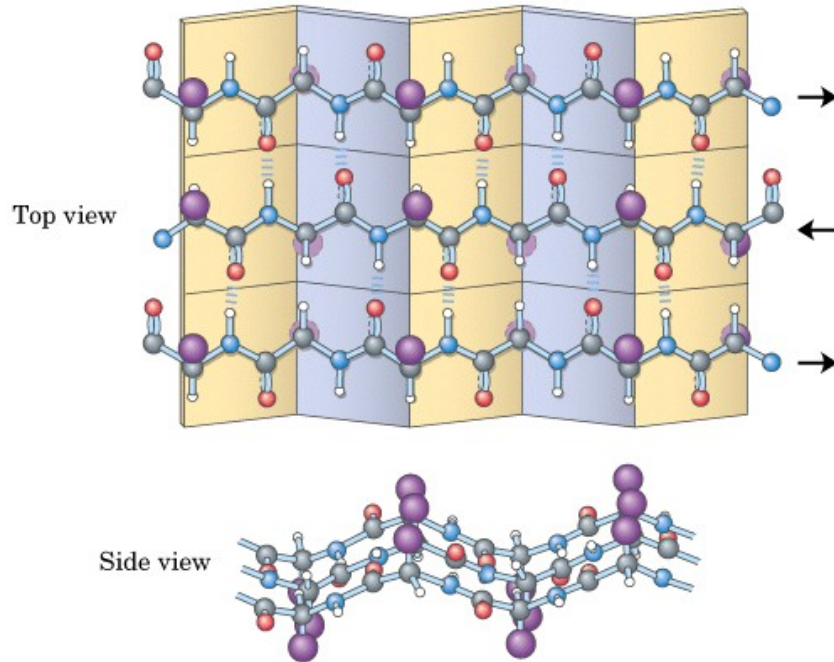
- β -strands are **elongated** peptide segments.
- Single β -strands are not stable structures but occur in association with neighboring strands.
- Regular **hydrogen bonds** are formed between the carbonyl oxygen and amide hydrogen between adjacent chains (look like a zipper).



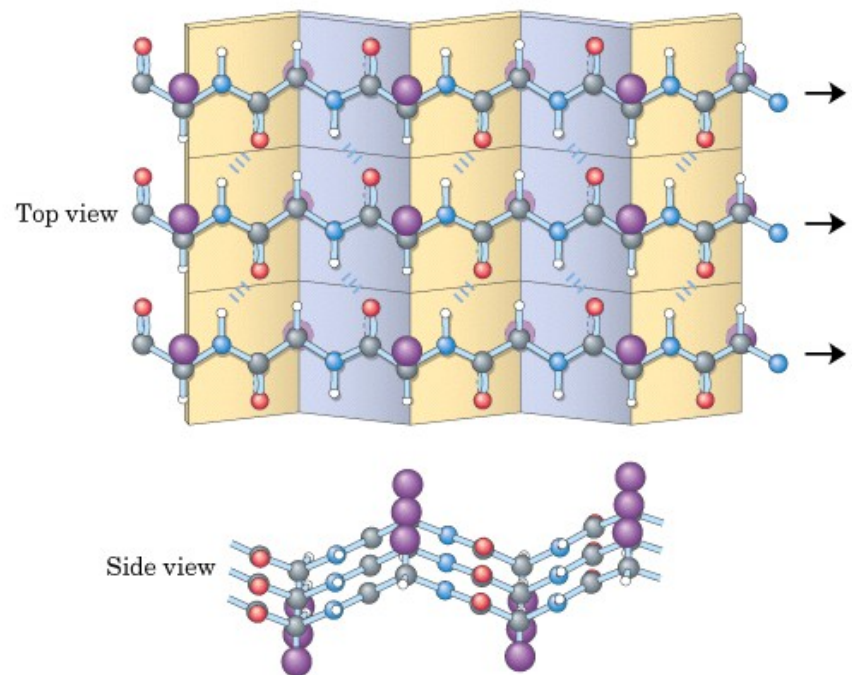
β -pleated sheet

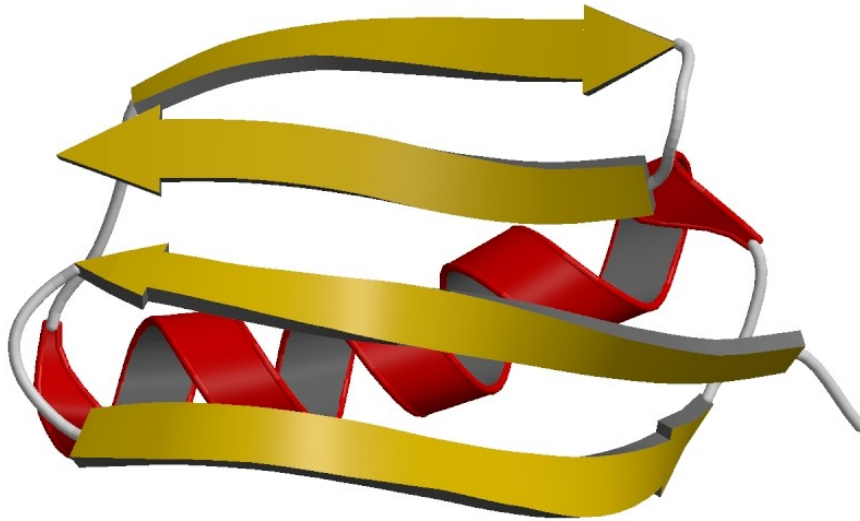
- β -pleated sheet can be found as either **parallel** (the same direction) or **anti-parallel** (the opposite direction).

(a) Antiparallel



(b) Parallel

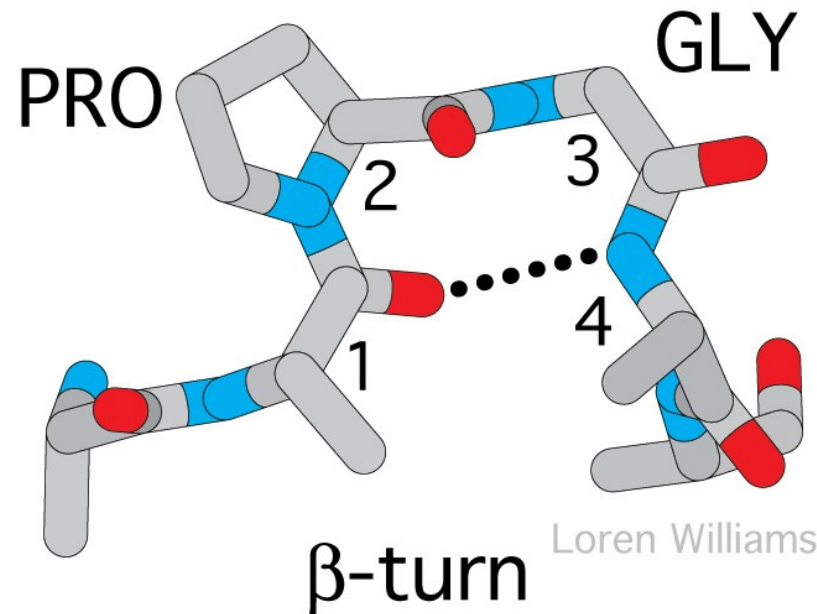
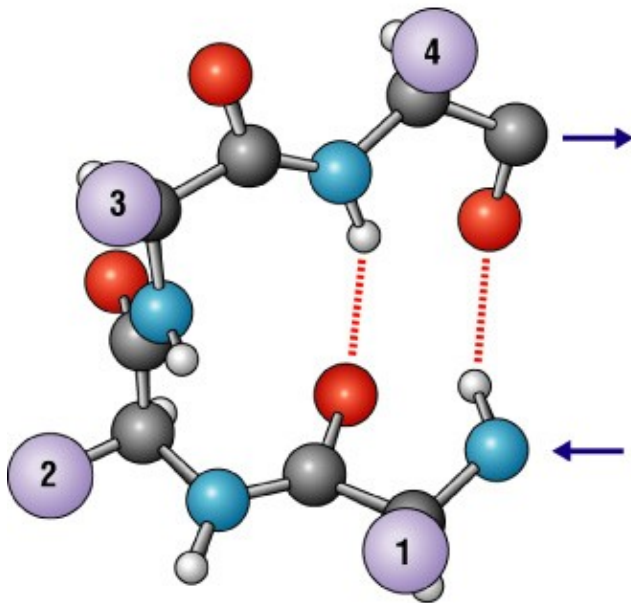




- **b**-Strands are often visualized as **broad arrows**.

- Ribbon diagram of the beta-1 domain of streptococcal protein G.
- **a**-helices are in **red** while **b**-strands are in **gold**.

- β turn is a tight turn of ~ 180 degrees involving **four amino acid residues**.
- The essence of the structure is the **hydrogen bonding** between the C=O group of residue **n** and the NH group of the residue **n+3**.
- **Gly** and **Pro** are often found in β turns.

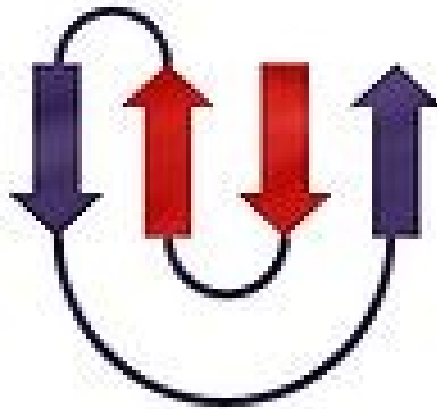


Supersecondary structures (or motifs)

- clusters of secondary structures that repeatedly appear in different proteins.
- include mainly $\beta\alpha\beta$ motif, Greek key motif, β -barrel, ...etc.



$\beta - \alpha - \beta$



Greek key



β barrel

5.2.3 Tertiary structure



Definition:

- The tertiary structure is the complete *three-dimensional structure of a polypeptide chain*.
- It is the combination of the elements of secondary structure linked by turns and loops.
- Tertiary structure is considered to be largely determined by the protein's primary structure, or the sequence of amino acids of which it is composed.

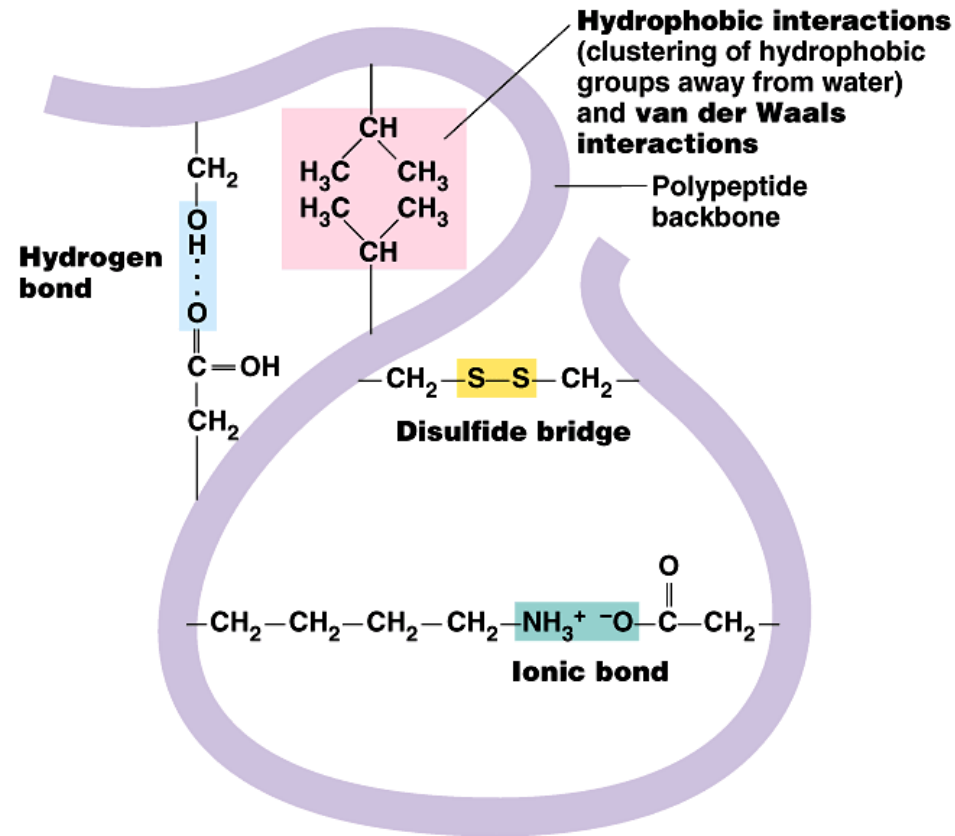
Interactions stabilizing tertiary structure



- Stability of the tertiary structure is determined by **Noncovalent interactions** & the **disulfide bond**.

- Noncovalent interactions**

- Hydrophobic interactions
- Hydrogen bonds
- Ionic bond
- Van der Waals interactions



Four Types of Noncovalent ("Weak") Interactions among Biomolecules in Aqueous Solvent

Hydrogen bonds

Between neutral groups



Between peptide bonds



Ionic interactions

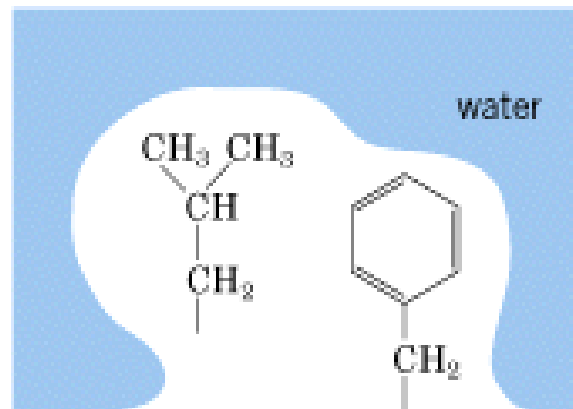
Attraction



Repulsion



Hydrophobic interactions



Van der Waals interactions

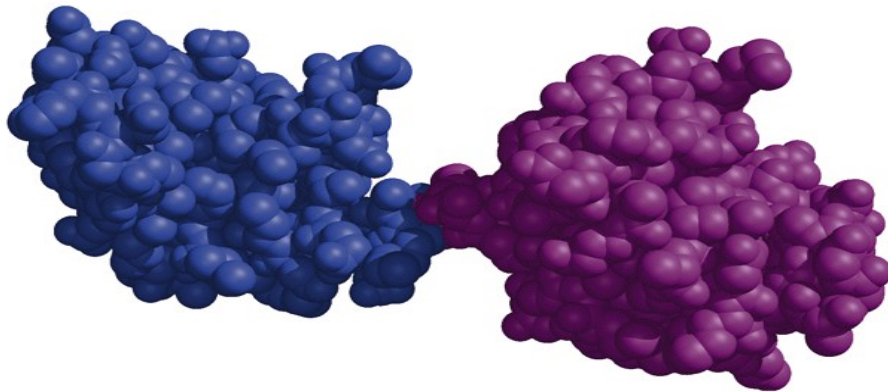
Any two atoms in close proximity

Three-dimensional arrangement of protein

- **AAs with nonpolar side chains** tend to be located **in the interior** of the polypeptide molecule.
- In contrast, **AAs with polar or charged side chains** tend to be located **on the surface** of the molecule in contact with the polar solvent.

Domain

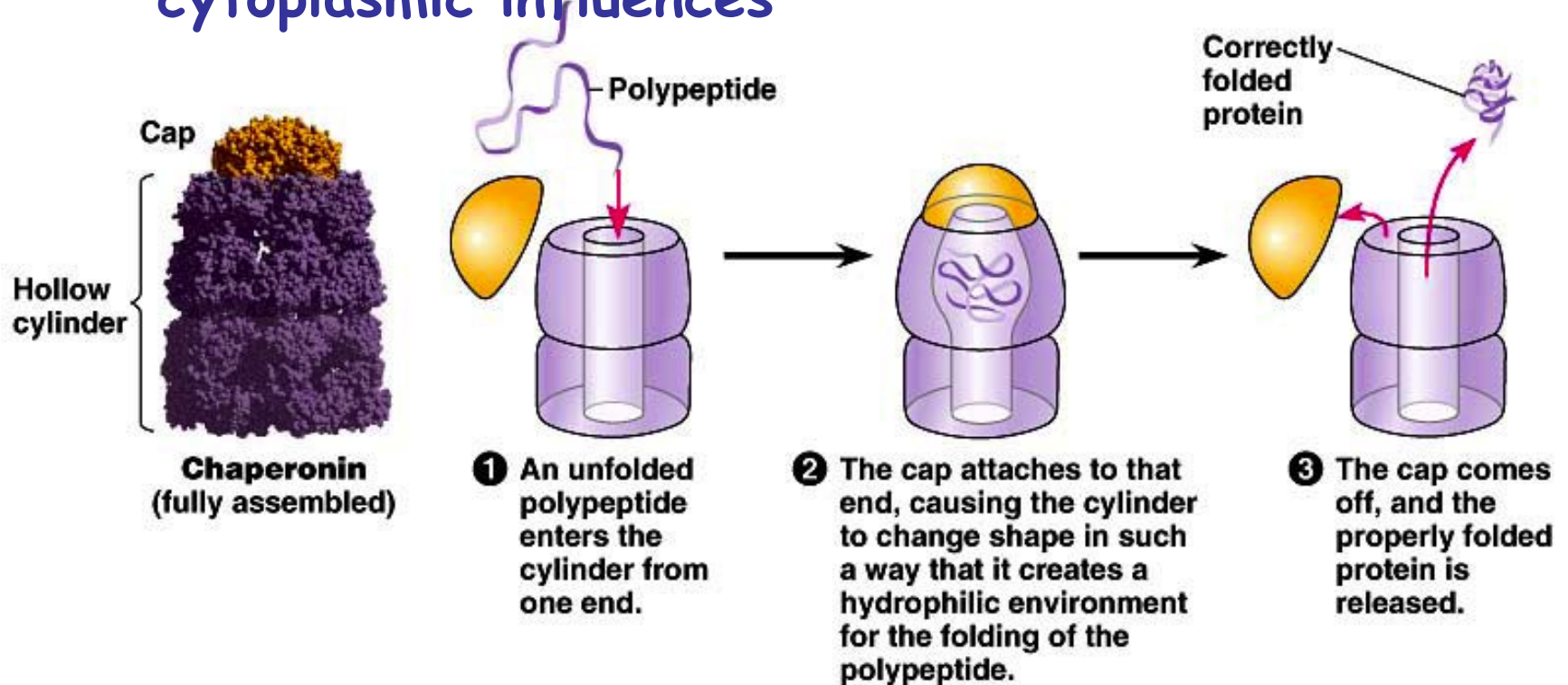
- Domains are **the fundamental functional and 3D structural units** of a polypeptide.
 - usually include less than 200~400 residues
 - folding based on secondary structure and supersecondary structure.
- Many domains **fold independently** into thermodynamically stable structures, and sometimes, have **separate functions**.



- Structural domains in the polypeptide troponin C, **two separate** calcium-binding domains

Chaperones in protein folding

- Guide protein folding
 - provide shelter for folding polypeptides
 - keep the new protein segregated from cytoplasmic influences

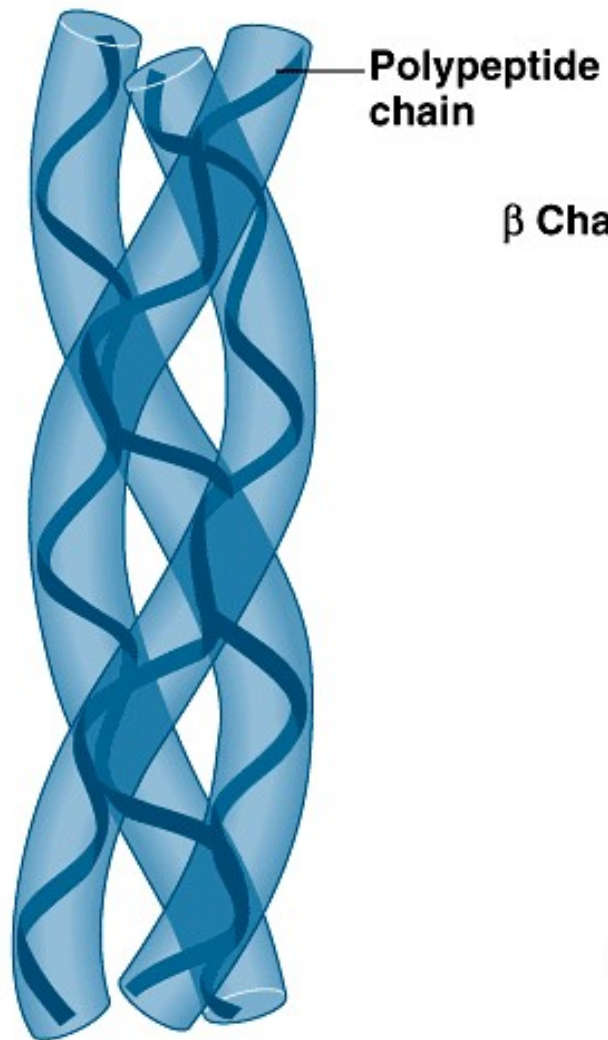


Role of chaperones in protein folding

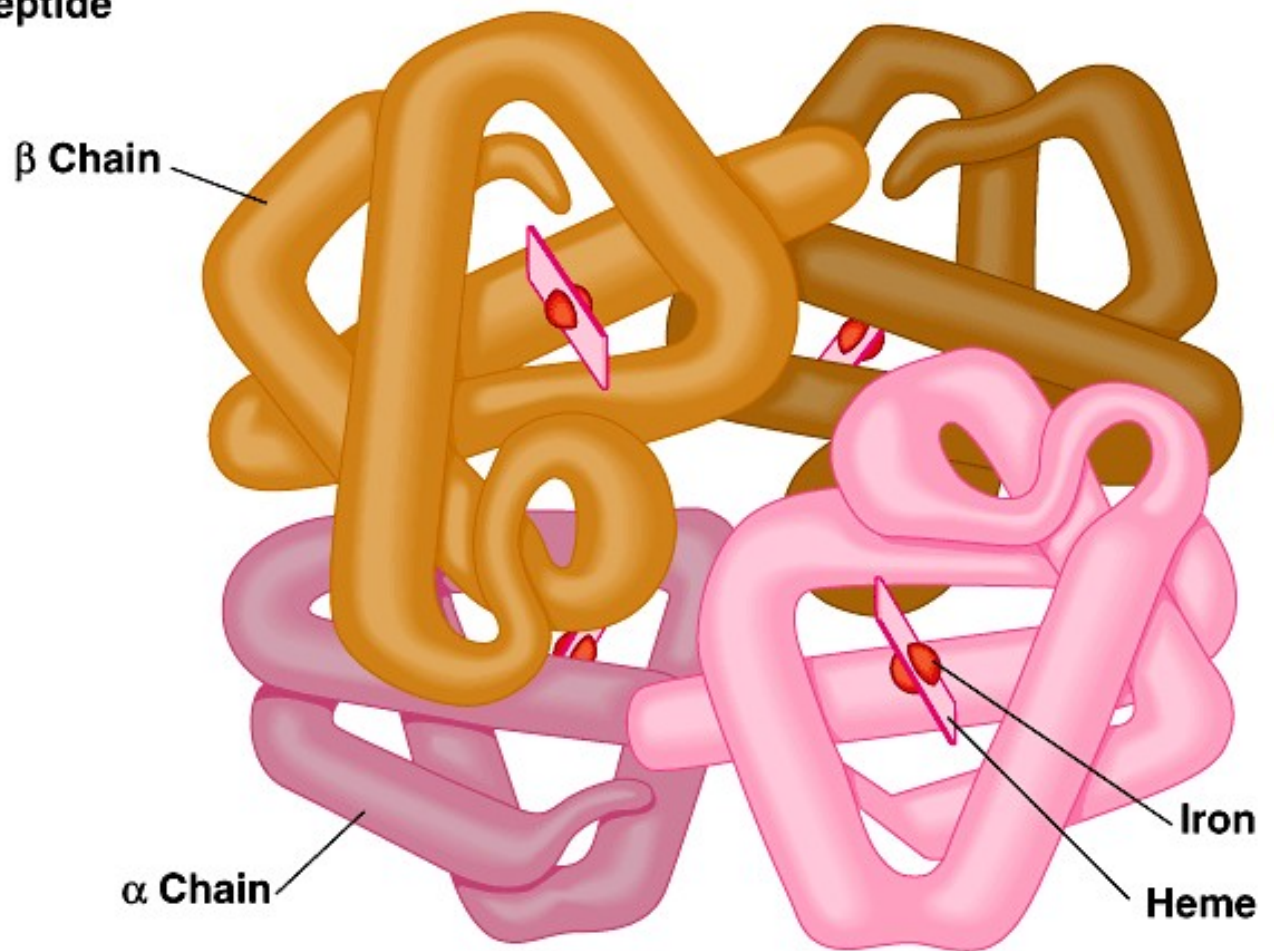
- The *information needed for correct protein folding*
 - is *contained in the primary structure* of the polypeptide.
- “**chaperones**,” are specialized group of proteins required for the proper folding of many species of proteins.
- also known as “**heat shock**” **proteins**—interact with the polypeptide at various stages during the folding process.
 - Some chaperones are important in *keeping the protein unfolded until its synthesis is finished*, or
 - act *as catalysts* by increasing the rates of the final stages in the folding process.
 - Others *protect* proteins as they fold so that their vulnerable, exposed regions do not become tangled in unproductive encounters.

5.2.4 Quaternary structure

- The quaternary structure describes the arrangement and position of each of the subunits in a multiunit protein.
 - Only those proteins containing more than one polypeptide chain exhibit.
 - Subunits may be identical or different.
 - only then it is a functional protein.
- Subunits are held together by many weak, noncovalent interactions (hydrophobic, electrostatic)

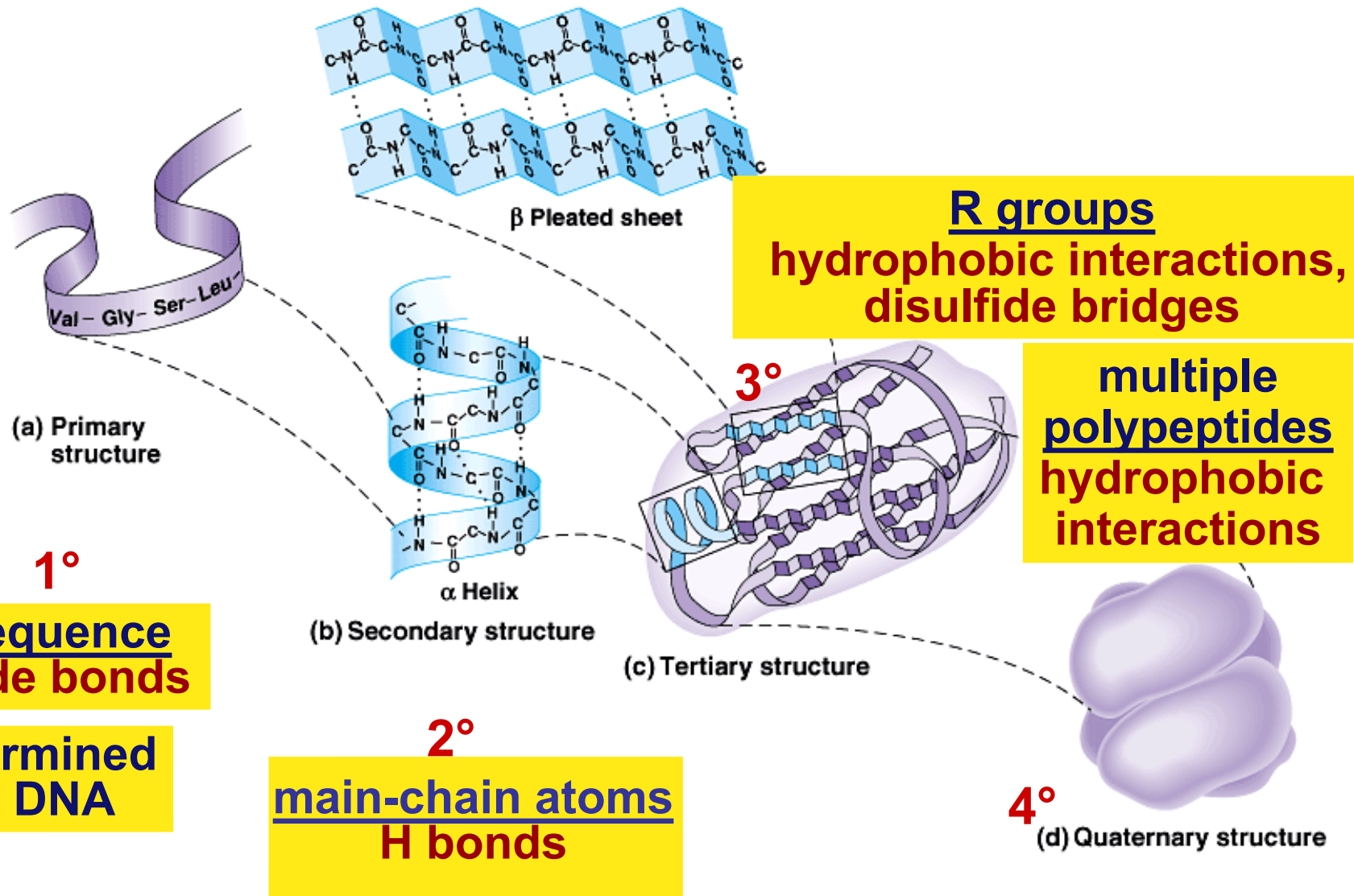


(a) Collagen



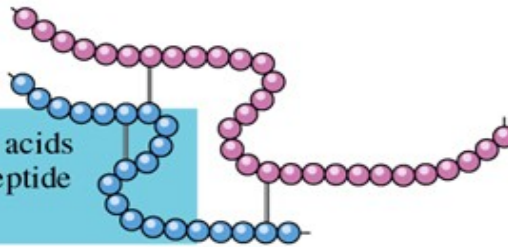
(b) Hemoglobin

Protein structure (review)



PRIMARY STRUCTURE

The sequence of amino acids present in a protein's peptide chain or chains



SECONDARY STRUCTURE

The regularly repeating ordered spatial arrangements of amino acids near each other in the protein chain, which result from hydrogen bonds between carbonyl oxygen atoms and amino hydrogen atoms

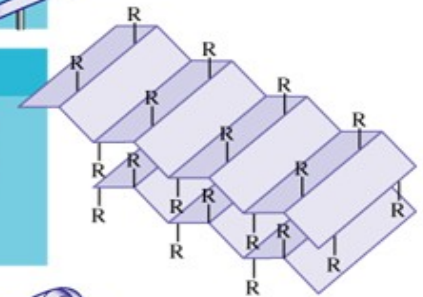
Alpha Helix

Hydrogen bonds between every fourth amino acid



Beta Pleated Sheet

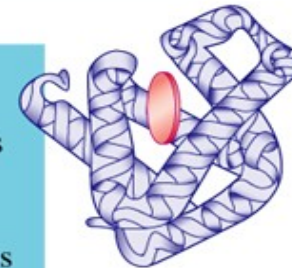
Hydrogen bonds between two side-by-side chains, or a single chain that is folded back on itself



TERTIARY STRUCTURE

The overall three-dimensional shape that results from the attractive forces between amino acid side chains (R groups) that are not near each other in the protein chain

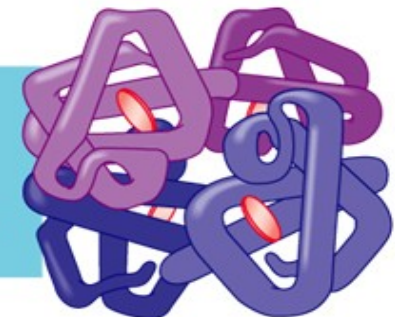
- Disulfide bonds
- Electrostatic interactions
- Hydrogen bonds
- Hydrophobic interactions



QUATERNARY STRUCTURE

The three-dimensional shape of a protein consisting of two or more independent peptide chains, which results from noncovalent interactions between R groups

- Electrostatic interactions
- Hydrogen bonds
- Hydrophobic interactions



Denaturation of proteins

- Denaturation is a process in which proteins lose their three-dimensional structure.
- Many means can cause protein to denature:
 - strong acid or base: disrupt the salt bridge;
 - Heavy Metal Salts: Hg^{2+} , Pb^{2+} , Ag^{+} ;
 - organic solvent (alcohol or chloroform): disrupt H-bonds;
 - Heat: disrupt H-bonds and non-polar hydrophobic interactions;
 - Solutes (urea, guanidine).

Denaturing Agent

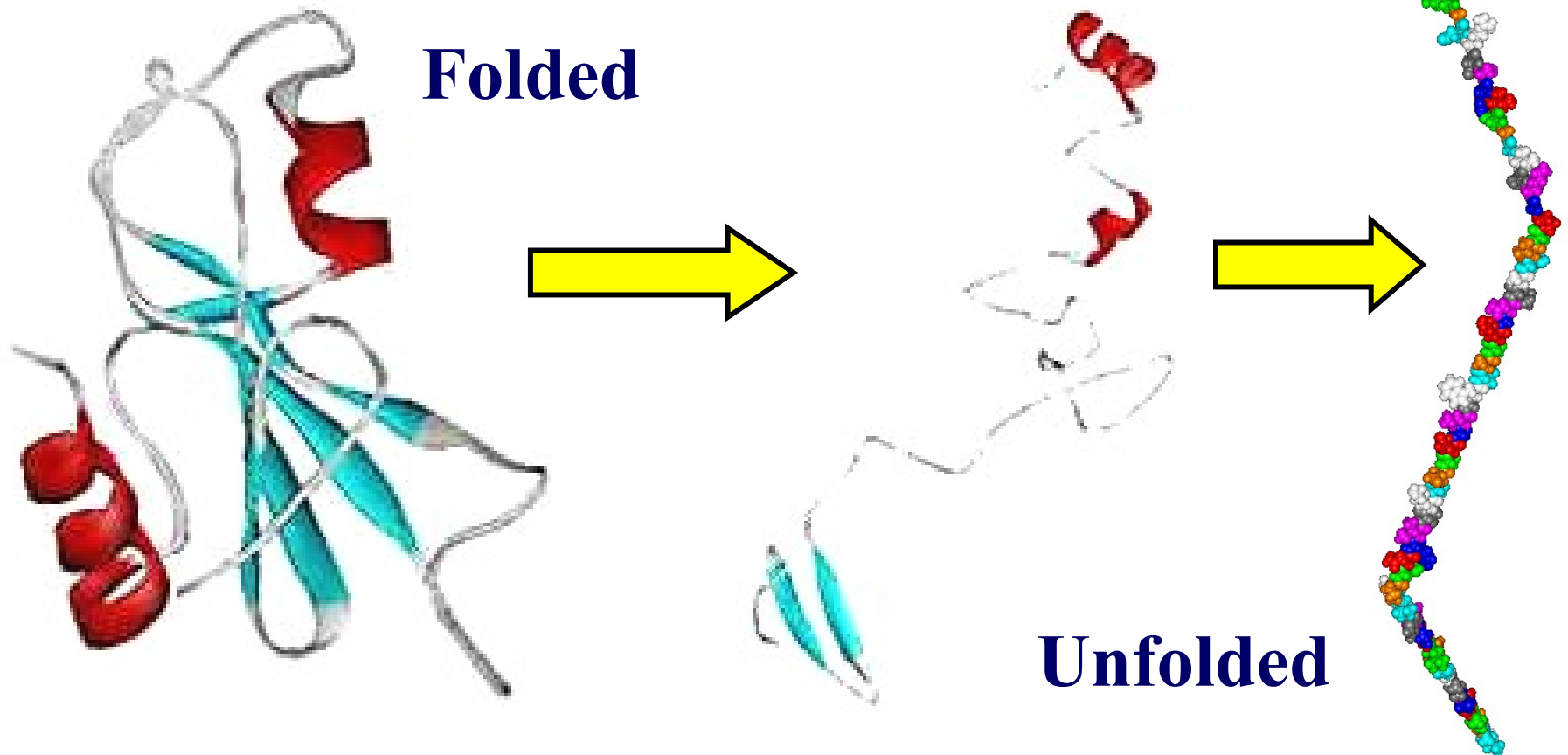
Mode of Action

heat	disrupts hydrogen bonds by making molecules vibrate too violently; produces coagulation, as in the frying of an egg
microwave radiation	causes violent vibrations of molecules that disrupt hydrogen bonds
ultraviolet radiation	operates very similarly to the action of heat (e.g., sunburning)
violent whipping or shaking	causes molecules in globular shapes to extend to longer lengths, which then entangle (e.g., beating egg white into meringue)
detergent	affects R-group interactions
organic solvents (e.g., ethanol, 2-propanol, acetone)	interfere with R-group interactions because these solvents also can form hydrogen bonds; quickly denature proteins in bacteria, killing them (e.g., the disinfectant action of 70% ethanol)
strong acids and bases	disrupt hydrogen bonds and salt bridges; prolonged action leads to actual hydrolysis of peptide bonds
salts of heavy metals (e.g., salts of Hg^{2+} , Ag^+ , Pb^{2+})	metal ions combine with —SH groups and form poisonous salts
reducing agents	reduce disulfide linkages to produce —SH groups

- Denaturation occurs because the bonding interactions responsible for the secondary structure (hydrogen bonds to amides) and tertiary structure are disrupted.
- Since denaturation reactions are not strong enough to break the peptide bonds, the primary structure (sequence of amino acids) remains the same after a denaturation process.

Protein Denaturation

- denature: loss of structure due to protein unfolding
- unfolding leads to loss of function



- Denatured proteins can exhibit a wide range of characteristics, from **loss of solubility** to **communal aggregation**.
- Most biological proteins **lose their biological function** when denatured.
- If proteins in a living cell are denatured, this results in disruption of cell activity and possibly cell death.
 - Medical supplies and instruments are **sterilized by heating** to denature proteins in bacteria and thus destroy the bacteria.
 - A **70% alcohol solution** is used as a disinfectant on the skin.

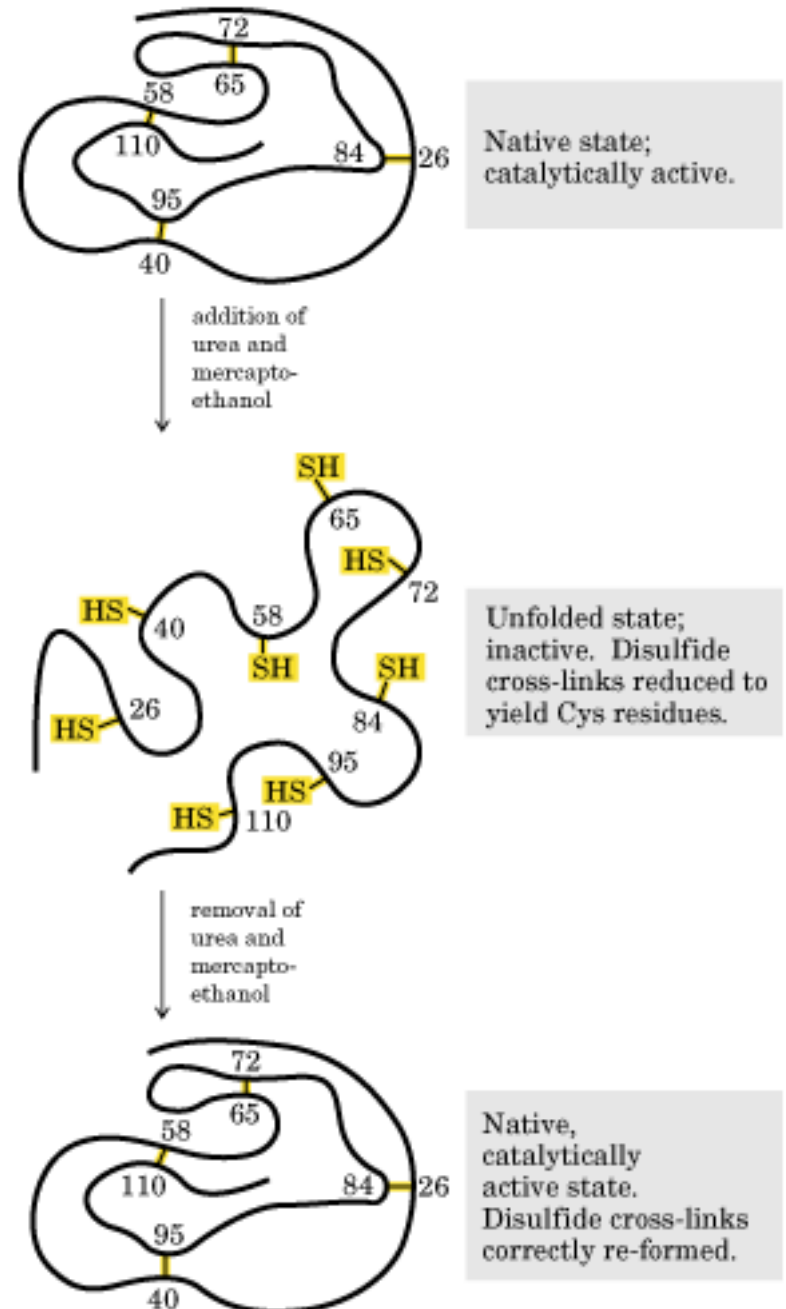
This egg's protein has undergone denaturation and loss of solubility, caused by the high temperature of the cooking process.



Reversibility and irreversibility

- In some proteins (unlike egg whites), denaturation is **reversible** (the proteins can regain their native state when the denaturing influence is removed).

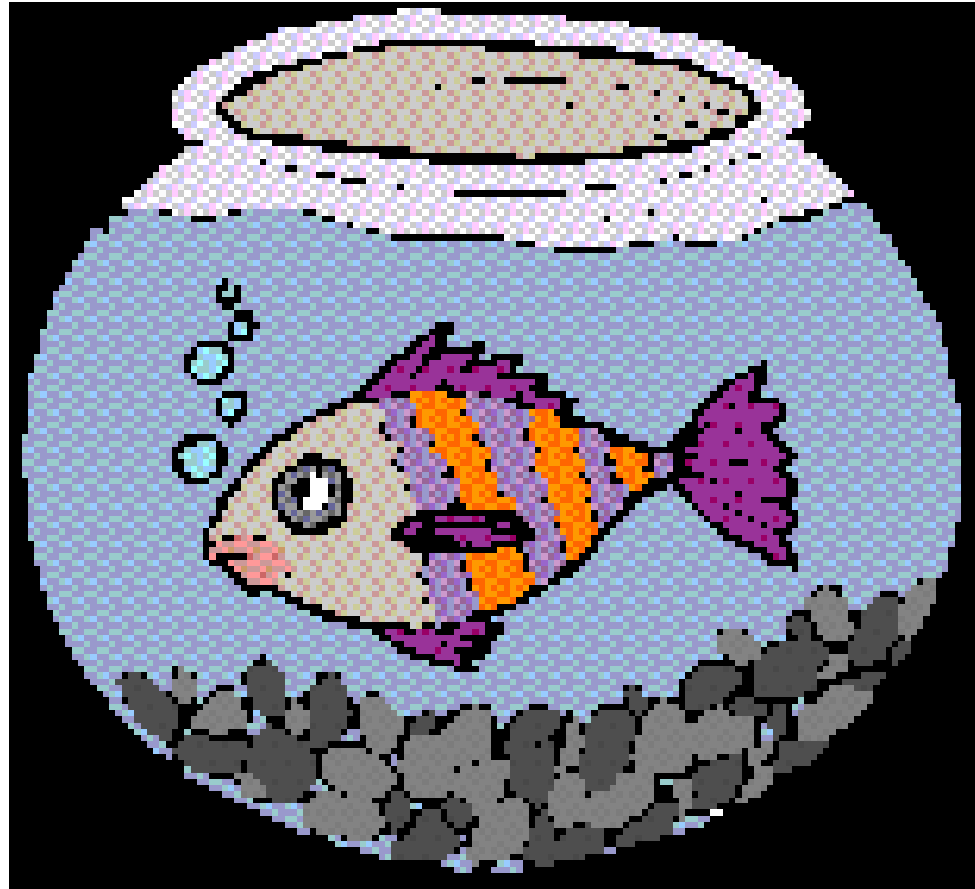
- Protein denaturation and renaturation



5.3 Two model families of clinically important proteins

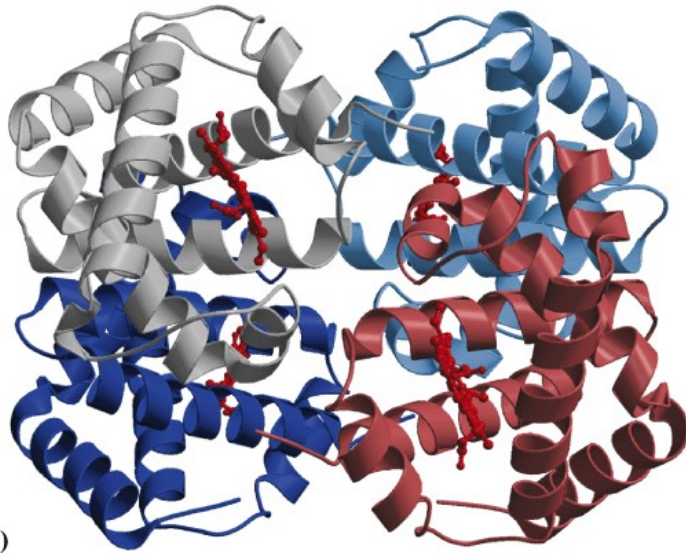
- Examine the relationship between structure and function for two model families of clinically important proteins
 - Globular hemoproteins
 - Fibrous structural proteins

5.3.1 Globular hemeproteins

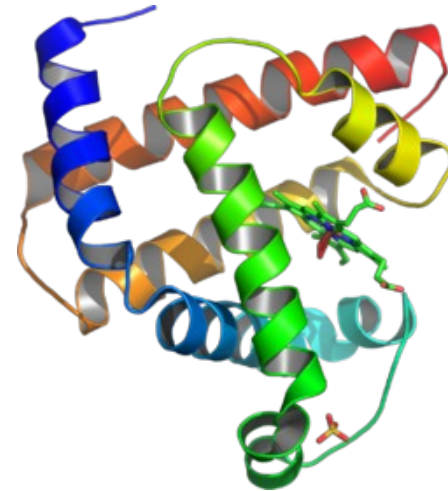


Problem: Oxygen has a low solubility in H_2O

Solution: Use proteins to transport oxygen from the lungs to other parts of the body



Hb is used to transport O_2 in the blood stream

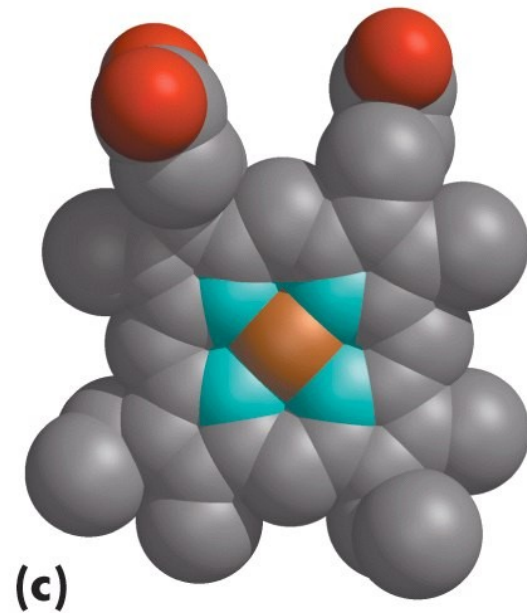
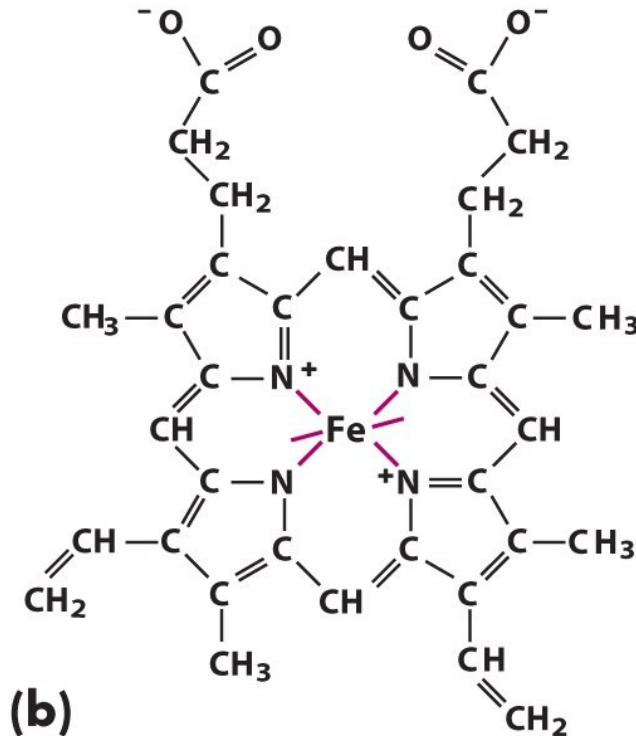
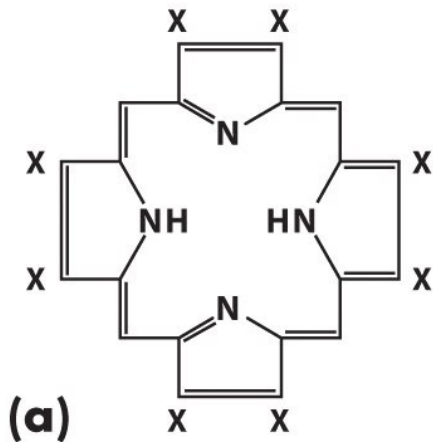


Myoglobin is used to **store** O_2 in muscle

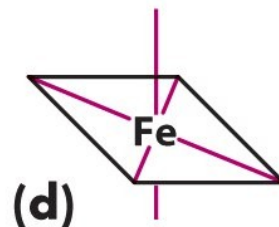
Hemeproteins

- Hemeprotein: proteins that contain heme (iron-porphyrin) as a tightly bound prosthetic group .
- In myoglobin and hemoglobin, the heme group serves to bind and deliver oxygen.

- **Heme** is made up of **protoporphyrin IX** ring structure with **an iron** atom in the ferrous (Fe^{2+}) oxidation state.

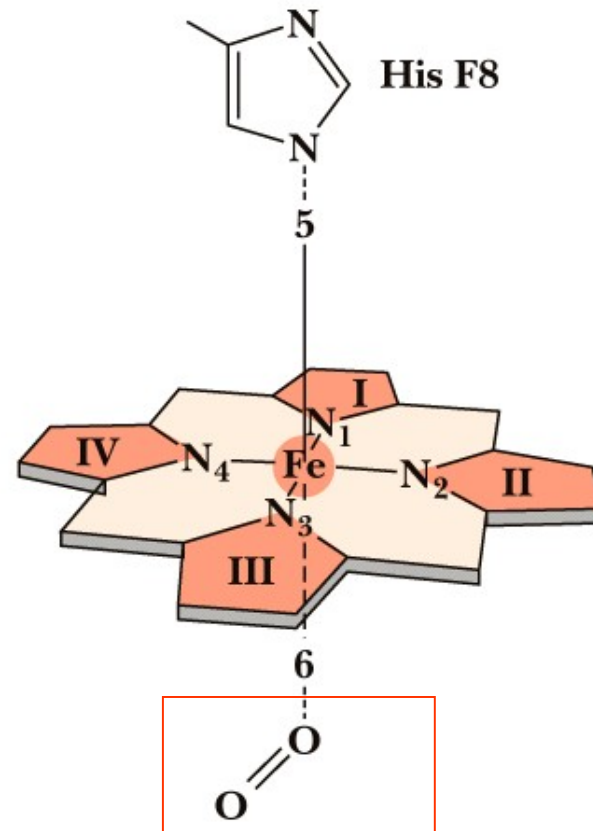
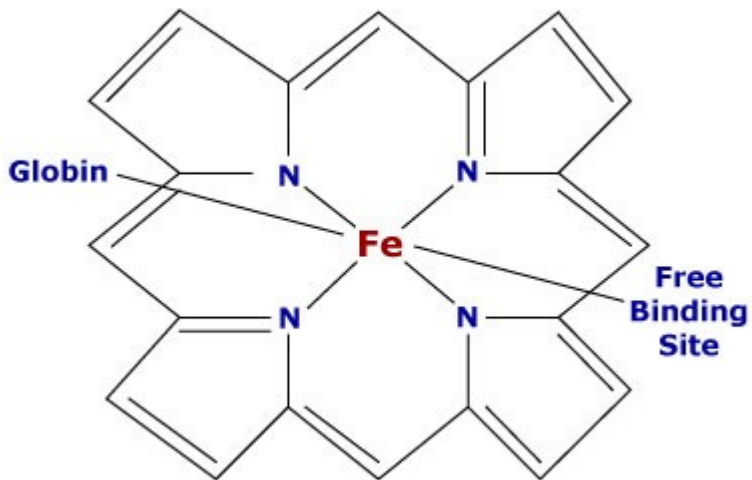


protoporphyrin IX
ring structure



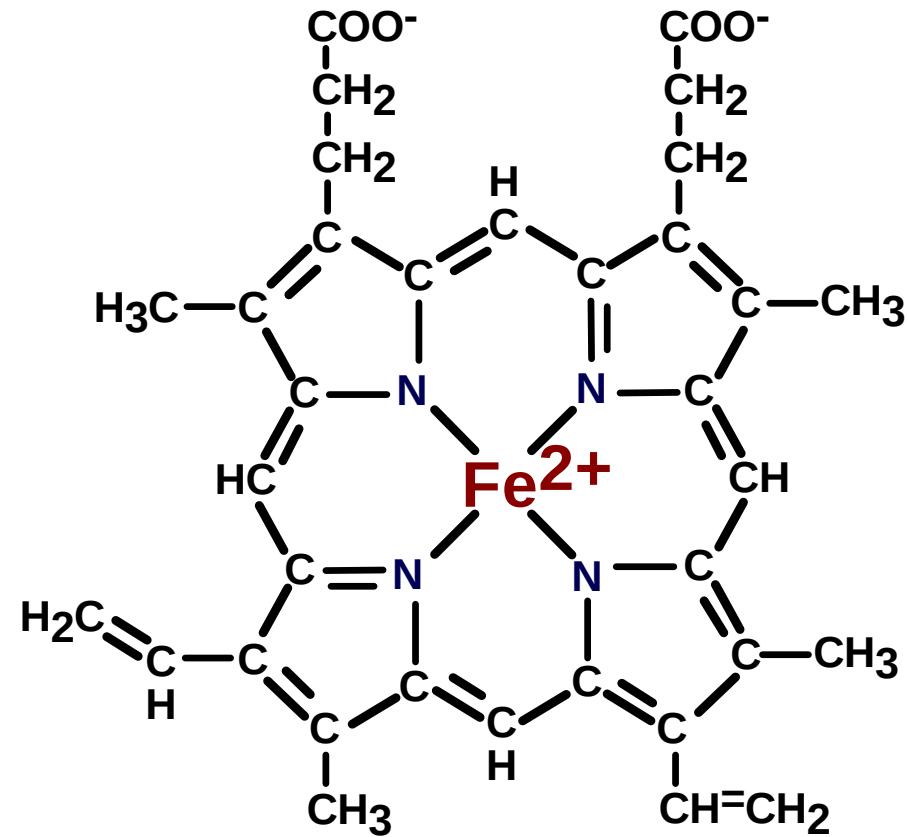
Fe^{2+} has six bonds

- Two additional bonds are on either side of the plane of the protoporphyrin ring.

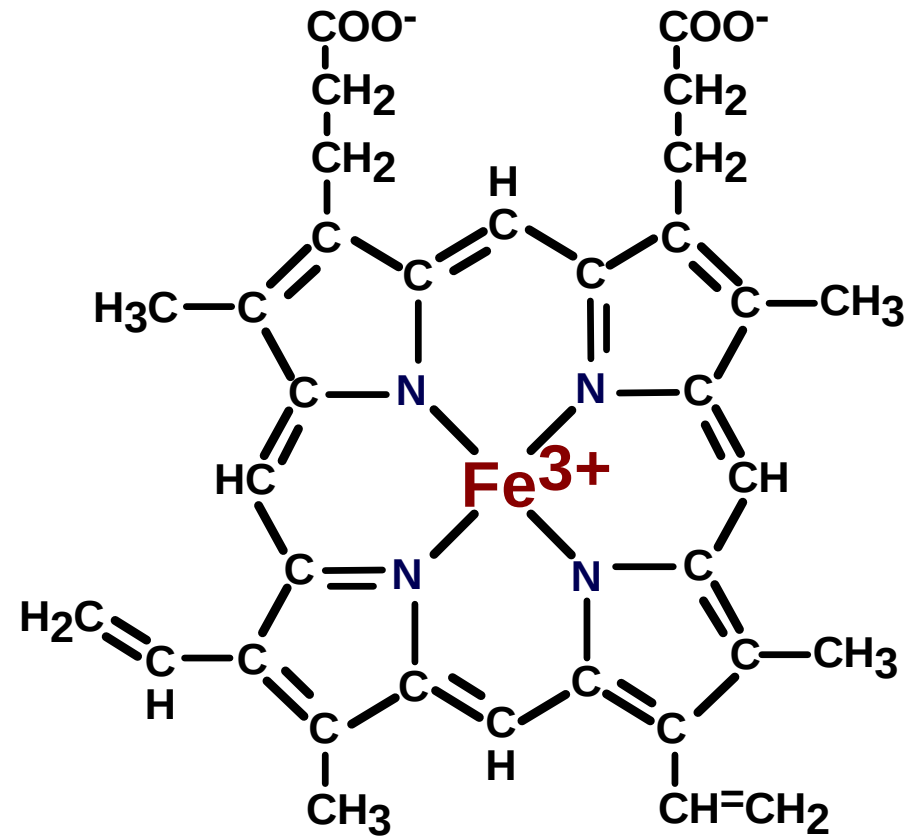


The heme plane

Iron has two oxidation states



Ferrohemoglobin
(Fe²⁺) binds O₂

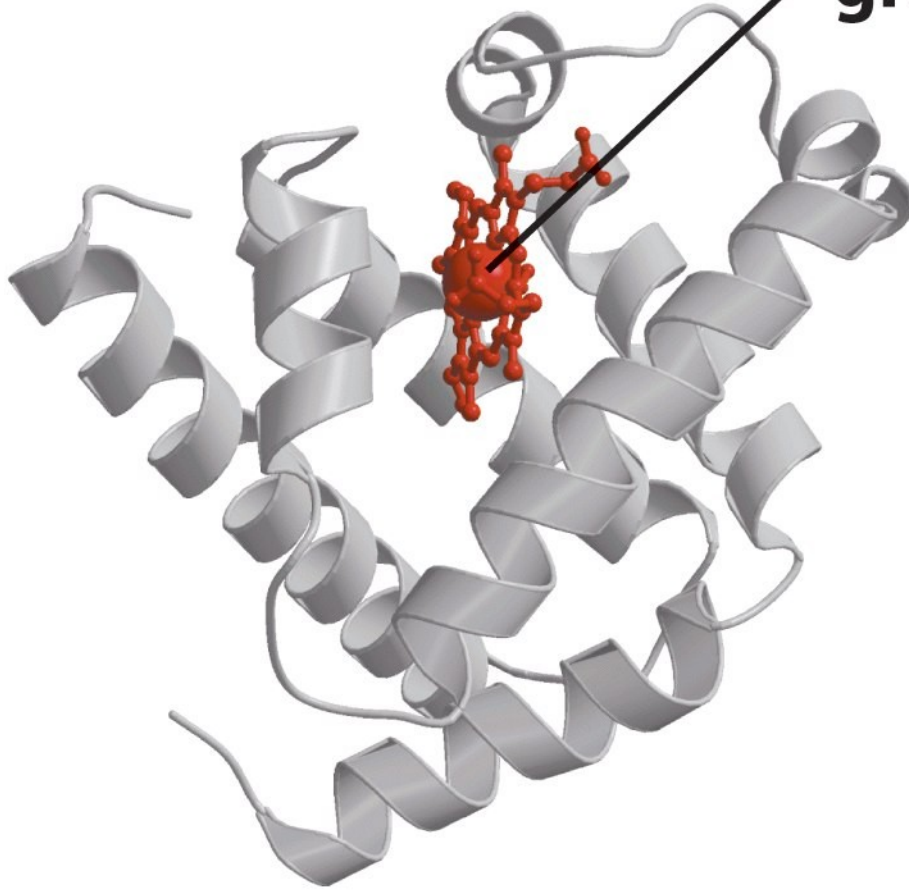


Ferrihemoglobin (Fe³⁺), also
known as methemoglobin,
does not bind O₂

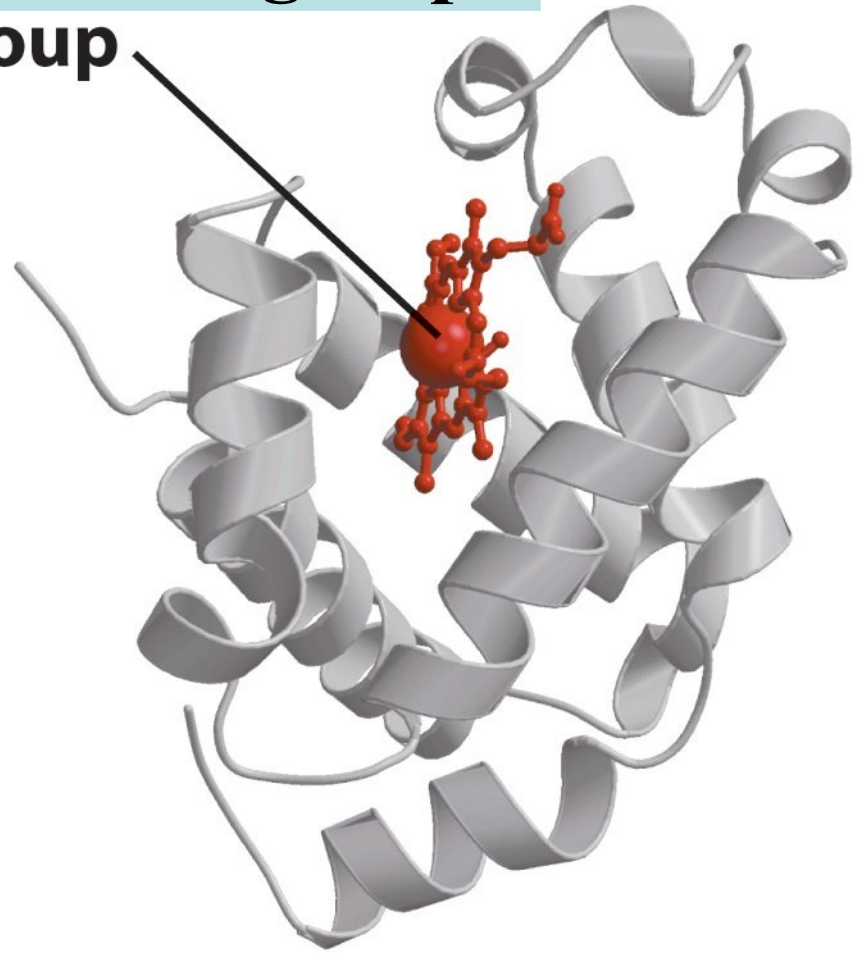
Myoglobin (Mb)

- Myoglobin is found in vertebrate muscle cells.
- A single-chain (monomeric), iron-containing protein, structurally similar to a single subunit of hemoglobin and having a higher affinity for oxygen than hemoglobin of the blood.
 - acts as a store of oxygen that can be used during strenuous exercise.
- 153 aa, MW: 17,200 Da

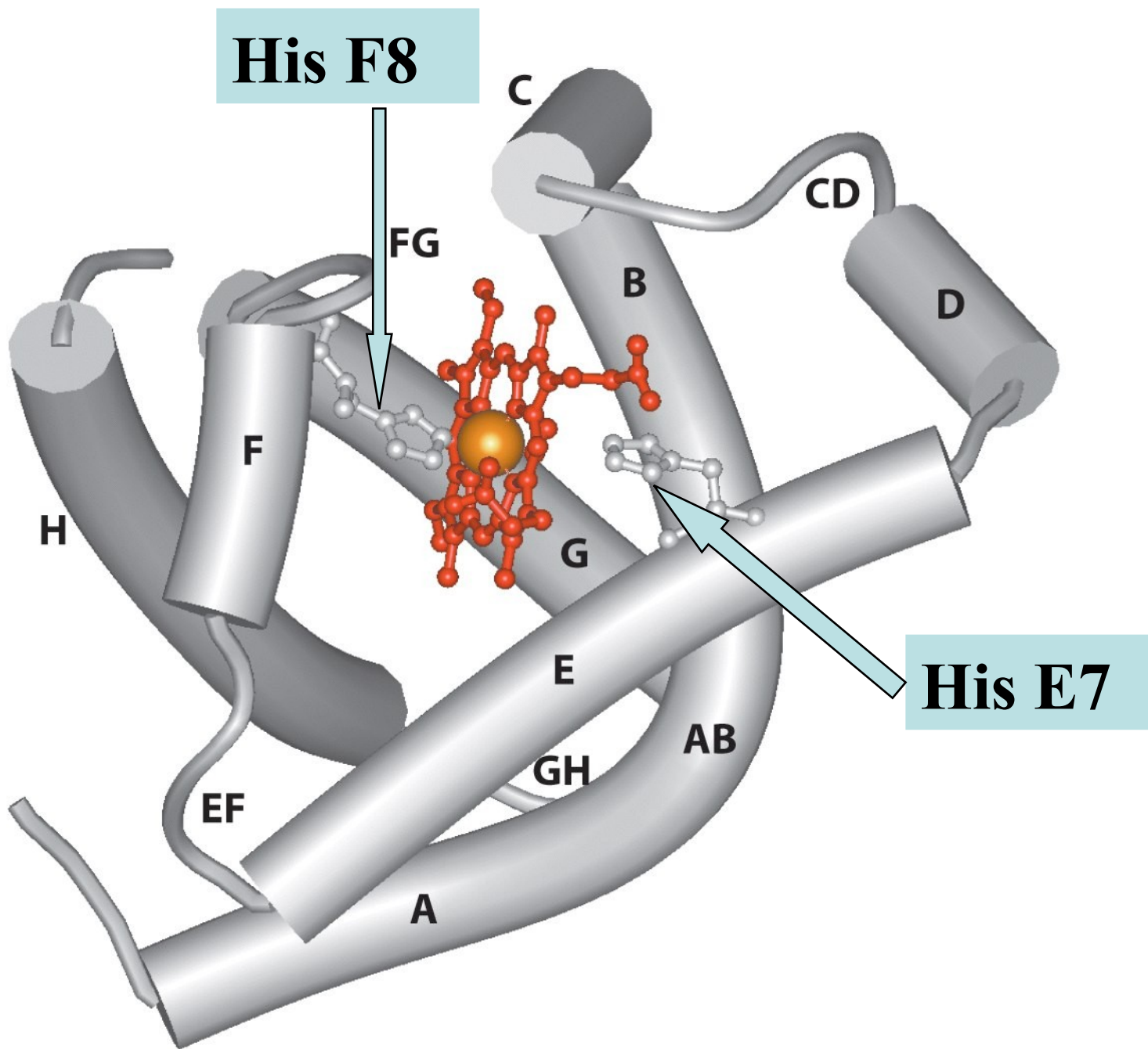
heme prosthetic group
group



Myoglobin



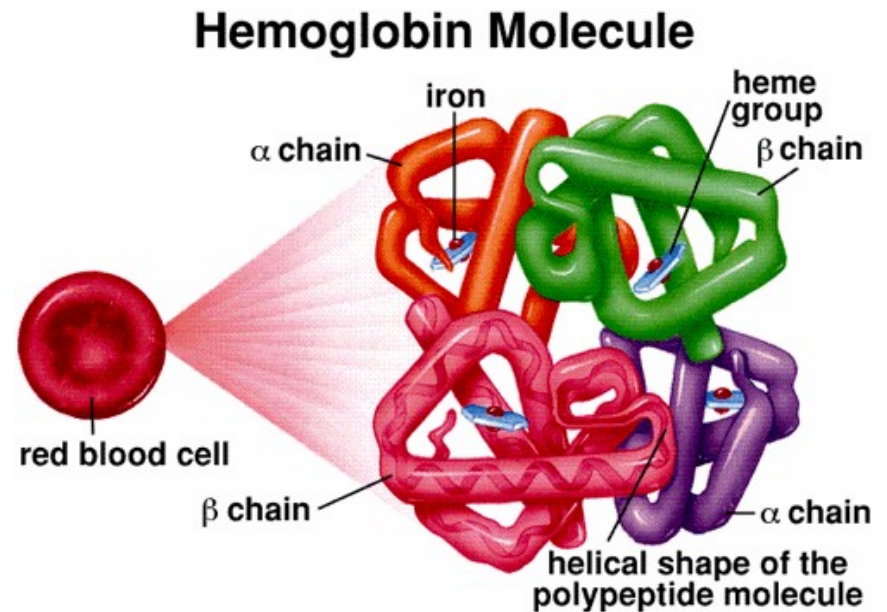
**β subunit of
hemoglobin**



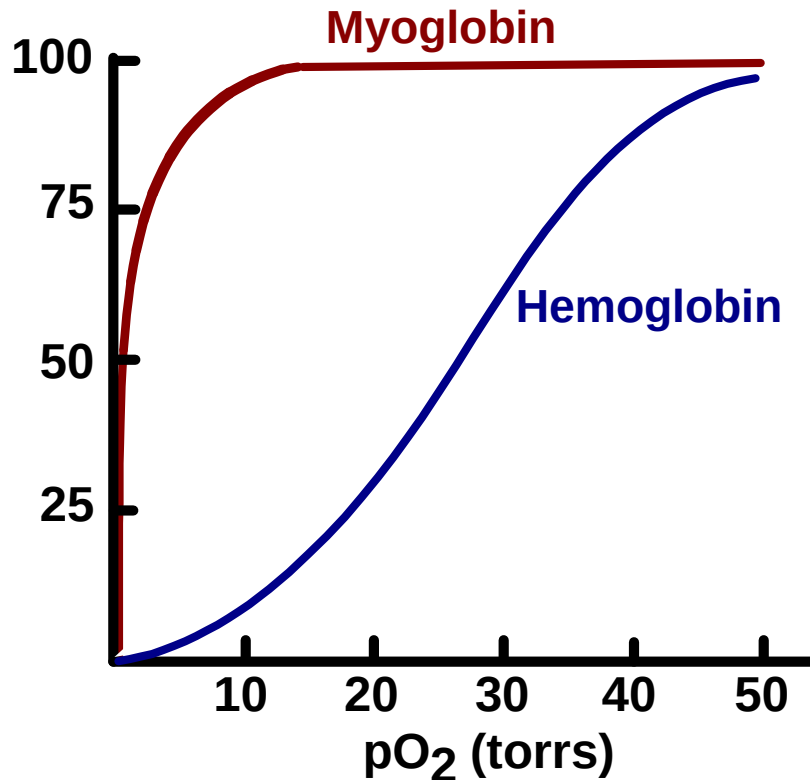
Hemoglobin

- Hemoglobin is found exclusively in red blood cells.
 - Function: transport oxygen from the lungs to the capillaries of the tissues.
- Hemoglobin A (HbA), major hemoglobin in adults, contains four polypeptide chains (**tetrameric**, two α chains and two β chains, $\alpha_2\beta_2$), each has a heme prosthetic group.
- α : 141 residues, β : 146 residues

• It is the **heme group** that gives blood their distinct **red color**



Oxygen binds to hemoglobin in a cooperative manner



- Oxygen dissociation curve:
 - shows the **percent saturation** of hemoglobin (or myoglobin) at various partial pressures of oxygen.

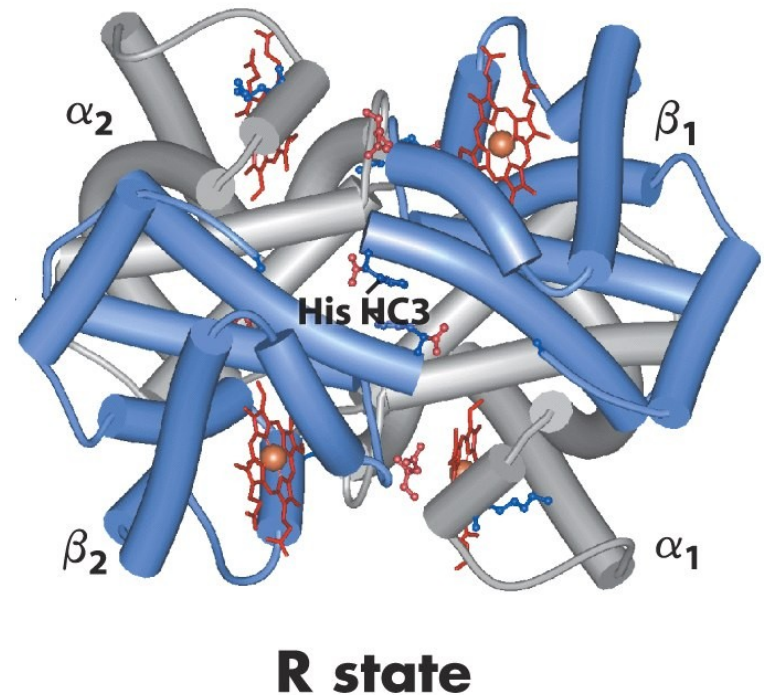
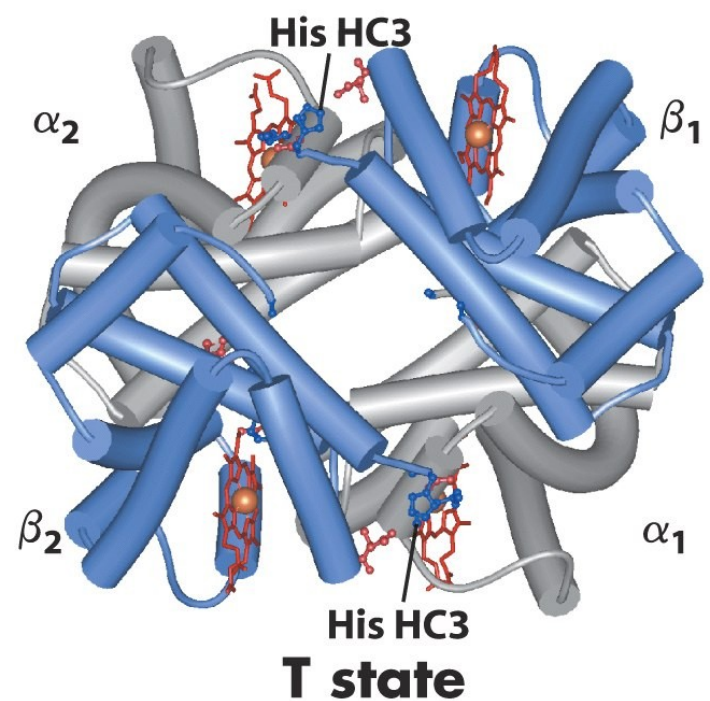
• The binding of oxygen to **myoglobin** follows a **hyperbolic curve**, while the binding of oxygen to **Hb** follows a **sigmoidal(S) curve**.

Allosteric effects

- The **S curve** suggests that the binding of one oxygen molecule to Hb increases its affinity for binding additional oxygen molecules.
- The oxygen binding sites in Hb “talk” to one another, so when one site binds O_2 , it makes it easier for the other sites to bind O_2 .
- This effect is known as **cooperative binding (allosteric effect)** and is often observed in multisubunit proteins.

Oxygen binding induces a conformational change in Hb

- In the **absence of bound O_2** , the Hb subunits are in a **low affinity state** (also known as the tense, or **T state**).
- The oxygen binding to Hb in the T-state triggers a conformational change to the R-state.
- **R state** (the relaxed state): **high affinity** state of hemoglobin.



Each subunit in hemoglobin can exist in either a high affinity (R) or low affinity (T) state



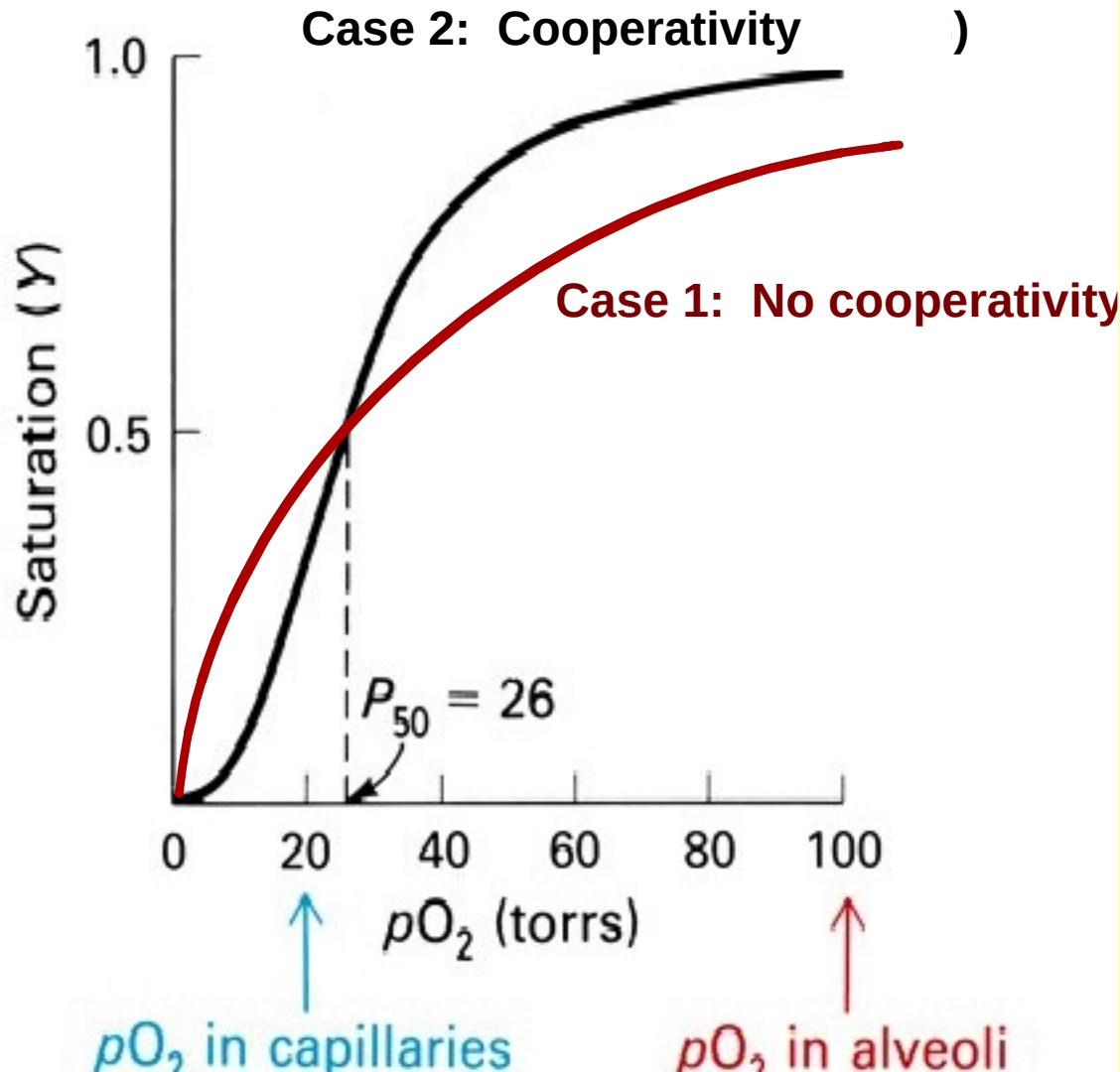
T state



R state

- T state = no O_2 , low affinity binding of O_2
- R state = high affinity binding of O_2

Significance of cooperativity in hemoglobin



Case 1: No cooperativity

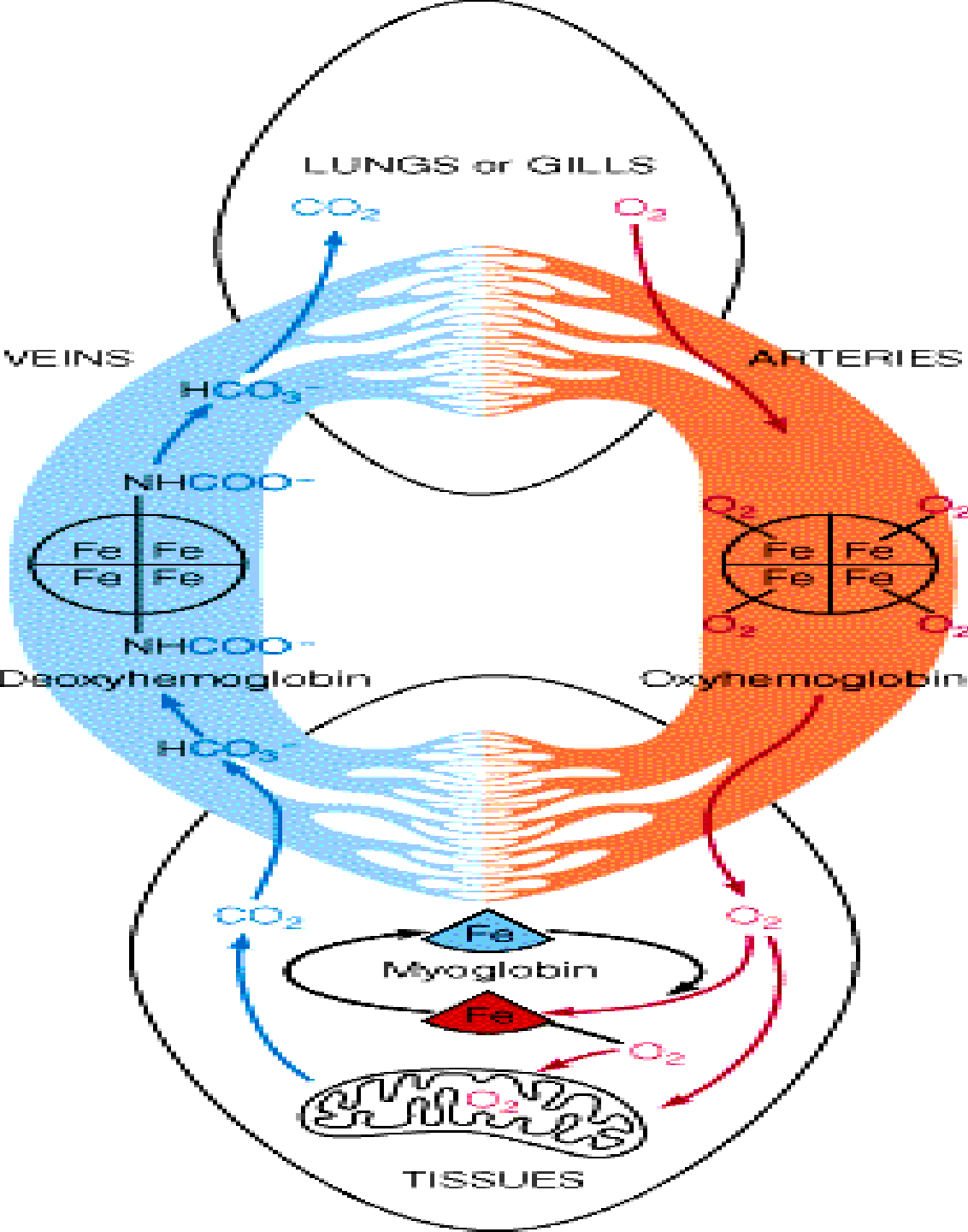
- In alveoli of lungs, $pO_2=100$, saturation, **79%**;
- In muscle, $pO_2=20$; saturation, **43%**;
- Fraction of O_2 delivered to muscle = $79\% - 43\% = \mathbf{36\%}$

Case 2: Cooperativity

- In alveoli of lungs, $pO_2=100$, saturation, **98%**;
- In muscle, $pO_2=20$, saturation, **32%**;
- Fraction of O_2 delivered to muscle = $98\% - 32\% = \mathbf{66\%}$

From the previous slide, we see that:

The cooperative binding of O_2 allows Hb to deliver 1.83 times more O_2 to the muscle cells under physiological conditions than would be delivered if the O_2 binding sites in Hb were independent of each other.



Hb transports O_2 and CO_2

Mb transports and stores O_2 in muscle tissues

Bohr effect

- *The release of oxygen* from hemoglobin is enhanced :
 - when the pH is lowered or
 - when the hemoglobin is in the presence of an increased partial pressure of CO₂.
- Both result in a decreased oxygen affinity of hemoglobin and, therefore, **a shift to the right** in the oxygen dissociation curve and both, then, stabilize the T state.
- This change in oxygen binding is called the **Bohr effect**.
- Conversely, raising the pH or lowering the concentration of CO₂ results in a **greater affinity for oxygen**, and *a shift to the left* in the oxygen dissociation curve.

Decrease in pH results in decreased oxygen affinity of hemoglobin and, therefore, a shift to the right in the oxygen dissociation curve.

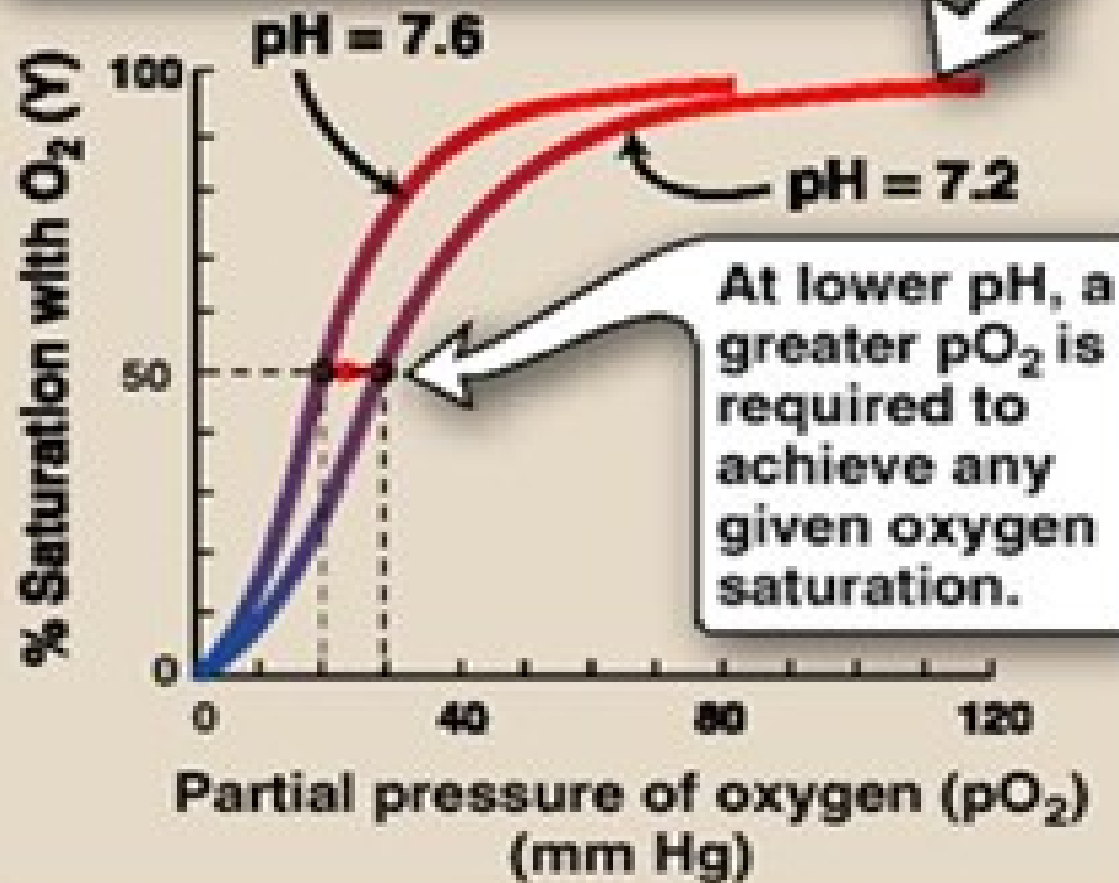


Fig. Effect of pH on the oxygen affinity of hemoglobin.
Protons are allosteric effectors of hemoglobin.

a) Source of the protons that lower the pH

- The concentration of both CO_2 and H^+ in the capillaries of metabolically active tissues is higher than that observed in alveolar capillaries of the lungs, where CO_2 is released into the expired air.
- Organic acids, such as lactic acid, are produced during anaerobic metabolism in rapidly contracting muscle.
- In the tissues, CO_2 is converted by carbonic anhydrase to carbonic acid:
$$\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3$$
- which spontaneously loses a proton, becoming bicarbonate (the major blood buffer):
$$\text{H}_2\text{CO}_3 \rightleftharpoons \text{HCO}_3^- + \text{H}^+$$

- The H^+ produced by this pair of reactions contributes to the lowering of pH.
- This differential pH gradient (lungs having a higher pH, tissues a lower pH) favours the unloading of oxygen in the peripheral tissues, and the loading of oxygen in the lung.
 - Thus, the oxygen affinity of the hemoglobin molecule responds to small shifts in pH between the lungs and oxygen-consuming tissues, making hemoglobin a more efficient transporter of oxygen.

b) Mechanism of the Bohr effect

- The Bohr effect reflects the fact that the deoxy form of hemoglobin has a greater affinity for protons than does oxyhemoglobin.
 - This effect is caused by ionizable groups, such as the N-terminal α amino groups, and specific histidine side chains that have higher pK_a s in deoxyhemoglobin than in oxyhemoglobin.
- Therefore, an increase in the concentration of protons (resulting in a decrease in pH) causes these groups to become protonated (charged) and able to form ionic bonds (also called salt bridges).
- These bonds preferentially stabilize the deoxy form of hemoglobin, producing a decrease in oxygen affinity.

- The Bohr effect can be represented schematically as:



- where an **increase in protons** (or a lower pO_2) shifts the equilibrium to the **right (favoring deoxyhemoglobin)**,
- whereas an **increase in pO_2** (or a decrease in protons) shifts the equilibrium to the **left**.

Effect of 2,3-bisphosphoglycerate on oxygen affinity:

- 2,3-BPG is an important regulator of the binding of oxygen to hemoglobin.
- It is the most abundant organic phosphate in the **red blood cell**, where its concentration is approximately that of hemoglobin.
- 2,3-BPG is synthesized from an intermediate of the glycolytic pathway .

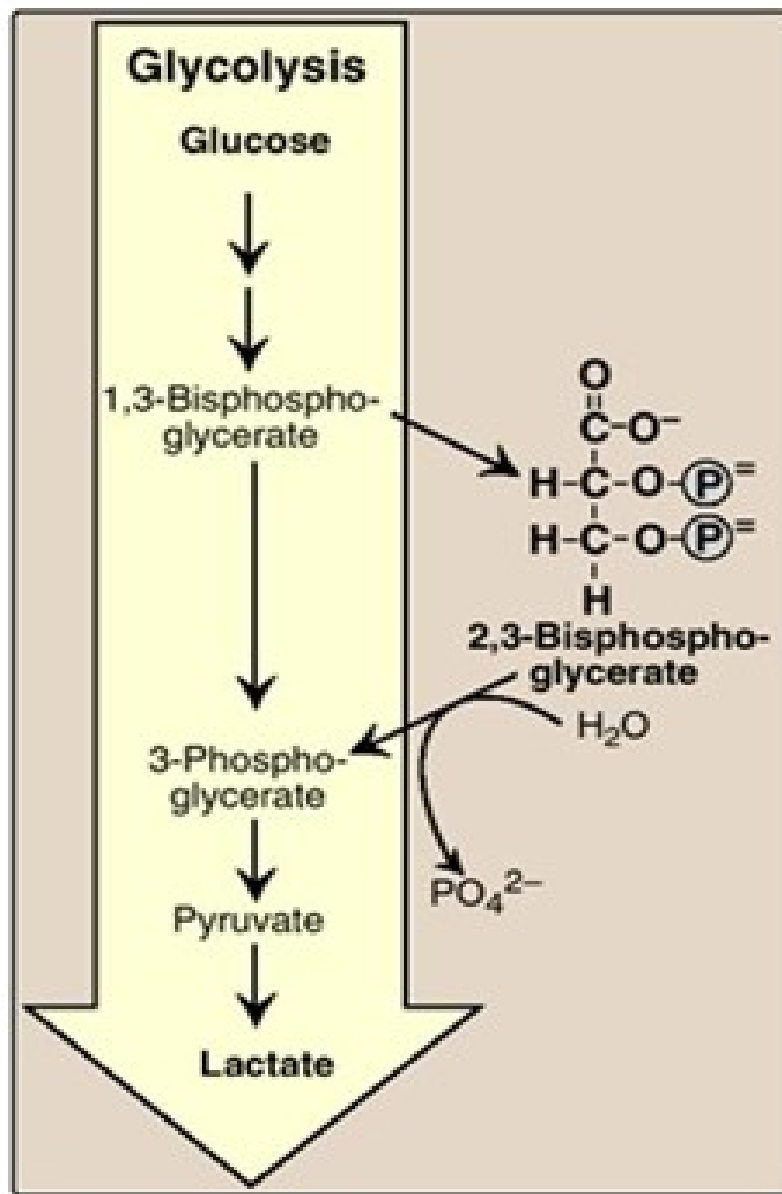


Fig. Synthesis of 2,3-bisphosphoglycerate.
[NB: P is a phosphoryl group]

a) Binding of 2,3-BPG to deoxyhemoglobin

- 2,3-BPG decreases the oxygen affinity of hemoglobin by binding to deoxyhemoglobin but not to oxyhemoglobin.
 - This preferential binding stabilizes the **taut conformation of deoxyhemoglobin**.
- The effect of binding 2,3-BPG can be represented schematically as:



b) Binding site of 2,3-BPG

- One molecule of 2,3-BPG binds to a pocket, formed by the **two β -globin** chains, in the center of the **deoxyhemoglobin** tetramer.
- This pocket contains several positively charged amino acids that form **ionic bonds** with the negatively charged phosphate groups of 2,3-BPG.
- 2,3-BPG is expelled on oxygenation of the hemoglobin.

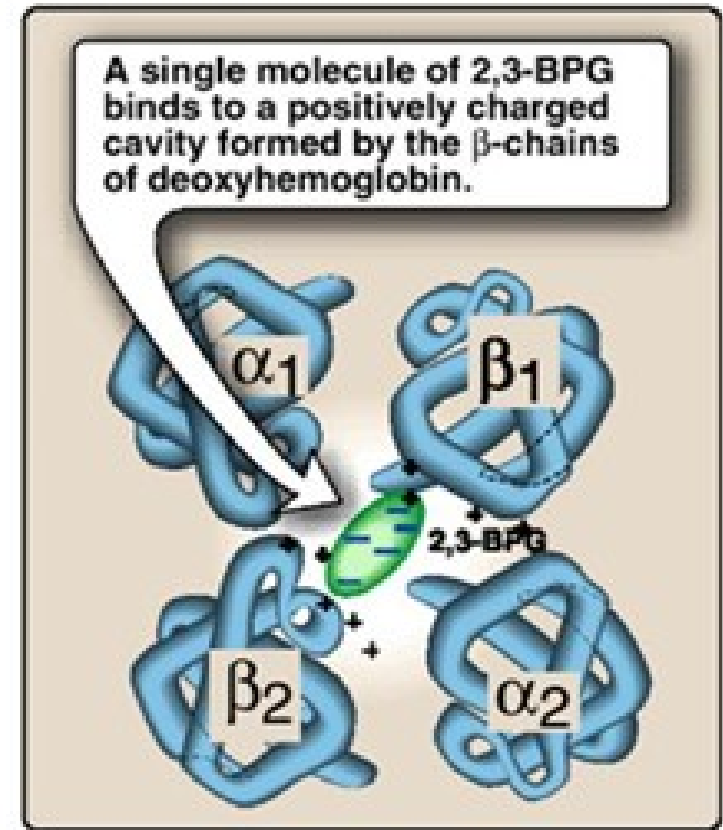


Fig. Binding of 2,3-BPG by deoxy-hemoglobin.

c) Shift of the oxygen dissociation curve

- Hemoglobin from which 2,3-BPG has been removed has a high affinity for oxygen.
- However, as seen in the red blood cell, the presence of 2,3-BPG significantly reduces the affinity of hemoglobin for oxygen, shifting the oxygen-dissociation curve to the right.
- This reduced affinity enables hemoglobin to release oxygen efficiently at the partial pressures found in the tissues.

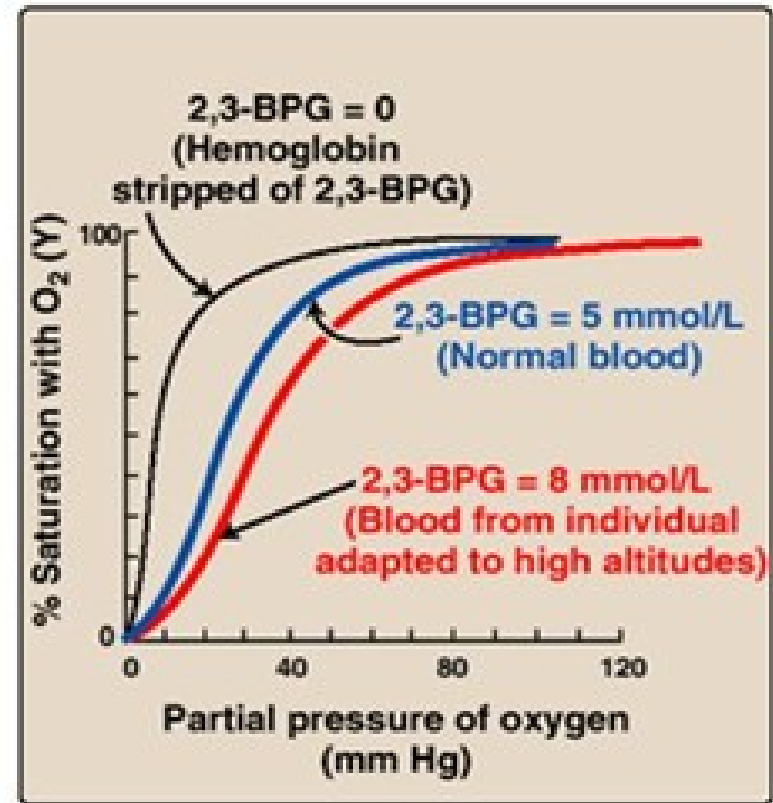


Fig. Allosteric effect of 2,3-BPG on the oxygen affinity of hemoglobin

Binding of CO₂

- Most of the carbon dioxide produced in metabolism is hydrated and transported as **bicarbonate ion** .
- However, some CO₂ is carried as carbamate bound to the **uncharged αamino groups** of hemoglobin (carbamino-hemoglobin),
- which can be represented schematically as follows:



- The binding of CO₂ stabilizes the **T (taut) or deoxy form of hemoglobin**, resulting in a **decrease in its affinity for oxygen**.
- In the lungs, CO₂ dissociates from the hemoglobin, and is released in the breath.

Binding of CO

- Carbon monoxide (CO) binds tightly (but reversibly) to the hemoglobin iron, forming carbon monoxyhemoglobin (or *carboxyhemoglobin*).
- When *carbon monoxide* binds to one or more of the four heme sites, *hemoglobin shifts to the relaxed conformation, causing the remaining heme sites to bind oxygen with high affinity.*
- This shifts the oxygen saturation curve to the *left*, and changes the *normal sigmoidal shape toward a hyperbola.*
- *As a result, the affected hemoglobin is unable to release oxygen to the tissues.*

- [**NB**: The affinity of hemoglobin for CO is 220 times greater than for oxygen. Consequently, even minute concentrations of carbon monoxide in the environment can produce **toxic** concentrations of **carbon monoxyhemoglobin** in the blood.
 - For example, increased levels of CO are found in the blood of **tobacco smokers**. Carbon monoxide toxicity appears to result from a combination of tissue hypoxia and direct CO-mediated damage at the cellular level.]
- ❖ Carbon monoxide poisoning is **treated** with 100 percent **oxygen therapy**, which facilitates the dissociation of CO from the hemoglobin.

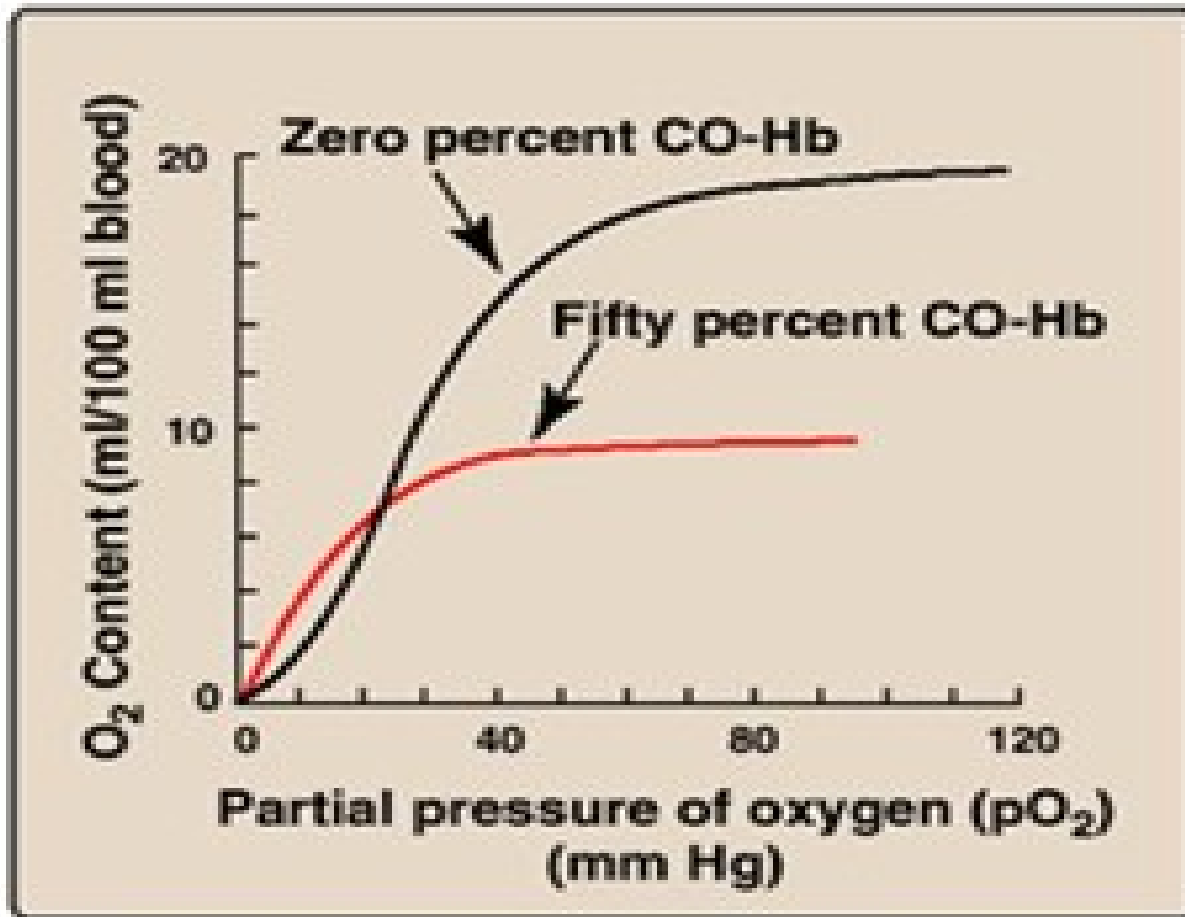
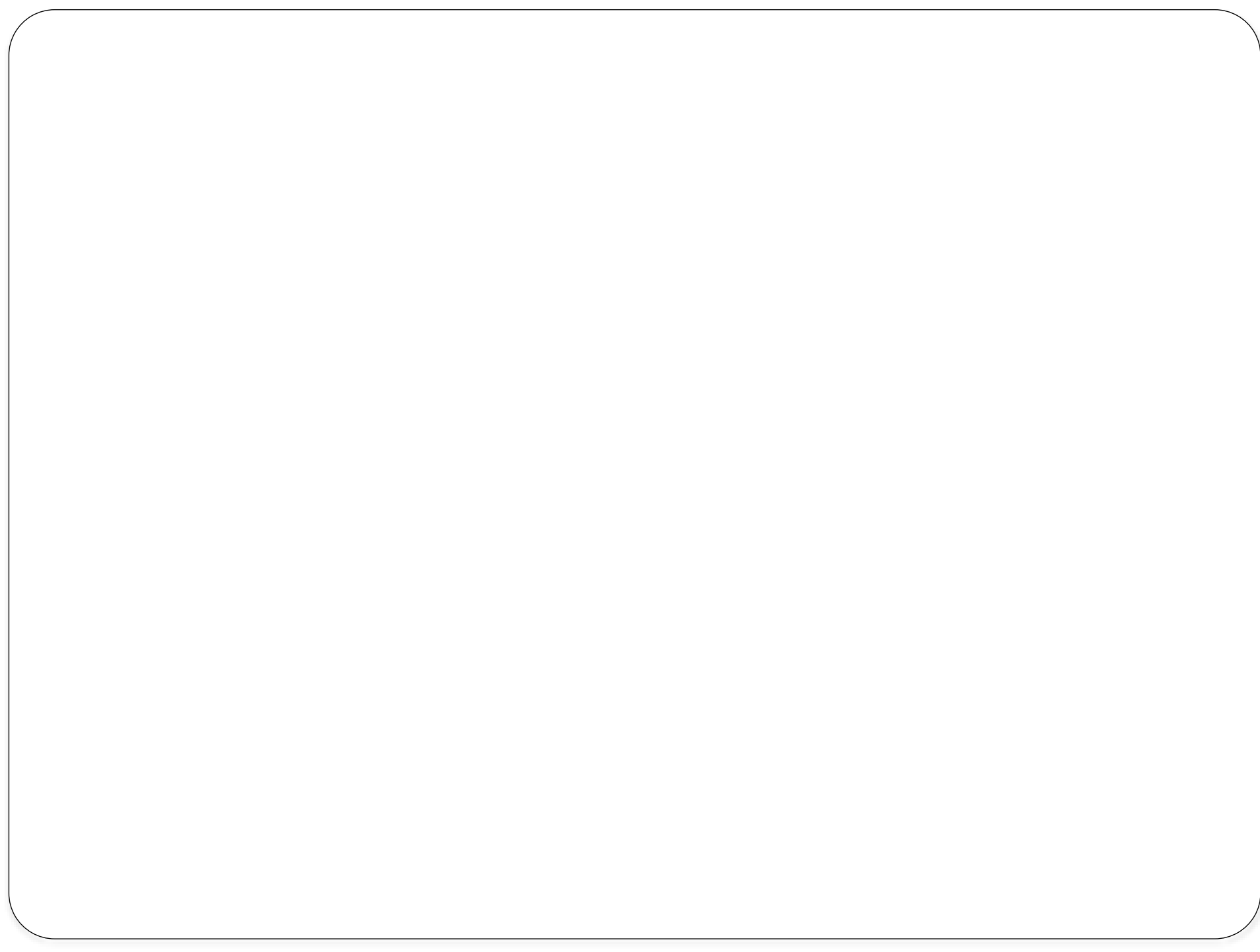
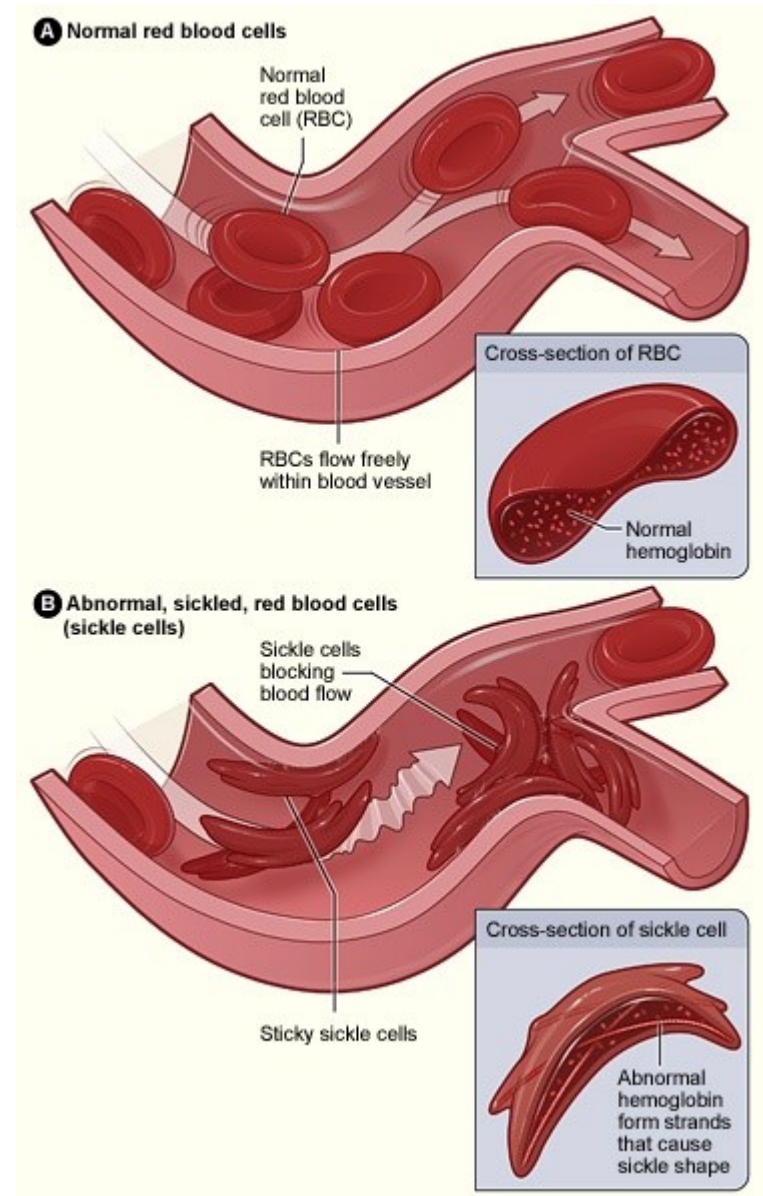
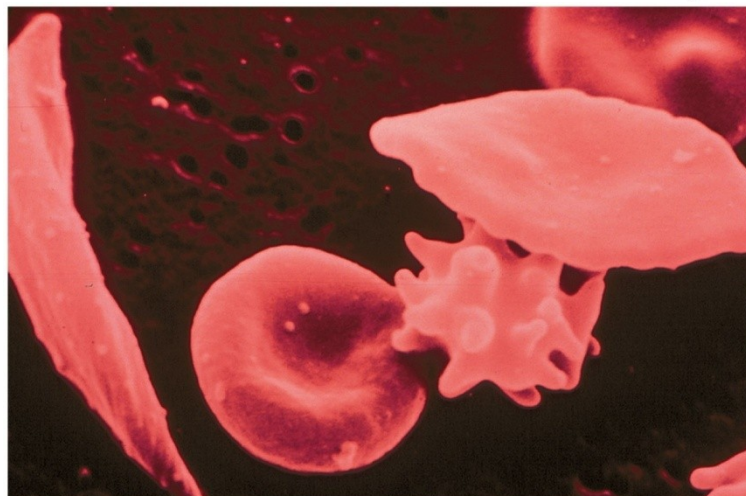
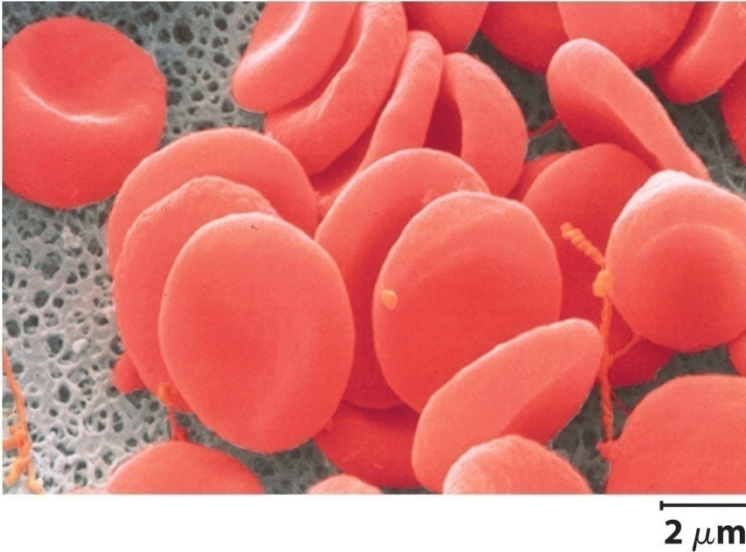


Fig. Effect of carbon monoxide on the oxygen affinity of hemoglobin. CO-Hb = carbon monoxy-hemoglobin.



Hemoglobinopathies: abnormal hemoglobins

Sickle-cell anemia (sickle-cell hemoglobin; HbS)



Occurrence of the disease

- Sickle-cell anemia is largely confined to tropical areas of the world.
- It primarily occurs in the black population.
- It is estimated that 1 in 500 newborn black infants in the USA are affected by sickle-cell anemia.

Worldwide:

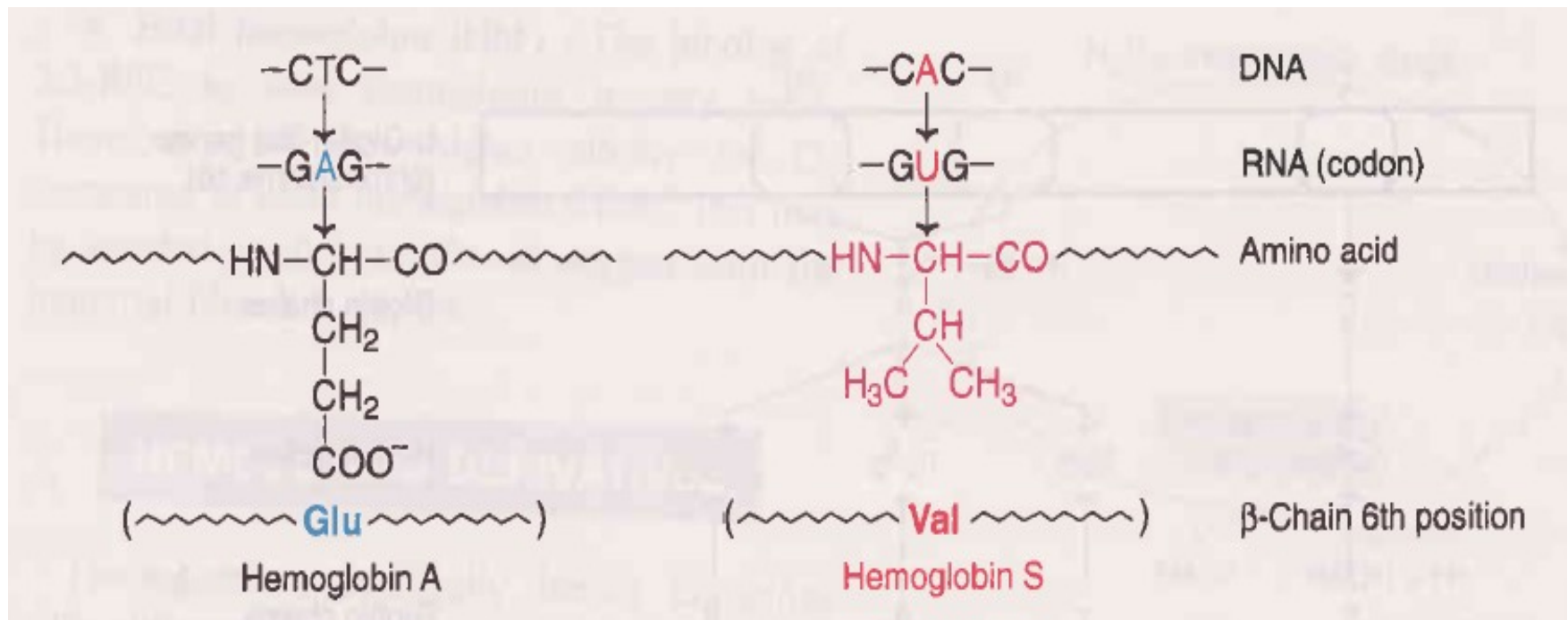
- 5% of world population harbor alleles for hemoglobinopathies
- 300,000 children born each year with hemoglobinopathy
- 200,000 children born yearly in Africa with Sickle Cell Disease
- **Areas of Prevalence**
 - Sub-Saharan Africa
 - S.C. trait frequency 10-40%
 - Highest rates in Ghana, Nigeria, Uganda

...cont'd

- In case of sickle-cell anemia, the hemoglobin (HbS) has:
 - two normal α -globin chains and
 - two abnormal (mutant) β -globin chains.
- This is due to a difference in a single amino acid.
- In HbS, glutamate at sixth position of β -chain is replaced by valine ($\text{Glu } \beta_6 \longrightarrow \text{Val}$).
- Sickle-cell anemia is due to :
 - a change (missense mutation) in the single nucleotide (thymine $\longrightarrow$ adenine) of β -globin gene.

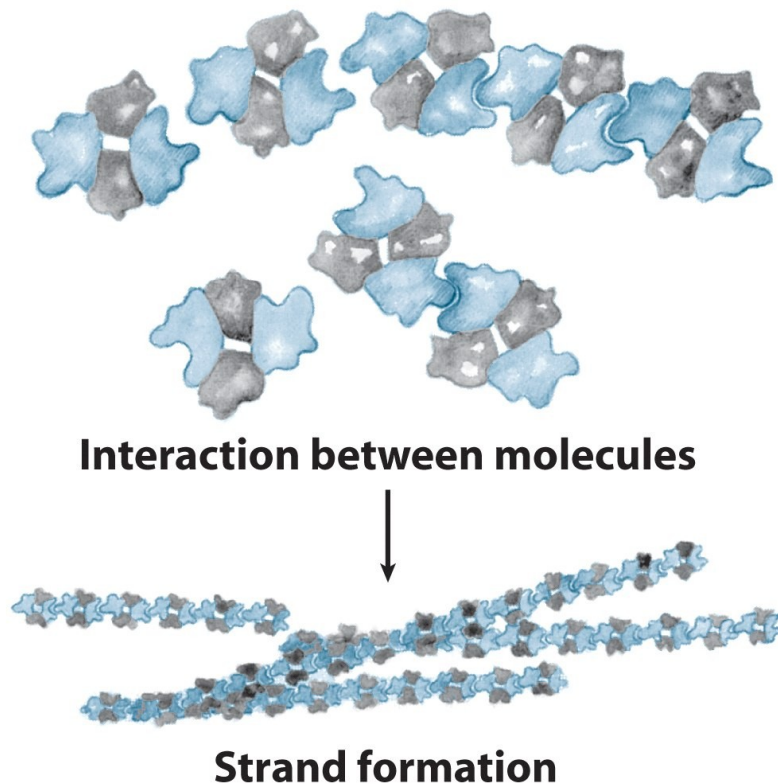
...cont'd

- This error causes:
 - the formation of **altered codon** (**GUG** in place of **GAG**)
 - which leads to the incorporation of **valine** instead of **glutamate** at the **sixth position in β -chain**.



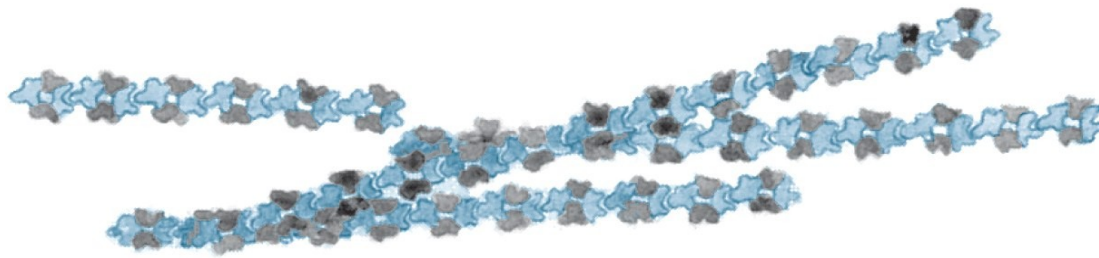
A change of a glutamic acid residue for a valine at position 6 of the β -chain.

	gene	peptide	property of AA
Hb A β	G A G	glutamic acid	polar
Hb S β	G T G	valine	hydrophobic

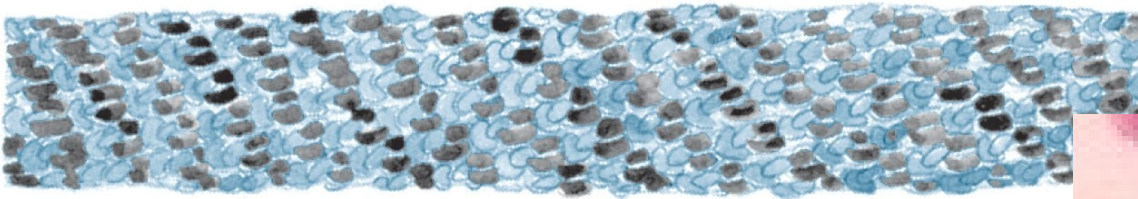


- The hydrophobic residues of the **valine** at position 6 of the beta chain in hemoglobin are able to associate with the hydrophobic patch.
- Cause HbS molecules to aggregate and form fibrous precipitates.

Molecular disease

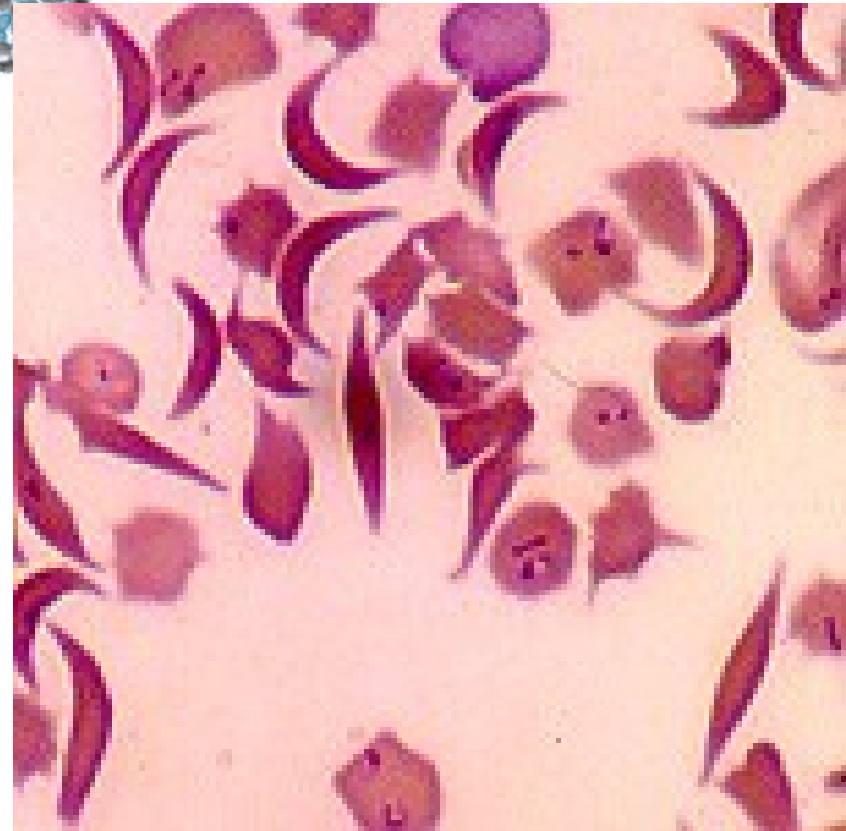


Strand formation



**Alignment and crystallization
(fiber formation)**

- Sickle-cell anemia is a genetically transmitted, hemolytic disease.

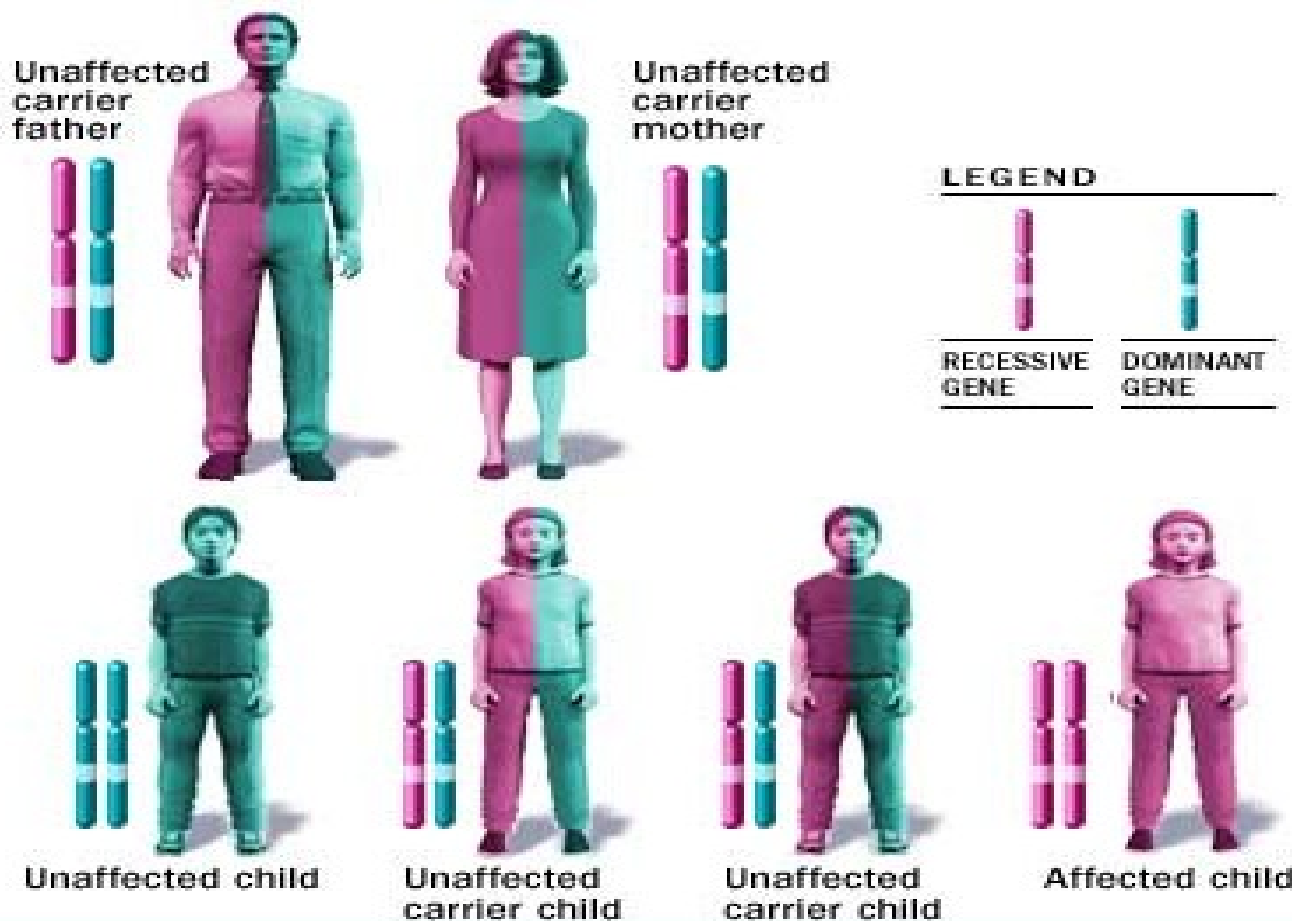


Homozygous and heterozygous HbS

- Sickle cell anemia is said to be homozygous,
 - if caused by inheritance of two mutant genes (one from each parent) that code for β -chains.
- In case of heterozygous HbS,
 - only one gene (of β -chain) is affected while the other is normal.
- The erythrocytes of heterozygotes contain:
 - both HbS and HbA and
 - the disease is referred to as sickle cell trait
 - which is more common in blacks (almost 1 in 10 are affected).
- The individuals of sickle-cell trait lead a normal life, and do not usually show clinical symptoms.
- This is in contrast to homozygous sickle-cell anemia.

Causes

- Normal hemoglobin is classified as Hemoglobin A, while sickle-cell patients produce Hemoglobin S.
- Sickle-cell anemia is caused by an **autosomal homozygous recessive trait**.
 - Thus, each parent's genotype must contain at least one recessive allele to produce affected offspring.
- The presence of only one allele is also potentially harmful,
 - as carriers of sickle-cell anemia produce both types of hemoglobin, and
 - thus often experience minor symptoms in environments of limited oxygen.



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Inheritance: Autosomal recessive.

Parents with sickle cell trait:

- Probability of child with hemoglobin AA: 25%
- Probability of child with sickle cell trait AS: 50%
- Probability of child with sickle cell disease SS: 25%

Abnormalities associated with HbS

1. Life-long hemolytic anemia :

- The sickled erythrocytes are fragile and their continuous breakdown leads to life-long anemia.

2. Tissue damage and pain :

- The sickled cells block the capillaries resulting in poor blood supply to tissues .
- This leads to extensive damage and inflammation of certain tissues causing pain.

3. Increased susceptibility to infection :

- Hemolysis and tissue damage are accompanied by increased susceptibility to infection and diseases.

4. Premature death :

- Homozygous individuals of sickle-cell anemia die before they reach adulthood (< 20 years).

Sickle-cell trait provides resistance to malaria

- The incidence of sickle-cell disease coincides with;
 - the high incidence of malaria in tropical areas of the world (particularly among the black Africans).
- Sickle-cell trait (heterozygous state with about 40% HbS) provides;
 - resistance to malaria
 - which is a major cause of death in tropical areas.
- Unfortunately, homozygous individuals, the patients of sickle-cell anemia (much less frequent than the trait), cannot live beyond 20 years.

1. Malaria is a parasitic disease caused by *Plasmodium falciparum* in Africa.
 - The **malarial parasite** spends a part of its **life cycle in erythrocytes**.
 - **increased lysis of sickled cells** is (shorter life span of erythrocytes) **interrupts the parasite cycle**.

2. More recent studies indicate that **malarial parasite increases the acidity of erythrocytes**(pH down by 0.4).
 - The **lowered pH increases the sickling of erythrocytes** to about 40% from the normally occurring 2%.
 - Therefore, the **entry of malarial parasite promotes sickling leading to lysis of erythrocytes**.
 - Furthermore, the concentration of K^+ is low in sickled cells which is unfavourable for the parasite to survive.

5.3.2 Fibrous protein

- Proteins that tend to be **insoluble and strong** and so **play a structural role** in organisms for support or protection.
- Types:
 - **Collagens**, the most abundant proteins in a vertebrate body, found in connective tissues such as cartilage.
 - **Keratins**, found in hair, fingernails, and bird feathers.
 - **Elastins**, found in ligaments, around blood vessels.

•The keratin in hair is a fibrous protein



Collagen

- Collagen is the **most abundant protein** in mammals.
 - About 25% of the total protein mass in mammals is collagen.
 - It is strong, extensible, insoluble and inert.
 - It is a major component of tendons, the extracellular matrix of the connective tissues (skin, bone matrix), and the cornea of the eye.

Collagen in different organisms

Tissue	Content
Bone	88.0
Calcaneal tendon	86.0
Skin	71.9
Cornea	68.1
Cartilage	46-63
Ligament	17.0
Aorta	12-24
Liver	3.9

Structure of collagen

- A typical collagen molecule is a long, rigid structure in which three polypeptides " α -chains" are wound around one another in a rope-like **triple-helix**.

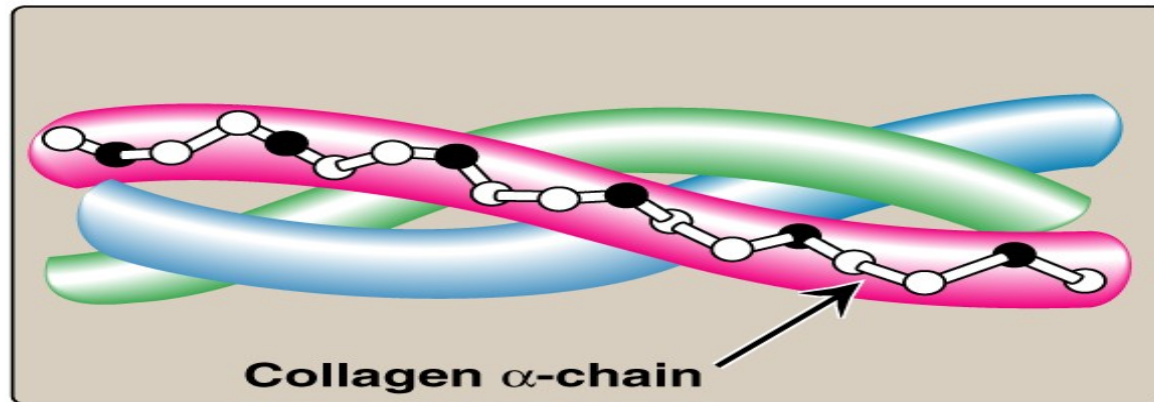


Figure 4.1
Triple-stranded helix of collagen.

- The three polypeptide α -chains are held together by **hydrogen bonds** between the chains.

Amino acid sequence of collagen

- The amino acid sequence of collagen is revealed to be **remarkable regular**.
 - Nearly every third residue is **Gly (G-X-Y)**.
 - It is abundant in Pro and Hyp(hydroxylproline).
 - The sequence Gly-X-Hyp/Pro recurs frequently.

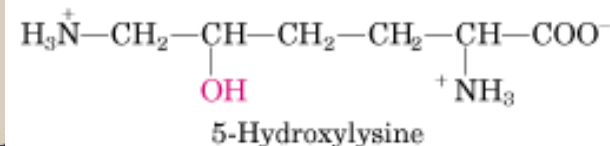
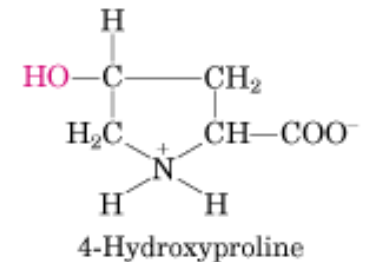
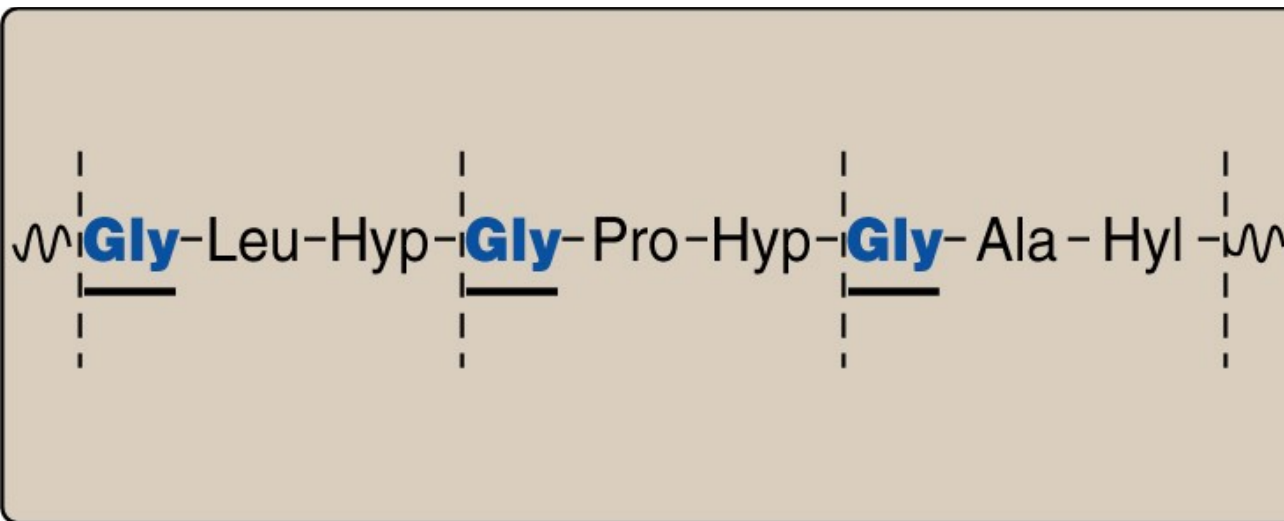
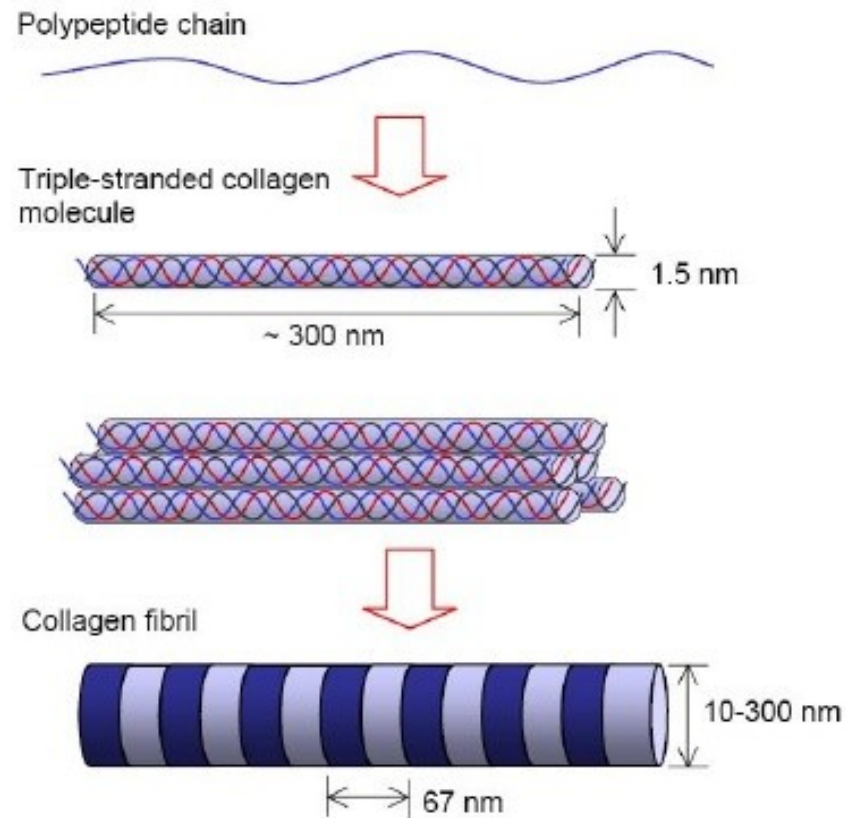
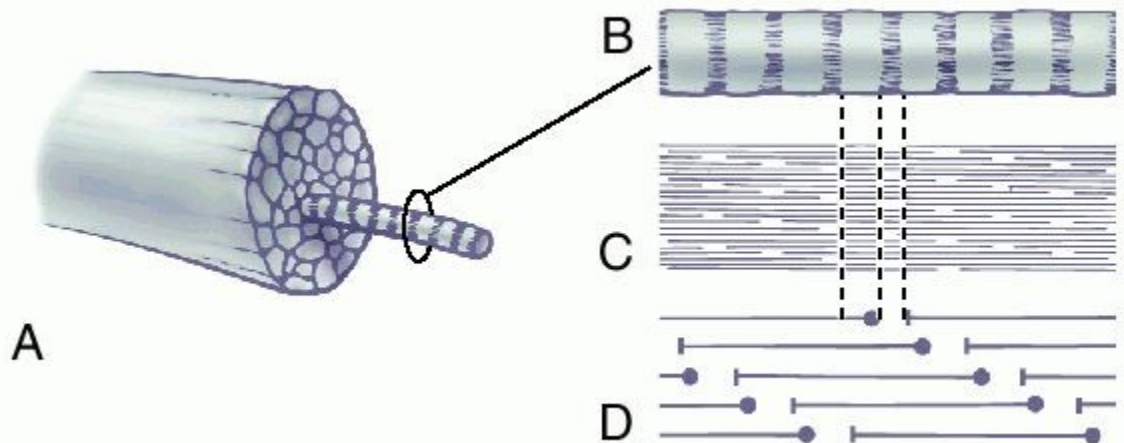


Figure 4.5
Amino acid sequence of a portion of the $\alpha 1$ -chain of collagen. [Note: Hyp is hydroxyproline and Hyl is hydroxylysine.]

- Collagen triple helices can form much larger collagen **fibrils** that further aggregate into collagen **fibers**.



- The unit fibrils themselves comprise triple helices molecules **in regular staggered arrays (C, D)**.

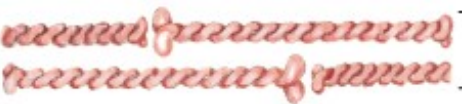


Keratins

- The chief structural constituent of hair, nails, horns, feathers and hooves.
- Keratins are rich in hydrophobic residues.
 - Phe, Ile, Val, Met, and Ala residues are rich, this makes the keratins insoluble in water.
- Two helical strands oriented in parallel are wrapped together to form a superhelix in keratin.
- The individual α -helices are cross linked by interchain disulfide bonds.

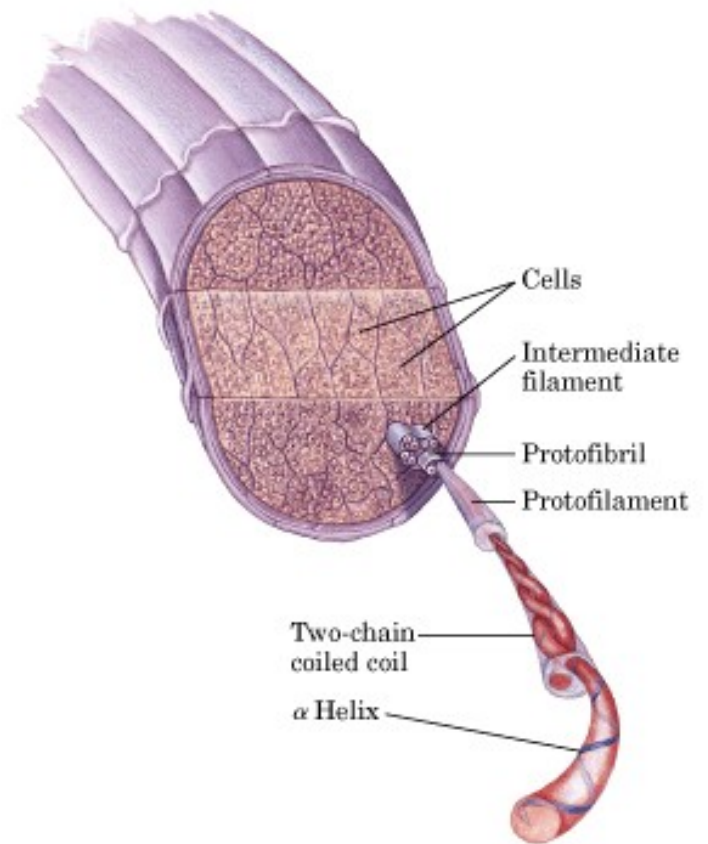
Keratin α helix — 

Two-chain coiled coil — 

Protofilament {  } 20–30 Å

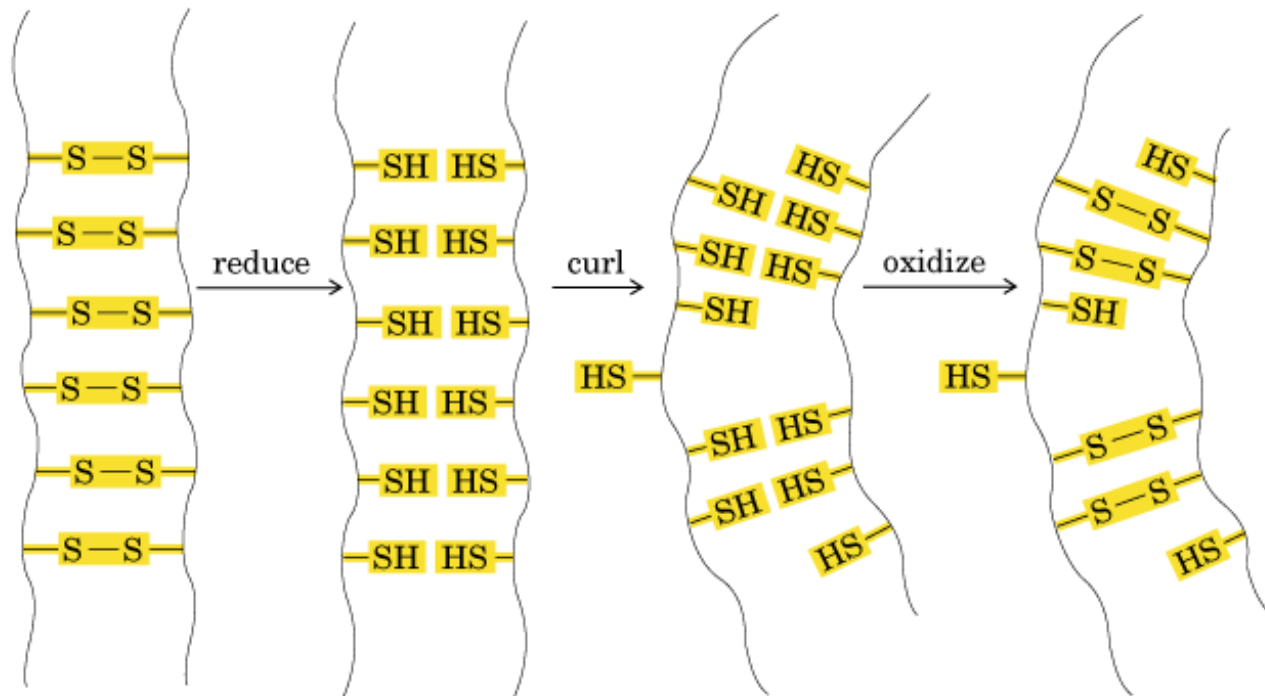
Protofibril {  }

(a)



Cross section of a hair
(b)

- Keratins contain higher number of **Cys**.
- Permanent waving of hair is biochemical engineering, where **disulfide bonds** between individual chains are **reduced**, **curled**, and **reoxidized**.



Points

- Functions of proteins
- Structure of Proteins (four levels)
 - Definition, bonds or interactions, character
- Myoglobin and hemoglobin
 - Structure, function
 - Allosteric effects
 - Sickle-cell anemia
- Fibrous protein
 - Collagen, keratin: structure
- Denaturation of proteins

Reading assignment

Biomedical Implications

- Myoglobinuria
- Anemias
- Thalassemias
- Glycated Hemoglobin (HbA1c)

Thank You!!